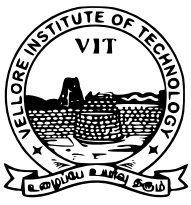
ENHANCED FEATURE REPRESENTATION FOR DEEP LEARNING-BASED CLASSIFICATION OF ECG SIGNALS

Submitted in partial fulfilment of the requirements for the degree of

Doctor of Philosophy

by

PREMANAND S



VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

October 2026

DECLARATION

I, **PREMANAND S** hereby declare that the thesis entitled **"Enhanced Feature Representation for Deep Learning-based Classification of ECG Signals"** submitted to Vellore Institute of Technology (VIT), Chennai for the award of the degree of *Doctor of Philosophy* is a record of bonafide work carried out by me under the supervision of **DR S SATHIYA NARAYANAN**, Associate Professor, School of Electronics Engineering, Vellore Institute of Technology, Chennai.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.



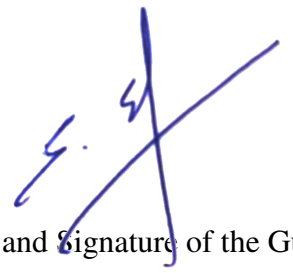
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ABSTRACT

Cardiovascular Disease (CVD) continues to be the primary cause of mortality worldwide and precise and timely diagnosis is vital, with Electrocardiogram (ECG) playing a key role. Machine Learning (ML) and Deep Learning (DL) techniques have pushed automated ECG analysis to new heights, but there are still several drawbacks, such as using homogeneous Activation Function (AF), modeling the long-range temporal dependencies poorly, and few features derived from a single ECG representation. This thesis offers three complementary answers to these challenges.

The first one is the Convolution-based Heterogeneous Activation Facility (CHAF), which seeks to enhance the ability of convolutional blocks to extract nonlinear features from raw ECG signals by introducing heterogeneous AFs. CHAF is evaluated on PTB and MIT-BIH arrhythmia database. CHAF is evaluated on PTB and MIT-BIH. The 99.55% / 99.68% / 0.008 s figures are for PTB, and 99.08% is for MIT-BIH.

The second contribution, ECG-ResViT, is a fusion of residual and dilated CNNs with a Vision Transformer (ViT) to simultaneously learn local morphological information and long-range temporal dependencies from the one-dimensional ECG signal. On a three-class ECG dataset, ECG-ResViT performs with a 99.13% accuracy and an ROC-AUC of 99.94%.

BlendNet is the third contribution, which fuses a Continuous Wavelet Transform (CWT)-based scalogram and its binarised version by parallel convolutional streams to merge spectral and structural features of the ECG. BlendNet correctly classifies all the test cases (mean 99.21 $\pm$ 1.05% under five-fold stratified cross-validation) on the PhysioNet ARR–CHF–NSR database for all train–test splits at α = 0.7.

In summary, the proposed architectures solve the problems of activation homogeneity, limited temporal modeling and single-representation feature constraints in ECG classification. High accuracy and computational efficiency are demonstrated on benchmark databases, providing a foundation for further clinical validation.

Keywords: *Electrocardiogram, Deep Learning, Convolutional Neural Network, Heterogeneous Activation Function, Machine Learning, Vision Transformer, Residual Network, Dilated Convolution, Continuous Wavelet Transform, Scalogram, Alpha Blending, Binary Image, Feature Extraction.*

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LIST OF SYMBOLS AND ABBREVIATIONS

AF	Activation Function
ARR	Arrhythmia
CVD	Cardiovascular Disease
CatBoost	Categorical Boosting
Acc	Classification Accuracy
CReLU	Concatenated Rectified Linear Unit
CHF	Congestive Heart Failure
CWT	Continuous Wavelet Transform
CHAF	Convolution-based Heterogeneous Activation Facility
CNN	Convolutional Neural Network
DT	Decision Tree
DL	Deep Learning
DALY	Disability-Adjusted Life Year
DWT	Discrete Wavelet Transform
ECG	Electrocardiogram
EEG	Electroencephalogram
Time	Execution Time
ELU	Exponential Linear Unit
XGBoost	Extreme Gradient Boosting
FN	False Negative
FP	False Positive
FFT	Fast Fourier Transform
FFN	Feed-Forward Network
GRU	Gated Recurrent Unit
GELU	Gaussian Error Linear Unit
GAN	Generative Adversarial Network
GAP	Global Average Pooling
GAF	Gramian Angular Field
GNN	Graph Neural Network

HRV	Heart Rate Variability
HGB	Histogram Gradient Boosting
Tanh	Hyperbolic Tangent
IoT	Internet of Things
KNN	K-Nearest Neighbour
LS	Label Smoothing
LightGBM	Light Gradient Boosting Machine
LSTM	Long Short-Term Memory
ML	Machine Learning
MTF	Markov Transition Field
MIT-BIH	Massachusetts Institute of Technology–Beth Israel Hospital Arrhythmia Database
MHSA	Multi-Head Self-Attention
MI	Myocardial Infarction
NLP	Natural Language Processing
NAS	Neural Architecture Search
Non-IID	Non-Independent and Identically Distributed
NSR	Normal Sinus Rhythm
1D	One-Dimensional
PPG	Photoplethysmography
PTB	Physikalisch-Technische Bundesanstalt
PCA	Principal Component Analysis
RF	Random Forest
ReLU	Rectified Linear Unit
ReLU-6	Rectified Linear Unit 6
RP	Recurrence Plot
RNN	Recurrent Neural Network
STFT	Short-Time Fourier Transform
SOTA	State-of-the-Art
SVM	Support Vector Machine
SMOTE	Synthetic Minority Oversampling Technique
t-SNE	t-distributed Stochastic Neighbour Embedding
TL	Transfer Learning

TN	True Negative
TP	True Positive
2D	Two-Dimensional
UMAP	Uniform Manifold Approximation and Projection
VCG	Vectorcardiogram
ViT	Vision Transformer
WPD	Wavelet Packet Decomposition
WHO	World Health Organization

Chapter 1

INTRODUCTION

1.1 Background and Motivation

In the world, the most common causes of illness and early deaths are CVD. The aggregate of CVD mortality in the world in 2022 was nudged to be around 19.8 million, with ischemic heart disease being the leading cause of CVD mortality with an age-standardized burden of 2,275.9 Disability-Adjusted Life Year (DALY) per 100,000 people (Mensah *et al.*, 2023). In addition to continued modifiable risk factors and an aging population, a new study estimates that by 2050, there will be an additional 73.4% increase in crude CVD deaths (to 35.6 million deaths / year) (Chong *et al.*, 2025). More than 75% of these deaths are in low and middle income countries with limited access to specialist cardiology care and health care (World Health Organization, 2023).

Electrical activity of the heart over time is measured by a low-cost, non-invasive clinical instrument called an ECG. For some of the cardiac conditions, like arrhythmia, Myocardial Infarction (MI), Congestive Heart Failure (CHF), and bundle branch blocks, ECG is a primary diagnostic procedure. In order to see precise indicators of cardiac electrical health, we need to understand the characteristic morphological features of an ECG waveform — the P wave (atrial depolarization), the QRS complex (ventricular depolarization), and the T wave (ventricular repolarization), along with the associated intervals (PR, QT, RR) and segments (ST) (Ansari *et al.*, 2023). Any deviation from normalcy in this regard is an indication of the underlying physiologic disorder that needs immediate medical attention.

Analysis of ECG is one of the most basic clinical procedures, especially during emergencies where prompt classification is necessary. Manual analysis is also needed to spot differences among observations, which is time-consuming and subjective (Wu and Guo, 2025). This issue has been further compounded by the availability of ECG monitoring systems like holter monitors, event recorders, and single-lead monitoring systems used in smartphones (Panwar *et al.*, 2025). All of these have increased the amount of ECG data, which cannot be interpreted

by humans. These developments have resulted in the need for automated computational procedures for analyzing ECG data.

1.2 Automated ECG Analysis: The ML-to-DL Evolution

Conventional signal processing techniques and rule-based systems based on manually defined parameters like RR interval statistics, waveform amplitude, and morphological templates were used in the early initiatives to automating ECG interpretation. However, to satisfy these computationally convenient requirements, they were often constrained in real-world use for different patients, different ECG recording methods and different lead configurations (Safdar *et al.*, 2024). In order to overcome these limitations, ML tools like KNN, Support Vector Machine (SVM), Random Forest (RF) and Extreme Gradient Boosting (XGBoost) are employed (Aziz *et al.*, 2021). These techniques improved the accuracy of classification by learning patterns in representations of structured ECGs. This has been the result in more than one benchmark study, where, in some cases, the Classification Accuracy (Acc) of tree-based ensemble models, such as XGBoost and RF, is above 98% (Premanand and Narayanan, 2021).

Although ML offers numerous advantages, it is still highly reliant on manual feature engineering, a process that demands a strong background in the field and can sometimes overlook the full complexity of the ECG signal (Jaya Prakash *et al.*, 2025). The restriction is more pronounced in non-stationary ECG signals with subtle inter-class differences, including distinguishing cardiac arrhythmias from Normal Sinus Rhythm (NSR) and discriminating between similar arrhythmia patterns.

In contrast to ML, by enabling models to learn hierarchical representations of features directly from raw or very little preprocessed data, DL has completely transformed automated ECG analysis. Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM) networks, and transformer-based models are just a few of the designs that have recently demonstrated outstanding performance in biological signal categorization (Hong *et al.*, 2020). CNN-based architectures have gained considerable popularity among these because of their high performance in obtaining discriminative local features from ECG signals (Xiao *et al.*, 2023). Even with signal noise and class imbalance, several studies have reported high classification accuracy of 99% or more on benchmark database like

MIT-BIH Arrhythmia database when DL models are directly applied to raw ECG waveforms (Ahmed, Ali, Abdullah and Malebary, 2023).

1.3 CNNs for Automated ECG Classification

CNNs have proven excellent capability in recognising local morphological features present in ECG data, such as acute QRS complexes, ST-segment alterations, and subtle anomalies in P-wave patterns, through multilayer convolution and pooling processes (Alzubaidi *et al.*, 2021). Multiple convolutional blocks made up of convolution layers, AFs, and pooling layers are typically found in a CNN architecture intended for One-Dimensional (1D) ECG classification. Fully linked layers are then used for final classification. Advanced CNN variants such as ResNet and architectures using depth-wise separable convolutions have further improved feature learning capability while reducing training difficulties associated with deep networks (Khan *et al.*, 2023).

Recently, there has been a lot of interest in hybrid architectures that combine traditional ML classifiers with CNN-based extraction of features. For computationally demanding fully connected layers, these techniques supplement lightweight classifiers such as SVM, KNN, or XGBoost. These architectures are appealing for real-world cardiac diagnostic applications since they can achieve competitive classification performance while drastically lowering inference time (Ghosh *et al.*, 2022). A major flaw that is still frequently overlooked in many of these hybrid CNN-ML approaches is the periodic usage of the same AF, frequently Rectified Linear Unit (ReLU), for all convolutional layers. This homogeneous activation strategy may limit the diversity of non-linear feature transformations found across the network, as well as the richness of resulting feature representations.

By adding non-linearity, affecting gradient propagation during training, and deciding how features are given priority at various network layers, AFs are essential to DL architectures (Hao *et al.*, 2020). Different AFs such as ReLU, Exponential Linear Unit (ELU), Hyperbolic Tangent (Tanh), Mish, Swish, SoftSign, Concatenated Rectified Linear Unit (CReLU), and Rectified Linear Unit 6 (ReLU-6) exhibit distinct nonlinear behaviors and can affect classification performance differently (Agarap, 2026). Nevertheless, most existing CNN-based ECG classification architectures continue to employ a single fixed AF across the entire architec-

ture, without exploring the potential advantages of heterogeneous activation learning. This limitation forms one of the key motivations for the first contribution of this thesis.

The limited receptive field of CNN architectures is another important drawback that hinders their capacity to identify long-range temporal relationships in ECG signals (Kim *et al.*, 2025). ECG recordings are multi-scale temporal signals that contain clinically significant information in both longer temporal relationships like RR interval variability and beat-to-beat rhythm patterns, as well as short-duration waveform characteristics like QRS breadth and T-wave morphology (Islam *et al.*, 2022). CNN kernels frequently have trouble accurately modeling these lengthy temporal relationships over several cardiac cycles since they only work over small local regions. This limitation has been highlighted in several recent studies and review works focusing on advanced ECG classification architectures (Ansari *et al.*, 2025).

1.4 Transformer Architectures for ECG Analysis

The Transformer architecture was originally introduced for Natural Language Processing (NLP) tasks and is based on a self-attention mechanism capable of modeling relationships between all elements in a sequence, regardless of their positional distance (Vaswani *et al.*, 2023). Unlike conventional sequential models, Transformers can capture both local and long-range dependencies simultaneously, making them highly suitable for analyzing biomedical time-series signals such as ECG recordings, where temporal context plays an important role in diagnosis. ViTs, which extend transformer architectures to image-based inputs through patch-wise tokenization, have also shown remarkable performance in various image classification tasks (Dosovitskiy *et al.*, 2021).

Recently, researchers have begun using the Transformer and ViT architectures for ECG classification tasks, and have achieved promising results. The fusion of CNN and Transformer-based models has been shown to significantly boost classification performance in several studies (Liu *et al.*, 2022). These hybrid designs rely on CNNs to learn fine-grained temporal and spatial patterns and relationships within the ECG signal, while Transformers focus on learning long-range temporal and contextual relationships (Tao *et al.*, 2023). The mixture of two types of architectures enables the strengths of both types of approaches to be combined, yielding better feature representation and classification ability (Kim *et al.*, 2025).

Although existing studies have shown the potential of CNN-Transformer hybrid models for ECG analysis, several important research gaps still remain. Many current approaches first convert ECG signals into Two-Dimensional (2D) image representations such as spectrograms before applying Transformer architectures, which may introduce additional computational complexity and possible loss of temporal information. Furthermore, modern models rely on classic CNN architectures without incorporating sophisticated mechanisms to enhance temporal feature extraction, such as residual learning and dilated convolutions. Also, there is still a lack of research on the systematic optimization of hybrid CNN-Transformer architectures for ECG classification. The second contribution in this thesis is motivated by these restrictions.

1.5 Two-Dimensional ECG Representations and Image Fusion

To leverage 2D CNNs' significant spatial feature-extraction capabilities and transfer learning from extensive image databases, a parallel line of study has focused on converting 1D ECG signals into 2D image representations (Qiu, Li, Qi and Li, 2024*a*). Because it can concurrently resolve both temporal and frequency properties at many scales, the CWT has become the most used time-frequency transformation method for creating scalogram images from ECG signals (Wang *et al.*, 2021). The scalogram encodes the magnitude of wavelet coefficients as pixel intensities across a 2D time-scale plane, providing a visually rich representation of the non-stationary, multi-frequency components of cardiac electrical activity.

The binarised version of the scalograms — obtained by applying a threshold operation to the gray-scale scalogram image — offers a structurally distinct, complementary representation that highlights the spatial morphology of dominant signal components whilst suppressing fine spectral gradients (Naz *et al.*, 2021). Studies deploying binary ECG images have demonstrated their utility in extracting QRS complex patterns, identifying ventricular tachyarrhythmia, and enabling morphological feature extraction suitable for Internet of Things (IoT) edge devices (Yoon and Kang, 2023). Importantly, the scalogram and its binary counterpart encode different and partially non-overlapping feature information: the scalogram captures graded spectral detail and frequency-amplitude relationships, whereas the binary image highlights the structural geometry and spatial distribution of dominant waveform features.

The standard method in image (scalogram / binary)-based ECG classification is to process only one image type, either the scalogram or the binarised version, in isolation, despite the proven usefulness of each separate representation. The complimentary information found in the parallel picture stream is not utilized by this single-representation paradigm, which limits the integrated richness of the feature set that can be used for classification. Typically used in remote sensing and medical imaging, multi-modal fusion techniques offer a rational way to combine diverse representations to create feature sets that go beyond the constraints of any one modality (Scarpiniti, 2024). However, the application of such fusion strategies to ECG scalogram and binary image representations has not been systematically investigated. This gap motivates the third contribution of this thesis.

1.6 Research Gaps and Objectives

The preceding review of the literature identifies three clearly delineated and interrelated research gaps in automated ECG classification using DL:

Research Gap 1 — AF Homogeneity in CNN-ML Hybrids Existing hybrid CNN-ML architectures for ECG classification uniformly apply a single, fixed AF across all convolutional blocks. Since different AFs capture distinct nonlinear transformations of the input, this homogeneous paradigm limits the diversity and discriminatory of the extracted feature space. No study has systematically investigated the deployment of heterogeneous AFs within the convolutional blocks of a CNN-ML hybrid architecture for ECG classification.

Research Gap 2 — Underexplored CNN-ViT hybridization for 1D ECG Signals While hybrid CNN-Transformer models have been applied to ECG images and other biomedical signals, the specific combination of residual and dilated CNN with ViT for processing ECG signals directly in their native 1D form is largely unexplored. Furthermore, the potential of systematic hyper-parameter optimization to enhance the performance of such hybrid architectures has not been fully exploited.

Research Gap 3 — Single-Representation Feature Constraint in scalogram-based classification approaches predominantly process a single image representation — either the scalogram or its binarised form — thereby forfeiting the complementary feature information available from the parallel representation. A principled, flexible mechanism for fusing the feature

maps extracted from both representations has not been proposed or validated.

Correspondingly, this thesis pursues three primary research objectives:

Objective 1: To design and validate a CHAF that assigns a distinct AF to each convolutional block within a CNN-ML hybrid architectures, and to evaluate its impact on ECG classification accuracy and inference efficiency on the Physikalisch-Technische Bundesanstalt (PTB) and Massachusetts Institute of Technology–Beth Israel Hospital Arrhythmia Database (MIT-BIH) database.

Objective 2: To develop and evaluate ECG-ResViT, a hybrid architecture that integrates residual and dilated CNN blocks for local feature extraction with a ViT for global temporal modeling, operating on 1D ECG signals, and to optimize the architecture through systematic hyper-parameter tuning.

Objective 3: To propose BlendNet, a DL architecture that employs alpha blending to generate a composite feature map from the scalogram and binarised scalogram representations of ECG signals, and to validate the superiority of this blended feature set over single-representation baselines.

1.7 Thesis Contributions

The proposed research contributes to the field of DL based ECG classification as follows, addressing the research gaps previously identified:

Contribution 1: While the state-of-the-art approaches to DL-based ECG classification make use of a single type of AF at all convolutional layers, this contribution proposes the design of CHAF – a heterogeneous framework for feature learning in deep neural networks that relies on different AFs to learn the discriminative features of multiple convolutional layers. Such a framework provides the opportunity for the network to detect various patterns from raw 1D ECG signals that might not have been revealed otherwise in homogeneous AF.

Contribution 2: In contrast to current methods that either consider the waveform morphology of ECGs in the local space or the temporal dependencies over time, we propose the architecture of ECG-ResViT that can learn the two types of features simultaneously from the raw ECG data. Our architecture includes the residual blocks for facilitating the learning of deeper features without suffering the problem of vanishing gradients and the dilated con-

volutions that can capture the multi-temporal features of heartbeats across different sizes of receptive fields without adding any additional parameters. Furthermore, our architecture integrates the local representations learned above with the ViT modules that model the temporal dependencies of the ECGs globally within multiple heartbeats.

Contribution 3: In contrast to the existing fusion methods, which generally use simple concatenation, attention, and even complicated fusion operations, this research presents a new method named BlendNet, which is a feature-level fusion architecture based on the fact that the roles played by different ECG representations in diagnostic reasoning are inherently dissimilar. The network used in this method includes two parallel streams of CNNs for learning distinctive features from the scalogram and binary structural representations separately, retaining their distinctive nature before being fused. Alpha-controlled blending operation is further introduced into the architecture for controlling the contributions of different representations at the feature level, allowing them to interact meaningfully with each other.

1.8 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 presents a detailed review of existing literature related to ECG classification, including traditional signal based techniques and rule based calculations, ML techniques, CNN-based DL models, hybrid CNN-ML architectures, Transformer and ViT approaches for biomedical signal analysis, and scalogram-based ECG image classification methods. Chapter 3 introduces the proposed CHAF architecture, covering the database details, AF selection process, architecture design, evaluation metrics, and experimental analysis performed on the PTB and MIT-BIH database. Chapter 4 presents the proposed ECG-ResViT architecture, including the residual and dilated CNN feature extraction modules, the ViT classifier, computational complexity analysis, ablation studies, and comparative performance evaluation. The proposed BlendNet architecture is described in Chapter 5, together with the CWT-based ECG image generating procedure, the alpha-controlled blending mechanism, flexibility analysis utilizing various classifiers and dense-layer configurations, and thorough performance comparisons. Chapter 6 concludes the thesis by summarizing main contributions, discussing their limits, outlining potential future research avenues, and offering suggestions for practical clinical implementation.

Chapter 2

LITERATURE REVIEW

2.1 Overview of ECG Analysis Methods

Numerous CVD issues are evaluated and diagnosed using ECG signals. It measure the heart's electrical activity and is non-invasive. Conditions like Arrhythmia (ARR), heart failure, CHF, MI, and coronary artery disease can be analyzed by ECG waveform analysis (Talukder *et al.*, 2025; Rahman *et al.*, 2023). Thus, timely diagnosis and treatment depend on precise ECG signal processing. Traditionally, manual ECG analysis has depended on skilled medical personnel's ability to decipher wave shape, intervals, and rhythm patterns in order to spot anomalies. Manual interpretation can be time consuming and can lead to inter-observer variability, but is still clinically relevant and important, particularly for very long recordings or large amounts of patient data (Silva *et al.*, 2025). The automated ECG analysis using DL techniques has recently been attracting a lot of attention, and several studies have shown that this approach is superior in terms of diagnostic performance and efficiency to traditional methods (Talukder *et al.*, 2025; Silva *et al.*, 2025; Petmezas *et al.*, 2022)

Digital healthcare technologies and wearable monitoring devices are rapidly advancing, and resulting ECG data has vastly increased in both clinical and remote healthcare settings (Petmezas *et al.*, 2022). With this increase in data, automated systems for analyzing data with accurate, reliable and efficient diagnostic support are in great demand (Petmezas *et al.*, 2022). Some early automated ECG analysis techniques used were based on the traditional signal processing, including filtering, wave-form detection, and feature extraction from ECG morphology (Ben Slama *et al.*, 2025). These methods were fairly straightforward and easily understood but were also influenced by noise, variation in signals and inter-patient differences.

Some statistical methods were also investigated as tools for automated ECG analysis, along with signal processing methods (Chen *et al.*, 2022; Khalid *et al.*, 2024). In these techniques, statistical parameters (Liu *et al.*, 2021; Xie *et al.*, 2020; Khoshnevis and Sankar, 2022; Pradhan *et al.*, 2023) like mean, variance, skewness, kurtosis, entropy, and higher order moments

were computed on the ECG signals to describe the behavior of signal and to detect abnormalities. These statistical characteristics gave numerical summaries of the ECG recordings and were frequently combined with rule-based classification systems (Silva *et al.*, 2025; Śmigiel *et al.*, 2021). However, complicated and non-linear correlations present in cardiac signals were usually difficult to capture with them.

In order to alleviate these drawbacks, ML methods were used to improve automated ECG classification. In contrast to the traditional rule-based method, ML algorithms learn decision boundaries directly from the databases, which can enhance the ability of pattern recognition from ECG signals. Some of the classical ML models that have been found to perform well in ECG classification problems include SVM, KNN, and various ensemble-based models (Khalid *et al.*, 2024; Ameen *et al.*, 2024; Al-Zaiti *et al.*, 2023; Matin Malakouti, 2023). These methods are compute efficient, and strong enough for real time applications. However, they remain strongly dependent on the quality of handcrafted features extracted from ECG recordings, and it is not always easy to design such features, for which a lot of domain expertise and prior clinical knowledge is required.

The emergence of DL (Ansari *et al.*, 2023; Ding *et al.*, 2025) has significantly advanced the field of ECG signal analysis by enabling models to automatically learn hierarchical feature representations directly from raw or minimally processed ECG signals. CNNs (Ahmed, Ali, Abdullah and Malebary, 2023; Ozaltin and Yeniay, 2023; Guhdar *et al.*, 2025; Daphin Lilda and Jayaparvathy, 2025) have demonstrated exceptional ability in recognizing local morphological patterns, such as QRS complexes and other waveform features, among many DL architectures. For ECG analysis, sequence modeling such as RNNs (Jafari and Jafari, 2026; Suyama *et al.*, 2025), LSTM (Reddy C. *et al.*, 2025; Lin *et al.*, 2025) networks, and Gated Recurrent Unit (GRU) (Mohammed *et al.*, 2026; Tirumalasetty *et al.*, 2025) have also been extensively investigated. Specifically, LSTM and GRU models were created to capture long-term temporal dependencies found in sequential cardiac impulses and to get around the vanishing gradient issue with traditional RNNs (Ba Mahel *et al.*, 2025; Alghieth, 2025).

The architecture that has drawn significant attention due to its capability of modeling long-range relations over several cardiac cycles is the attention mechanism and transformers. Despite the effectiveness of DL algorithms, their clinical use is restricted because of the re-

quirement for a large amount of training data, computationally expensive nature, and complex design.

The choice of AFs plays an important role in determining how well the DL models perform (Chaubey *et al.*, 2026; Abubakar *et al.*, 2025). Non-linearity brought in by AFs allows neural networks to capture the intricate dependencies that exist in ECG signals. Most of the existing architectures offer a homogeneous architecture where one AF is used for multiple layers (Chaubey *et al.*, 2026). While this simplifies the implementation of DL models, this might reduce the diversity of features learned during training because of the fact that different AFs behave in unique ways to different kinds of signals.

In an effort to capitalize on the strengths of both ML and DL approaches, many contemporary studies have started to give more emphasis to hybrid models (Selvam *et al.*, 2025; Alyahya *et al.*, 2026; Liu, 2025; Ye *et al.*, 2025). The purpose of such architectures is to achieve high accuracy in diagnosing heart disease through efficient feature extraction in addition to accurate classification, without compromising computational efficiency. Hybrid models offer the flexibility of incorporating multiple feature sets and hence improving their generalization capacity with regards to various ECG databases.

Another rapidly growing area in research is exploring different interpretations of ECG signals for classification purposes. An ECG signal can be represented in several ways, including time domain, frequency domain, and time-frequency representations (Tian and Zhang, 2025). This includes the transformation of ECG signals into images, e.g., scalograms (Ozaltin and Yeniay, 2023; Yoon and Kang, 2025), spectrograms (Daphin Lilda and Jayaparvathy, 2025), recurrence plots (Behera *et al.*, 2025), Gramian Angular Fields (Yoon and Joo, 2025), and Markov Transition Fields (Sam *et al.*, 2025). Each of these characterizations captures different aspects of heart activity that can provide additional information for diagnosis

Besides that, latest developments in transformer-based (Alghieth, 2025; Ansari *et al.*, 2025) structures have highlighted the importance of simultaneously gathering local morphological features and long-range temporal relationships in ECG data.Convolution-based designs are excellent at identifying local waveform structures, but transformer models provide robust methods for learning global links within sequential data. Integrating these complementary capabilities (Xiong *et al.*, 2025) can therefore result in more robust and comprehensive ECG

classification models.Together, these advancements show how ECG analysis methods have moved from conventional signal processing to hybrid architectures.

This chapter provides a thorough study of the literature on automated ECG signal analysis in context with recent advancements. Traditional processing techniques, statistical feature-based methods, ML techniques, DL architectures, and newly developed hybrid models are all discussed. This chapter addresses the key research gaps that drive the approaches suggested in the next chapters of this thesis by evaluating their advantages and disadvantages in terms of metrics, computing efficiency, feature representation, and practical usability.

2.2 Traditional ECG Analysis Methods and their Limitations

Traditionally, statistical methods and signal processing have been used to analyse ECG signals (Chen *et al.*, 2022; Khalid *et al.*, 2024; Liu *et al.*, 2021; Xie *et al.*, 2020; Khoshnevis and Sankar, 2022; Pradhan *et al.*, 2023; Śmigiel *et al.*, 2021). Early computational techniques were primarily created to assist physicians by deriving clinically significant data from ECG waveform, such as heart rate, rhythm regularity, and morphological features of the P wave, QRS complex, and T wave (de Chazal *et al.*, 2004; Jang *et al.*, 2025; Cano *et al.*, 2025; Baig *et al.*, 2025). These methods converted clinical knowledge into computational processes that could help identify heart anomalies.

One of the most widely used methodologies was time-domain analysis (Cano *et al.*, 2025; Baig *et al.*, 2025; Rehman *et al.*, 2025; Li, Chen, Chen, Liu, Xu and Lu, 2025), which extracted parameters such as RR intervals, QRS duration, PR intervals, and QT intervals directly from ECG recordings. These features are clinically interpretable and closely related to the diagnostic criteria that cardiologists regularly use. Time-domain methods are effective in detecting many cardiac abnormalities, including conduction issues and arrhythmia (Baig *et al.*, 2025; Rehman *et al.*, 2025). However, signal noise, baseline drift, and variations in ECG recording quality often impair their performance in real-world healthcare situations.

Frequency-domain techniques were later created in order to avoid some of these limitations. Techniques like the Fast Fourier Transform (FFT) (Kumar and Chakrapani, 2022; Fathail and Bhagile, 2022; Sathya *et al.*, 2026) analyse ECG signal spectrum characteristics and assist in determining the important frequency components linked to heart activity. Fea-

tures in the domain can improve resistance to particular types of noise and are useful in rhythm analysis. However, solely frequency-based methods frequently fail to catch transient waveform fluctuations and localized signal abnormalities that are crucial for a correct diagnosis since ECG signals are intrinsically non-stationary.

Time-frequency analysis techniques were created to concurrently capture the temporal and spectral aspects of ECG data in order to overcome this problem. ECG signals can be examined in both the time and frequency domains using techniques like Short-Time Fourier Transform (STFT), (Ma *et al.*, 2025; Huang *et al.*, 2019), Discrete Wavelet Transform (DWT), (Singh *et al.*, 2017; Chen *et al.*, 2025), and CWT (Gupta *et al.*, 2021; Rahman *et al.*, 2025; Kharzi *et al.*, 2025). Wavelet-based have drawn a lot of attention among these strategies due to their capacity to efficiently localize ECG characteristics across various scales. They are therefore perfect for tasks like QRS complex identification, noise reduction, and multi-resolution feature extraction. While complex techniques like Wavelet Packet Decomposition (WPD) (Ali, 2025; Pradhan *et al.*, 2022) offer better frequency resolution, the S-transform (Ari *et al.*, 2013; Sul *et al.*, 2025) improves signal representation by combining phase information and time-frequency analysis. Nevertheless, the efficiency of such methods is largely dependent upon choosing the right wavelet functions and decomposition parameters.

ECG signal analysis has extensively employed statistical and higher order attributes (Pradhan *et al.*, 2023; Liu *et al.*, 2021; Xie *et al.*, 2020; Khoshnevis and Sankar, 2022; Karthikeyani *et al.*, 2025), alongside conventional time and frequency domain methods. Features that are utilized include mean value, variance, skewness, kurtosis, entropy, and higher order statistics. While such statistical measures may offer concise numerical representation of the ECG signals, they are not always adequate in describing their morphology and dynamics.

The reliance of conventional ECG analysis techniques on manually created feature extraction is one of their main drawbacks. Handcrafted features are clinically interpretable and computationally efficient, but their performance is frequently quite susceptible to inter-patient variances (Pradhan *et al.*, 2023; Karthikeyani *et al.*, 2025; Cano *et al.*, 2025; Kumar and Chakrapani, 2022), noise, and variations in signal quality. Furthermore, creating efficient handmade features sometimes lacks the potential to generalise across many databases, clinical circumstances and necessitates a high level of domain expertise.

The restricted flexibility of traditional techniques is another significant issue. Many signal processing are less efficient at handling hidden patterns or slight morphological changes (Karthikeyani *et al.*, 2025) in ECG signals because they rely on predefined assumptions, manually chosen thresholds, and established analytical criteria. As a result, their use is restricted in contemporary healthcare settings where completely automated analysis systems and constant monitoring are becoming more and necessary.

Given these drawbacks, the development of automated cardiac diagnostic systems relied heavily on conventional ECG analysis methods. They cleared the path for further developments in ML (Khalid *et al.*, 2024; Ameen *et al.*, 2024; Al-Zaiti *et al.*, 2023; Matin Malakouti, 2023) and DL-based (Ansari *et al.*, 2023; Ding *et al.*, 2025; Ahmed, Ali, Abdullah and Malebary, 2023; Ozaltin and Yeniay, 2023; Guhdar *et al.*, 2025; Lin *et al.*, 2025; Jafari and Jafari, 2026; Suyama *et al.*, 2025; Mohammed *et al.*, 2026; Tirumalasetty *et al.*, 2025) ECG analysis and defined a number of clinically significant signal features. However, researchers were compelled to adopt more data-driven methods due to the restricted scalability of handcrafted feature-based and their incapacity to capture complicated nonlinear interactions.

ML approaches, which seek to automatically extract discriminative patterns from ECG data, were adopted as a result of this shift. However, a lot of ML techniques still use manually created features, which restricts their capacity to accurately capture intricate signal properties. The following section discusses ML approaches for ECG classification and examines how they address some of the limitations associated with traditional ECG analysis methods.

2.3 Machine Learning-Based ECG Classification

As the limitations of handcrafted signal processing methods became more evident, ML techniques emerged as a data-driven approach for automated ECG classification. Unlike traditional rule-based systems, ML models learn patterns and decision boundaries directly from labeled data, making them more adaptable to signal variability and noise. Consequently, several ML algorithms have been explored for ECG analysis, particularly in applications where computational efficiency and interpretability are important.

A two-stage pipeline comprising feature extraction and classification was typically used by early ML-based ECG algorithms. Features such as RR intervals, wavelet coefficients, spectral

energy measurements, entropy values, and other statistical descriptors (Liu *et al.*, 2021; Xie *et al.*, 2020; Khoshnevis and Sankar, 2022; Pradhan *et al.*, 2023; Karthikeyani *et al.*, 2025) were extracted from time-domain (Cano *et al.*, 2025; Baig *et al.*, 2025; Rehman *et al.*, 2025; Li, Chen, Chen, Liu, Xu and Lu, 2025), frequency-domain (Kumar and Chakrapani, 2022; Fathail and Bhagile, 2022; Sathya *et al.*, 2026), and time-frequency representations (Ma *et al.*, 2025; Huang *et al.*, 2019; Singh *et al.*, 2017; Chen *et al.*, 2025; Gupta *et al.*, 2021; Rahman *et al.*, 2025; Kharzi *et al.*, 2025) of ECG data. Classifiers that could differentiate between normal and abnormal cardiac conditions were subsequently trained using these extracted features.

There have been many ML approaches; out of them, SVMs were quite famous because of their superior generalization abilities and handling high-dimensional spaces very efficiently (Ben Slama *et al.*, 2025; Ahmad *et al.*, 2025; Venkatesan *et al.*, 2018). Models based on SVMs performed very well in terms of classification of ECGs when combined with proper feature extraction and suitable kernels. The only issue with them was that their scalability was limited in case of large data sets.

Because of the simplicity and ease of implementation of KNN (Saini *et al.*, 2013; Nagendrudu *et al.*, 2025), they were extensively used as well. The operation of the KNN classification algorithm involves comparing the feature vectors and classifying the input data according to similarity to neighboring vectors. Despite the decent classification performance of the KNN classifier, it is sensitive to scaling and expensive to implement in practice.

Methods utilizing Decision Tree (DT) (Ashkir *et al.*, 2025; Kailan *et al.*, 2025) also became popular because of their interpretability and moderate computational complexity. Nevertheless, standalone DTs tend to suffer from overfitting. In order to mitigate this issue, ensemble learning techniques like RF started being used. RF (Tara *et al.*, 2025; Liu, Zhu, Deng and Chen, 2025) models achieve increased robustness and generalization capability through the use of many DTs that were trained on different subsets of the input data. The models demonstrated high proficiency in ECG classification, particularly when dealing with nonlinear interactions of features.

Further performance improvement was achieved through the use of gradient boosting. The algorithms such as XGBoost (Botros *et al.*, 2025; Huang *et al.*, 2025), Light Gradient Boosting Machine (LightGBM) (Jain *et al.*, 2025; Jun *et al.*, 2025), and Categorical Boosting (Cat-

Boost) (Wu *et al.*, 2025; Hamid *et al.*, 2025) correct the errors made by previous learners and create their models sequentially. Often, such methods provide good metrics, efficient learning process, and proper handling of unbalanced ECG databases are frequently offered by these methods.

Tree-based ensemble methods have some promise in real-world clinical use due to their potential ability to achieve competitive ECG classification performance while being relatively inexpensive computationally-wise, based on multiple studies. Moreover, various feature oversampling techniques such as the Synthetic Minority Oversampling Technique (SMOTE) (Sai *et al.*, 2024; Trigka and Dritsas, 2025) are frequently used together with ML algorithms in order to resolve the problem of class imbalance that is common in ECG datasets.

Nonetheless, classical ML approaches continue to mostly depend on handcrafted feature engineering (Ameen *et al.*, 2024; Al-Zaiti *et al.*, 2023; Matin Malakouti, 2023). The efficiency of classification will heavily depend on the retrieved features' quality and relevance, and thus the construction of features may prove to be difficult and require considerable expertise on the subject and extensive validation. Furthermore, the long-range dependencies and local morphological characteristics in ECG recordings may be missed by handcrafted features.

One additional significant disadvantage of the conventional ML models lies in the fact that they cannot automatically discover hierarchical representations of features based on raw ECG signals (Ben Slama *et al.*, 2025; Ameen *et al.*, 2024; Al-Zaiti *et al.*, 2023; Matin Malakouti, 2023; Ahmad *et al.*, 2025). While ensemble-based models are able to detect nonlinear associations between variables, they lack sufficient representational capabilities required for the adequate analysis of ECG data, particularly in the case of large and diverse databases.

Nevertheless, ML algorithms remain relevant in automated ECG analysis due to their robustness when working with relatively small databases, transparency, and computational efficiency. The current research into combining ML classifiers with DL models aimed at feature extraction has been inspired by these advantages. Hybridization of the two techniques can be considered as a natural step towards creating advanced ECG analysis systems.

2.4 Deep Learning-Based ECG Classification

The introduction of DL techniques for ECG signal analysis (Ansari *et al.*, 2023; Ding *et al.*, 2025; Ahmed, Ali, Abdullah and Malebary, 2023; Ozaltin and Yeniay, 2023; Guhdar *et al.*, 2025; Jafari and Jafari, 2026; Suyama *et al.*, 2025; Lin *et al.*, 2025; Mohammed *et al.*, 2026; Tirumalasetty *et al.*, 2025) was prompted by the shortcomings of handmade feature-based ML systems, including their reliance on manual feature engineering and limited representation capability. Unlike traditional ML techniques, DL models may automatically create hierarchical feature representations from raw or minimally processed ECG signals. This improves scalability for large databases and does away with the need for manual feature extraction.

CNNS (Ahmed, Ali, Abdullah and Malebary, 2023; Ozaltin and Yeniay, 2023; Guhdar *et al.*, 2025; Kalatehjari *et al.*, 2025; Panwar *et al.*, 2025) are amongst the popular DL models for classifying ECG signals currently. Since CNNS employ convolutional filters for analyzing small portions of the data, they are highly effective in extracting local features. T-wave, P-wave, and QRS complex are a few of the critical wave features which the CNNS are capable of extracting through their convolutional filters. The stacking of convolutional layers allows CNNs to extract more and more complex representations of the features.

A number of CNN architectures that are currently considered state-of-the-art include AlexNet (Rawshani *et al.*, 2025), VGG (Thite and Jagtap, 2025), ResNet (Yang, Zou, Qin, Cao, Zhang and Deng, 2025)[84], and DenseNet (Saranya *et al.*, 2025) which have all been analyzed through time in an effort to optimize feature detection and classification. For example, residual connections used by architectures such as ResNet enable deeper neural

CNN-based ECG analysis is commonly implemented using two major methods: 1D CNNs (Guhdar *et al.*, 2025) and 2D CNNs. By processing raw ECG data directly in the time domain, 1D CNN models preserve their original temporal structure. This approach usually keeps the computational complexity lower and does not require the conversion of ECG databases into graphical representations.

However, 2D CNN-based (Sau *et al.*, 2025) approaches first convert ECG data into time-frequency formats like spectrograms or scalograms using transformations like the CWT. These representations are then analysed as images using traditional CNN architectures. This method may sometimes change the original temporal properties of the ECG signal and adds extra

preprocessing needs, even if it can collect more spectrum information.

To increase CNN's performance in ECG analysis, a number of architectural changes have also been suggested. Deeper designs are supported and the vanishing gradient issue is resolved with the aid of residual connections. Dilated convolutions (Mahmood and Hasan, 2026) allow the network to capture more contextual information by expanding the receptive field without significantly increasing the number of trainable parameters. Additionally, multi-scale CNN (Wei and Li, 2025) architectures have been created to improve the identification of small waveform fluctuations by extracting features at various temporal resolutions.

CNNs still have trouble representing long-range temporal dependencies, even if they are quite good at learning local features. ECG signals frequently include correlations between heartbeats and rhythm-related patterns that go beyond convolutional filters' receptive field. Deeper CNN architectures typically increase computing complexity and may still have trouble properly modeling global temporal linkages, even if they can partially overcome this problem.

RNNs (Jafari and Jafari, 2026; Suyama *et al.*, 2025) and its variations, such as LSTM (Reddy C. *et al.*, 2025; Lin *et al.*, 2025) and GRU networks (Mohammed *et al.*, 2026; Tirumalasetty *et al.*, 2025), have also been investigated for ECG analysis in order to better capture temporal dependencies. These designs may learn correlations across signal sequences since they are specifically made for sequential data and preserve internal memory states. LSTM-based models have demonstrated a good ability to identify beat-to-beat fluctuations and rhythm anomalies in ECG data.

Recurrent designs do, however, have some drawbacks. Gradient-related issues may still arise when processing very long sequences, and training RNN-based models can be computationally demanding. In addition, the sequential nature of recurrent processing limits parallel computation, resulting in longer training times when working with large ECG databases.

Transformer-based architectures and attention techniques have been presented for biomedical signal analysis more recently. Transformers provide more efficient modeling of long-range temporal dependencies by using self-attention mechanisms to learn correlations between various points within a sequence. Transformers process input sequences in parallel, which increases training efficiency and scalability in contrast to recurrent models. They are ideal for ECG classification jobs because of their capacity to concentrate on key areas of a signal.

For ECG analysis, a number of additional DL models have also been investigated in addition to these fundamental designs. While Generative Adversarial Network (GAN) (Li, Xu, Habib *et al.*, 2025) have been utilized for data augmentation and class imbalance issues, autoencoder-based (Lomoio *et al.*, 2025) techniques are frequently used for feature learning and noise reduction. Capsule networks have been researched for maintaining hierarchical feature associations, whereas Graph Neural Network (GNN) (Guo *et al.*, 2025) have been investigated for simulating links between multi-lead ECG signals. These methods are not as popular as CNNs, recurrent networks, and Transformer-based designs, despite the fact that they have demonstrated promise in particular applications.

Attention mechanisms (Guhdar *et al.*, 2025; Najia and Faouzi, 2025; Yang, Jiang, Zhu, Wu, He and Wang, 2025) and Transformer models (Alghieth, 2025; Ansari *et al.*, 2025) have recently gained considerable attention in ECG signal analysis because of their ability to capture global relationships within sequential data. These models maintain effective parallel processing while identifying dependencies across various signal regions using self-attention. Their applicability for automated ECG categorization is further enhanced by their capacity to concentrate on clinically significant waveform segments.

ViT models (Kilimci *et al.*, 2025; Liu *et al.*, 2026) expand the Transformer architectures to structured inputs by processing data as sequential tokens after breaking it up into smaller patches. ViT-based methods have been modified for biomedical signals, such as ECG databases, while being initially created for image analysis. Through positional encoding methods, these models aid in capturing global contextual information while maintaining temporal connections.

Although the above models are efficient, their major weaknesses include being computationally expensive and the requirement for big databases during training. Moreover, the many learnable parameters in the model could lead to overfitting if there isn't enough training data. Finally, using the attention mechanism alone may not be enough to capture all the local waveform features.

Considering everything discussed above, it can be concluded that through the implementation of end-to-end learning and hierarchical feature extraction, DL models have improved automated analysis of ECG data. The CNN model can detect local waveforms while the

RNN model captures temporal dependencies, and the transformer model captures global dependencies. However, each of the above models comes with some drawbacks when applied independently.

These complementary advantages and disadvantages have prompted the creation of hybrid architectures that integrate attention-based methods with convolutional operations. These methods are a crucial area of study for contemporary ECG classification systems since they seek to combine local and global feature learning while preserving computational efficiency.

2.5 Hybrid and Advanced ECG Architectures

It became more and clear as automated ECG classification research progressed that neither independent DL architectures nor conventional ML techniques could fully handle the complexity of ECG data processing. Although ML techniques mostly rely on manually created feature extraction, they are typically interpretable and computationally efficient. DL models, on the other hand, are capable of automatically learning intricate feature representations, however they frequently need for huge databases and significant processing power. These complimentary advantages and disadvantages have prompted academics to create hybrid architectures that use several modeling techniques.

One widely explored hybrid approach (Alghieth, 2025) combines CNNs with classical ML classifiers (Jahangir *et al.*, 2025; Dhandapani *et al.*, 2025). In these models, classifiers like SVM, KNN, or gradient boosting algorithms carry out the final classification task, while CNNs are mostly utilized to extract discriminative features from ECG data. This approach maintains the effectiveness of traditional machine learning techniques while taking advantage of deep networks' effective feature capabilities.

Numerous research have demonstrated that using ML classifiers in place of fully connected dense layers can lower computing cost and, in certain situations, enhance model generalization. When used with CNN-based feature extraction, ensemble techniques like RF and XGBoost have proven to perform exceptionally well, particularly for imbalanced ECG databases. These CNN-ML hybrid architectures are appropriate for real-time clinical applications because they strike a balance across feature representation capacity and inference efficiency. Nevertheless, a lot of current implementations still rely on homogeneous architectural de-

signs, where models are trained using a single representation and identical AFs are applied across all convolutional layers. The depth and diversity of the taught feature space may be limited by such methods.

Another important approach in hybrid modeling is combining convolutional networks with Transformer-based designs (Ansari *et al.*, 2025; Chaubey *et al.*, 2026) and attention mechanisms (Najia and Faouzi, 2025). While Transformer models are more suited for capturing long-range connections and global temporal linkages, convolutional layers excel at capturing local morphological elements in ECG data. Consequently, CNN–Transformer hybrid models were developed, in which convolutional layers collect local waveform information and Transformer modules capture greater number of temporal associations throughout the signal.

Because cardiac signals contain both localized waveform features and rhythm-related patterns dispersed over several heartbeats, this hybrid architecture is especially well-suited for ECG interpretation. CNN-Transformer models have shown encouraging classification performance, however they frequently result in higher processing costs and architectural complexity. Therefore, to guarantee their suitability for actual clinical application, thorough design and optimization are required.

Another new area of study that goes beyond architectural hybridization is representation-level fusion. There are several ways to analyse ECG data, including time-domain, frequency-domain, and time-frequency representations like spectrograms and scalograms. Different aspects of heart action are highlighted in each portrayal. Waveform morphology is preserved by time-domain signals, rhythmic behavior is captured by frequency-domain features, and localized spectrum information is provided by time-frequency representations.

Most existing approaches rely on a single signal representation, which may limit the amount of data the model can collect. To overcome this limitation, researchers have investigated feature fusion techniques that combine several representations. Early input fusion, middle feature map fusion, and late classifier output fusion are among the various stages of the model where fusion can take place. Research on effective and adaptable fusion procedures is still ongoing, despite the encouraging outcomes of multi-representation approaches.

Another topic which received considerable attention is that of feature diversity in CNNs. By making the neural networks nonlinear, AFs have an effect on feature learning. Most DL

architectures rely on the same AF in all layers, which is usually ReLU or one of its variations. This allows the model building to become simple; however, it may prevent the variety of learned features from being large. Especially when it comes to sophisticated biomedical signals, such as ECG, using different AFs in the layers of a network can help learn more attributes and improve the classification accuracy.

In summary, advanced and hybrid approaches signify a major step forward in the field of automatic ECG classification. They combine the benefits of multiple methods of modeling, such as successful classification, diverse signal representation, and feature learning both locally and globally. The main difficulty in designing such systems is to create models which are extremely accurate and computationally efficient. Overcoming this problem, which remains an important aspect of ongoing research, serves as a basis for the approaches suggested in the next sections of this thesis.

2.6 Research Gaps

The accuracies in Table 2.1 should not be compared directly, because the studies differ in several respects. First, the datasets differ in size, class composition and acquisition conditions; a three-class task on a small balanced subset is considerably easier than a five-class AAMI beat-classification task. Second, the data-splitting strategy strongly affects the result: beat-level or segment-level random splits allow samples from the same patient to appear in both training and test sets and typically yield higher accuracy than patient-level (inter-patient) splits, which better reflect clinical use. Third, studies report different metrics, and accuracy alone can be misleading when classes are imbalanced, since a majority-class predictor can score highly. Fourth, class imbalance is handled differently (class weights, oversampling such as SMOTE, or none), and oversampling applied before splitting can leak information into the test set. Finally, preprocessing, such as resampling, segmentation, and normalisation, varies across studies. Consequently, differences of a few percentage points between studies may reflect experimental design rather than model quality, and claims of superiority are justified only where the datasets and protocols are comparable.

As mentioned in the previous sections, there are certain notable developments in the application of conventional ML and DL methods to automated ECG categorization in the literature.

Table 2.1 Comparative Summary of Representative ECG Classification Studies

S.No	Study	Method	Dataset	Reported Accuracy	Limitation
1	Thongsuwan et al. (2021)	CNN + XGBoost	PTB	99.38%	Homogeneous activation; 1.23 s execution time
2	Çınar and Tuncer (2021)	AlexNet features + SVM	ARR/CHF/NSR (PhysioNet)	96.77%	Pretrained image CNN used as feature extractor; single representation
3	Sabeenian and Sree Janani (2023)	Transfer learning (ResNet18, DenseNet)	ARR/CHF/NSR (PhysioNet)	98.81% (raw ECG), 97.05% (wavelet)	Transfer learning from image models; accuracy varies with input representation
4	Katal et al. (2023)	CNN vs. GoogLeNet	ARR/CHF/NSR (PhysioNet)	91.20%	Lower accuracy; no local–global modelling
5	Malleswari et al. (2024)	CWT scalogram + SqueezeNet	ARR/CHF/NSR (PhysioNet)	98.7%	Single 2D representation; extra preprocessing
6	Kolhar and Al Rajeh (2024)	CQ-NSGT + AlexNet	ARR/CHF/NSR (PhysioNet)	98.82%	Single time–frequency representation
7	Saeed and Yousif (2021)	Slantlet transform + SVM	MIT-BIH (900 segments; ARR/CHF/NSR)	92.9%	Handcrafted features; signal domain only

Note: The splitting protocols of the compared studies are not uniformly reported or comparable; hence the accuracies are indicative only.

While traditional ML methods rely on manually designed feature extraction and classification, recent DL architectures enable end-to-end feature learning, thereby improving performance in diagnosing across a range of ECG datasets.

However, there are still several research challenges to be addressed. Numerous ML-based methods are efficient but fail to capture the complex nonlinear characteristics of ECG data. While DL models may be able to classify better, they are more resource-intensive, require more time, and need more training, making real-time applications in healthcare challenging.

Second, there are difficulties in modelling local morphological features and long-range temporal correlations. While convolutional architectures showed great performance in identifying local waveform elements such as T waves, QRS complexes and P waves, they have difficulties in establishing global contextual relations across cardiac cycles. On the other hand, attention-based architectures can capture long-range dependencies but might not be able to capture fine-grained local features. This suggests the need to introduce local and global feature representation into architectures.

Third, most of the existing studies employ a single ECG representation (time-frequency images or raw signal). Different representations bring out complementary signal properties, however, and discriminative feature learning may be impeded by underutilisation of such complementary information.

The first gap, the reliance on homogeneous activation functions in CNN and CNN–ML hybrid models, motivates CHAF (Chapter 3). The second, the trade-off between local feature extraction, global contextual modelling and attention-related computational cost, motivates ECG-ResViT (Chapter 4). The third, the dependence on a single ECG representation, motivates BlendNet (Chapter 5). Since the reviewed studies differ in datasets and evaluation protocols, the comparisons in this thesis are made under the stated experimental conditions, and no claim of general superiority over other methods is made.

The extensive use of similar activation functions in deep networks is another drawback. Being convenient for computation, uniform activation behaviour might be restrictive in terms of the diversity of features that the network can express and learn diverse nonlinear patterns that are present in the ECG signals.

Finally, in evaluating models, accuracy of classification is the main focus, whereas costs

of computation and inference are relatively unimportant. These are necessary for the practical implementation of advanced computing, wearable technology and continuous cardiac monitoring systems.

ECG classification systems that simultaneously achieve high diagnostic accuracy, computational efficiency, representatives of features and efficient capturing of the local and global dependency of the signal are thus necessary. The main driving force for the approaches suggested in this thesis is to address these issues.

Chapter 3

CONVOLUTION-BASED HETEROGENEOUS ACTIVATION ARCHITECTURE FOR ECG SIGNAL CLASSIFICATION

3.1 Chapter Overview

The key techniques used for ECG classification were discussed in Chapter 2, which covered the entire pipeline from feature extraction along with final classification to database identification and preprocessing. Traditional ML, DL, and hybrid architectures have all been studied. Because they can combine rich representation learning with effective and comprehensible decision-making mechanisms, hybrid approaches—where DL models are used for feature extraction and ML algorithms are used for classification—have shown promising performance among these (Murali, 2025; Gupta *et al.*, 2024; Kashou *et al.*, 2023). Despite developments, a common drawback of many current approaches is the dependence on a uniform AF across the network.

While the choice of AFs is frequently regarded as a default design decision, architectural depth, filter configuration, and layer composition are usually given significant consideration in DL. But by adding non-linearity to the block (Dubey *et al.*, 2022; Ramadevi *et al.*, 2024), AFs play a technically important role in representation learning, allowing the model to learn complex patterns that go beyond linear transformations. Additionally, certain AFs have unique mathematical and functional properties that can have a direct impact on the kind and calibre of features that are learned across various network levels.

When characteristics of the ECG waveforms, such as the P-wave, QRS complex, and T-wave, become important for ECG signal analysis(Luthra, 2020; Boloor, 2022). The amplitude, smoothness, direction, and temporal behavior of these traits vary. As a result, not all waveform characteristics can be recorded by a single AF. The network's capacity to extract a diversity of distinctive, discriminative, and physiologically significant features from ECG signals may be

limited by the use of a uniform AF across all convolutional layers. (Nair and Hinton, 2010; Glorot *et al.*, 2011; Montúfar *et al.*, 2014; Sindhu *et al.*, 2023; Devasenapathy *et al.*, 2024; Ye *et al.*, 2025; Adeli *et al.*, 2025; Ebrahimi *et al.*, 2020; Nayyab *et al.*, 2025; Abubakar *et al.*, 2025; Śmigiel *et al.*, 2021)

Motivated by this limitation, this chapter proposes the CHAF. The proposed architecture incorporates multiple AFs across different convolutional blocks to facilitate more diverse nonlinear feature extraction from ECG waveforms. By promoting activation diversity within the CNN-based feature extraction process, CHAF aims to enhance representational richness while maintaining computational efficiency. In addition, the extracted deep features are combined with lightweight ML classifiers to achieve accurate and efficient ECG waveform classification.

The remainder of this chapter is organized as follows. Section 3.2 presents the background and related work on AFs and their variants. Section 3.3 discusses the identified research gaps, followed by the research objectives in Section 3.4. Section 3.5 describes the database and preprocessing methodology adopted in this study. Section 3.6 details the proposed CHAF architecture and its architectural analysis. Finally, Section 3.7 concludes the chapter.

3.2 Related Work

There are many different types of AFs that have been created to enhance gradient propagation, computational efficiency, and representational power in DL models. ReLU, Sigmoid, and Tanh are some of the most popular AFs in ECG classification problems (Abubakar *et al.*, 2025; Balaji *et al.*, 2023; Zhang *et al.*, 2024). The sigmoid function restricts itself to the interval (0,1) and so is ideally suited for probabilistic interpretation in a binary classification problem, while Tanh maps the inputs to (-1,1) and retains both positive and negative responses due to the zero-centered output. But both functions suffer from gradient saturation for high input values, resulting in difficulties for effective learning in deeper networks.

Although there are several AFs that have different nonlinear characteristics, most CNN-based ECG classification models are based on homogeneous activation strategy, which utilizes a single AF for the entire network. While simple to compute, this approach only allows for a limited set of nonlinear transformations in the feature extraction stage, which can lead to less ability to capture the complex morphological variations that exist in ECG signals.

Recently, hybrid DL–ML architectures have been investigated (Madan *et al.*, 2022; Shah *et al.*, 2025; Kolhar and Al Rajeh, 2024; Ozaltin and Yeniay, 2023; Zhang *et al.*, 2025) where the CNNs are used as feature extractors and the ML algorithms are used as classifiers. These methods have shown to be effective in ECG analysis by exploiting the hierarchical feature learning and flexibility of ML classifiers. However, they still make use of largely homogeneous activation configurations for their feature extraction stages.

Based on the literature review, it is clear that the potential of heterogeneous AFs for ECG feature learning has not been fully explored. To address this need, the proposed CHAF proposes activation diversity in the convolutional blocks to facilitate complementary nonlinear feature learning and richer ECG representations for better classification.

The application of AFs in CNNs is crucial for the addition of non-linearity and the regulation of feature propagation through layers. Not only do AFs allow the identification of complex patterns but also determine their relative significance and impact their quality as they can be emphasized or diminished to a certain degree.

The function of AFs becomes especially relevant in the domain of ECG classification as cardiac waves vary greatly in their morphology. For instance, sharp and high-amplitude QRS complexes are distinctly different from smooth and low-amplitude P and T waves. Thus, different activation functions react differently to signal patterns. ReLU functions are able to accentuate prominent local features like QRS complexes (Śmigiel *et al.*, 2021; Ramachandran *et al.*, 2022; Agrawal *et al.*, 2023) while other functions such as Tanh and ELU can help maintain polarity sensitivity and low amplitude waveforms (Ashtaiwi *et al.*, 2024; Wang *et al.*, 2020; Rajkumar *et al.*, 2019; Sindhu *et al.*, 2023).

These observations suggest that relying on a single AF throughout the network may limit its ability to capture the full spectrum of ECG morphology. Therefore, incorporating activation diversity offers the potential to learn richer and more representative ECG features than conventional homogeneous activation strategies.

3.3 Common Activation Functions in Deep Learning

Due to the crucial function that AFs play in deep neural architectures, a variety of AFs that are tailored to particular learning challenges have been developed as a result of significant

study. Sigmoid and Tanh functions were widely used in early neural network models because of their smooth and limited properties, which enable robust output representations. However, with increasing network depth, these functions were found to suffer from gradient saturation, leading to vanishing gradients and consequently hindering effective training.

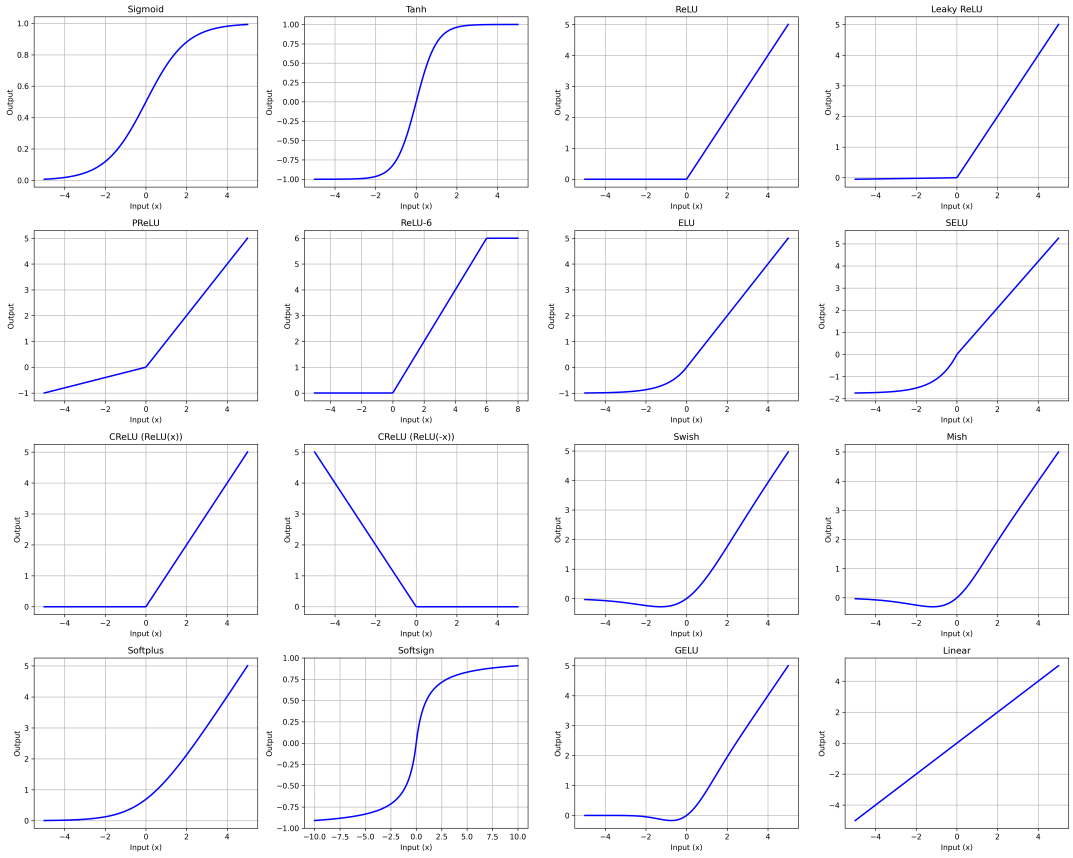


Fig. 3.1 Illustrations of Common Activation Functions and Their Nonlinear Characteristics

The ReLU and its variations became popular options as DL architectures advanced. Because to its favourable convergence behaviour, effective gradient propagation, and ease of computing, ReLU became widely used. By mitigating the vanishing gradient problem, ReLU-based AFs enabled the successful training of deeper convolutional networks and have since become standard in many signal and image processing applications.

More recently, advanced AFs such as Swish and Mish have been proposed to further enhance representational capacity through smooth and non-monotonic transformations. These functions dynamically regulate feature responses and help maintain stable gradient flow during training. Such properties are particularly beneficial in complex learning tasks, including biomedical signal analysis, where subtle waveform variations and nonlinear dependencies are critical for accurate classification.

In the context of ECG signal analysis, the effectiveness of an AF depends not only on its mathematical formulation but also on its ability to capture distinct morphological characteristics of the cardiac cycle. The selection of AFs is crucial for robust feature extraction since different AFs have different sensitivity to signal amplitude, polarity, and smoothness. The distinctive response curves of widely used AFs are shown in Fig 3.1, emphasising their operating ranges and nonlinear behaviours, in order to highlight these distinctions.

3.4 Limitations of Homogeneous Activation Functions

Different AFs exhibit different nonlinear response characteristics, making them suitable for capturing different types of ECG waveform patterns. While some AFs are effective for sharp and high-amplitude components, others better preserve smooth and low-amplitude variations. Therefore, no single AF is universally optimal across all stages of ECG feature extraction.

Nevertheless, most existing DL and hybrid ECG classification models use a homogeneous AF across all convolutional layers. This restricts the network's ability to adapt to the multi-scale and diverse morphology of ECG signals, thereby limiting feature diversity and representational flexibility. Consequently, homogeneous activation strategies may reduce the model's ability to learn subtle but clinically significant waveform variations. This limitation motivates the development of a heterogeneous activation-based architecture for improved ECG classification.

Consequently, homogeneous AF configurations may lead to suboptimal learning of clinically relevant waveform patterns, particularly subtle variations such as inverted waves and ST-segment deviations. These limitations motivate the need for a more flexible and heterogeneous activation strategy for ECG classification

Although there are many AFs with different mathematical properties, most of the current ECG classification methods still utilize the same activation strategy in each layer, and do not examine the activation strategy to better match the morphologic aspects of the ECG signals. There are many different AFs that have different features in their waveforms, that are inherently appropriate to capture different features, which existing models have not done in a single architecture, focusing instead on architectural complexity and not on careful AF selection. To the best of our knowledge, there has been no systematic study that explored using heteroge-

neous AFs to extract features from ECGs and other field, calling for an adaptive activation strategy for ECGs that takes into account the hierarchical and morphological characteristics of it.

The primary objective of this study is to investigate the effect of heterogeneous AF integration within CNN-based and hybrid learning architectures for ECG waveform classification. The study seeks to improve feature extraction and morphological representation by moving beyond conventional homogeneous activation strategies.

The specific objectives of this work are as follows:

1. For analyzing the effect and behavior of various AFs in ECG feature extraction.
2. For designing and developing an innovative CHAF through the combination of multiple AFs within convolutional layers.
3. For testing the capability of the developed CHAF architecture to detect diverse features of the ECG waveform, such as abrupt changes, minor low amplitude variations, and polarity-based features.
4. For performing comparative analysis between the developed CHAF based architecture and other DL as well as hybrid architectures for ECG classification.
5. For studying whether activation-based architectural design improves the classification of ECG while keeping computation efficiency.

3.5 Proposed CHAF Architecture

DL models have proven to be highly effective in ECG analysis due to their inherent capability of learning hierarchical representations of features automatically. Hybrid models have also been recently developed, which make use of deep feature extraction along with machine learning classifiers for enhanced performance. Nevertheless, the majority of the CNN based and hybrid models make use of homogenous activations, where a single activation function is utilized throughout the entire model, possibly hampering the diversity of the representations learned by the model.

Nonlinearities are observed within the ECG signal in several morphological aspects, such as P waves, QRS complex, T waves, and ST segments. As each AF behaves uniquely with respect to the pattern within the signal, using a single AF may limit the network from learning

all possible ECG characteristics (Zhou and Fang, 2025).

For overcoming this drawback, the proposed CHAF model adopts the heterogenous activation scheme, where different AFs are utilized at consecutive convolutional blocks.

3.5.1 Dataset and Preprocessing

For evaluating the suggested CHAF model, experiments are performed on two commonly used benchmark datasets for ECG classification tasks, i.e., the PTB Diagnostic ECG Dataset and the MIT-BIH Arrhythmia Dataset (Bousseljot *et al.*, 2009; Goldberger *et al.*, 2000; Moody and Mark, 2001). Publicly accessible datasets help perform experiments in an unbiased way and allow for a comparison with the state-of-the-art techniques. The PTB dataset is utilized for binary classification, and the MIT-BIH dataset is used for the five-class heartbeat classification problem. The most important features of both of the datasets are summarized in Table 3.1.

Table 3.1 Summary of ECG Databases Used in This Study

S.No	**Features**	**PTB Database**	**MIT-BIH Arrhythmia Database**
1.	Source	Provided by PTB, the National Metrology Institute of Germany	BIH laboratory between 1975–1979
2.	Recordings	549 records	24-hour recordings from Holter ECG (48 half-hour snippets)
3.	Total number of subjects	549 ECG records from 290 subjects (209 men, 81 women), age range: 17–87 years	47 subjects
4.	Conditions covered	Myocardial Infarction (148), Cardiomyopathy (18), Bundle Branch Block (15), Dysrhythmia (14), Myocardial Hypertrophy (7), Valvular Heart Disease (6), Myocarditis (4), Miscellaneous (4), Healthy people (52)	Normal sinus rhythm, ventricular arrhythmias, supraventricular arrhythmias, junctional arrhythmias, bundle branch block, heart block, other rhythm disorders, conduction abnormalities
5.	Total number of leads	15 (12 standard leads and 3 Frank leads)	2 (modified limb lead II and precordial lead)
6.	Sampling frequency	1000 Hz	360 Hz

ECGs are processed in accordance with (Kachuee *et al.*, 2018). First, only the Lead II records are taken into account, then resampled to 125Hz, segmented into individual beats by

derivative-based R-peak detection and normalized to minimize the subject-to-subject variability. Each beat is obtained with respect to the median R-R intervals and zero-padded to have a constant size of 187 samples. As far as the classification is concerned, the PTB database is divided into normal and abnormal cases, and the MIT-BIH dataset is split into five AAMI EC57 classes: N, S, V, F, and Q. A fixed 70:30 split was applied at beat level; each beat was normalised using its own statistics; classifiers were fitted on training data only. Patient-level partitioning was not applied.

3.5.2 Workflow of the Proposed CHAF Architecture

The suggested CHAF-based architecture provides an improvement in the field of ECG classification by using heterogeneous AFs in each subsequent convolutional block instead of using one AF across the entire neural network architecture. It allows the network to be trained to have different nonlinear representations of ECG morphology, as well as to have more discriminative features. The general architecture of the model is shown in Fig 3.2.

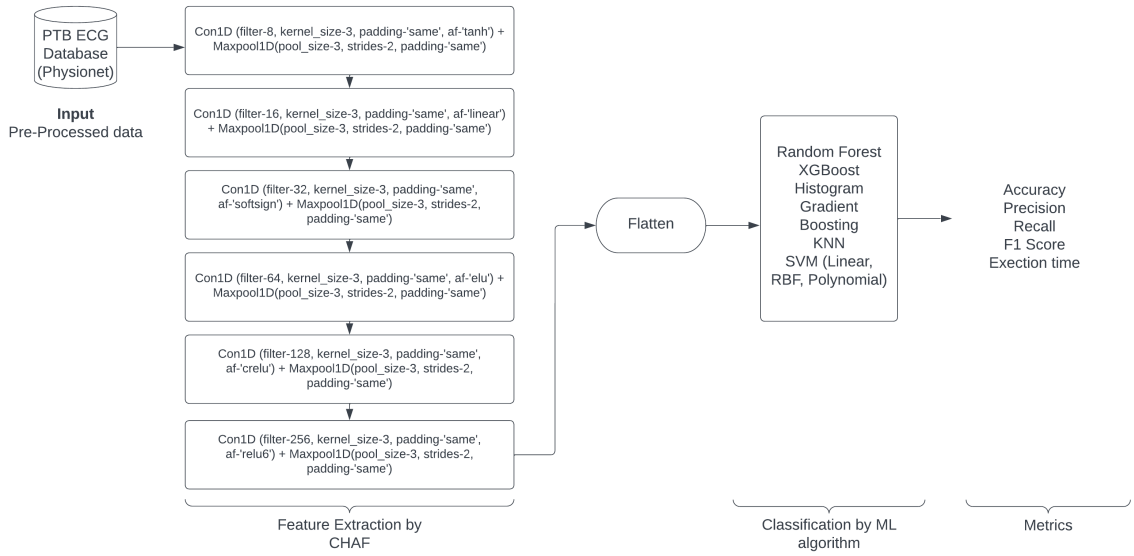


Fig. 3.2 Architecture of the CHAF-based ECG Classification Architecture Using Machine Learning Algorithms

The input signal to the model is a segment of the beat-level ECG signal which is presented as a fixed-length 1D signal denoted by $x(t)$. The feature extraction stage includes a sequence of one-dimensional convolutional layers, where the learning of local temporal patterns is accomplished by convolution (LeCun *et al.*, 1998; Goodfellow *et al.*, 2016):

$$y_k(t) = \sum_{i=1}^{N} x(t+i), w_k(i) + b_k, \tag{3.1}$$

where $w_k(i)$ and b_k are the weights and the bias of the kth filter, correspondingly.

In contrast to the typical CNNs, where each convolutional block uses the same activation function, in the proposed model, each convolutional layer uses different AFs. The activation process is as follows:

$$z_k(t) = f_k!\left(y_k(t)\right), \tag{3.2}$$

where $f_k(\cdot)$ is the activation function of the kth convolutional block. Such an approach increases the variety of features since it allows different layers to learn different nonlinear features of the input signal.

To decrease the temporal redundancy but keep the important features, max-pooling is used after each convolutional block (Goodfellow *et al.*, 2016):

$$p(t) = \max_{t \in \Omega} z(t), \tag{3.3}$$

where Ω is the pooling window. The obtained feature maps can be described as follows:

$$\mathbf{P} \in \mathbb{R}^{T' \times C}, \tag{3.4}$$

where T' is the size of the temporal dimension after applying the max-pooling operation and C is the number

3.5.3 Block-wise Feature Representation Analysis

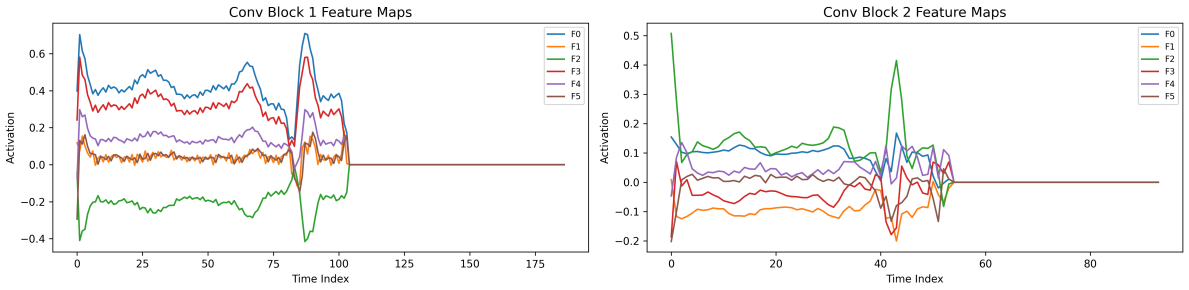


Fig. 3.3 Feature Maps Generated by Convolution Blocks 1 and 2

As depicted in Fig 3.3, 3.4, and 3.5, activations are plotted within sequential convolutional blocks to investigate the feature learning process of the proposed CHAF architecture. As can be observed, the original ECG waveform is highly similar to the feature maps generated within the initial convolutional blocks (Fig 3.3) where mostly local information such as peak points,

transitions, and amplitudes are captured. In other words, the representation gets increasingly smooth and organized as the signal traverses the intermediate blocks (Fig 3.4), meaning that local information is transformed into important features of the waveform.

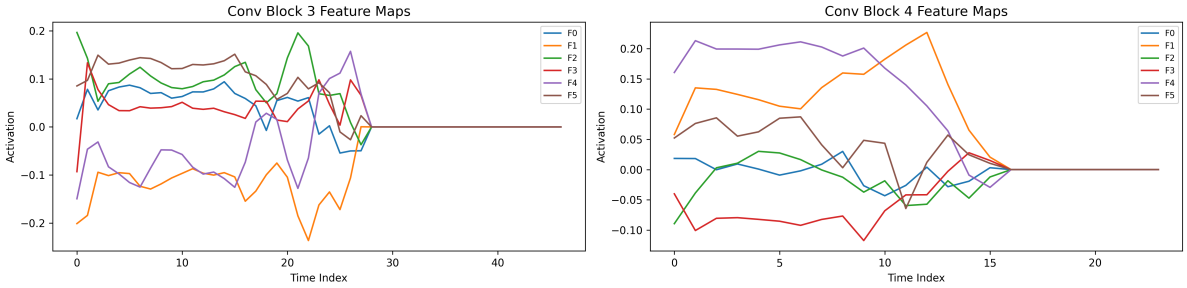


Fig. 3.4 Feature Maps Generated by Convolution Blocks 3 and 4

The feature maps get more selective and compact within deeper blocks (Fig 3.5) in such a way that only the most discriminatory information is kept for the classification task. This step-by-step transformation from the least amount of signal information to the advanced abstract representations demonstrates the effectiveness of CHAF in extracting meaningful ECG features for classification purposes.

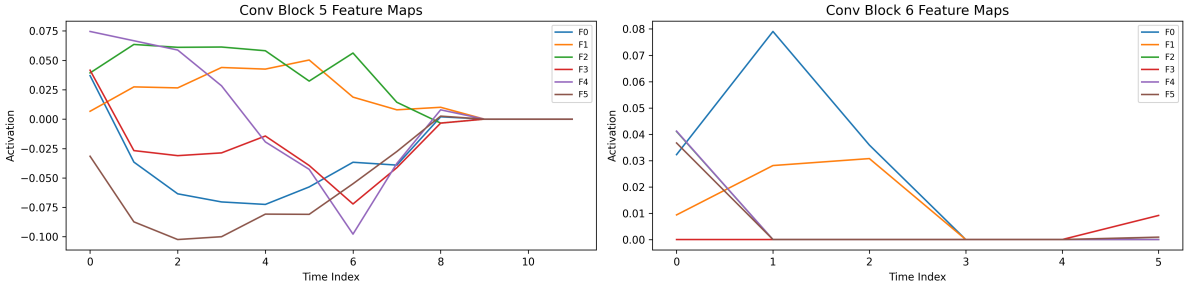


Fig. 3.5 Feature Maps Generated by Convolution Blocks 5 and 6

3.5.4 Statistical Analysis of Feature Distribution

In this section, we conduct quantitative study on the activation distributions that can be learned through the proposed CHAF framework, extending the analysis of feature representations at section 3.5.3. Statistical analysis would provide more understanding about the changing feature responses along the depth of the network, while feature map visualization provides qualitative insights. We analyze the evolution of feature refinement, noise reduction, and representation stabilization during learning through the analysis of the activation distributions and descriptive statistics in each convolutional block.

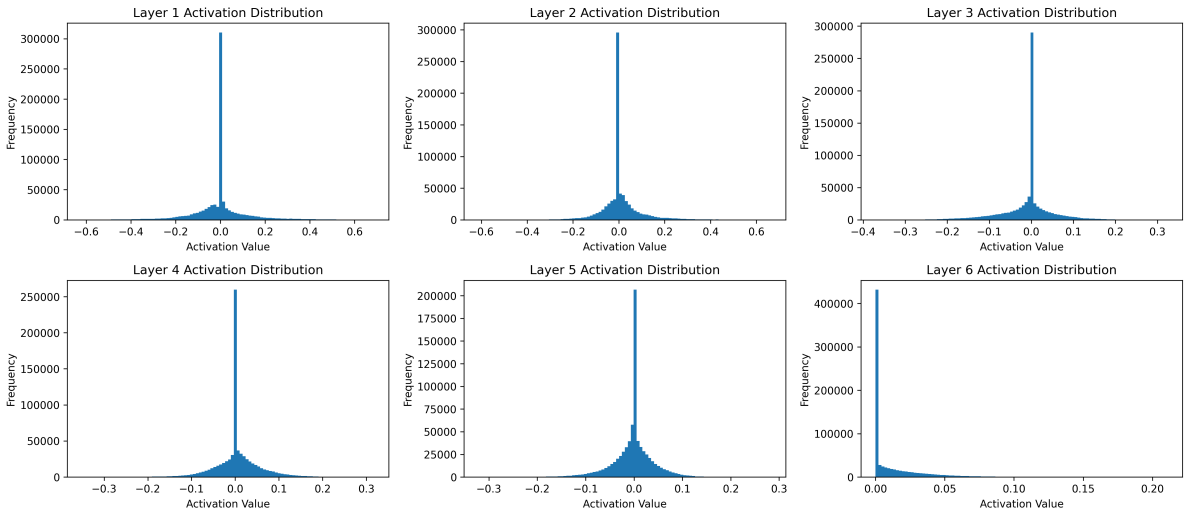


Fig. 3.6 Activation Distribution Histograms Across Different Convolutional Layers

The histogram of the activation distributions for each convolutional block is shown in Fig 3.6. The earlier layers react more sensitively to changes in the basic ECG signal, as shown from their more spread out and higher variance distribution. The distributions get more concentrated as the depth increases, reflecting the gradual removal of irrelevant changes and noise.

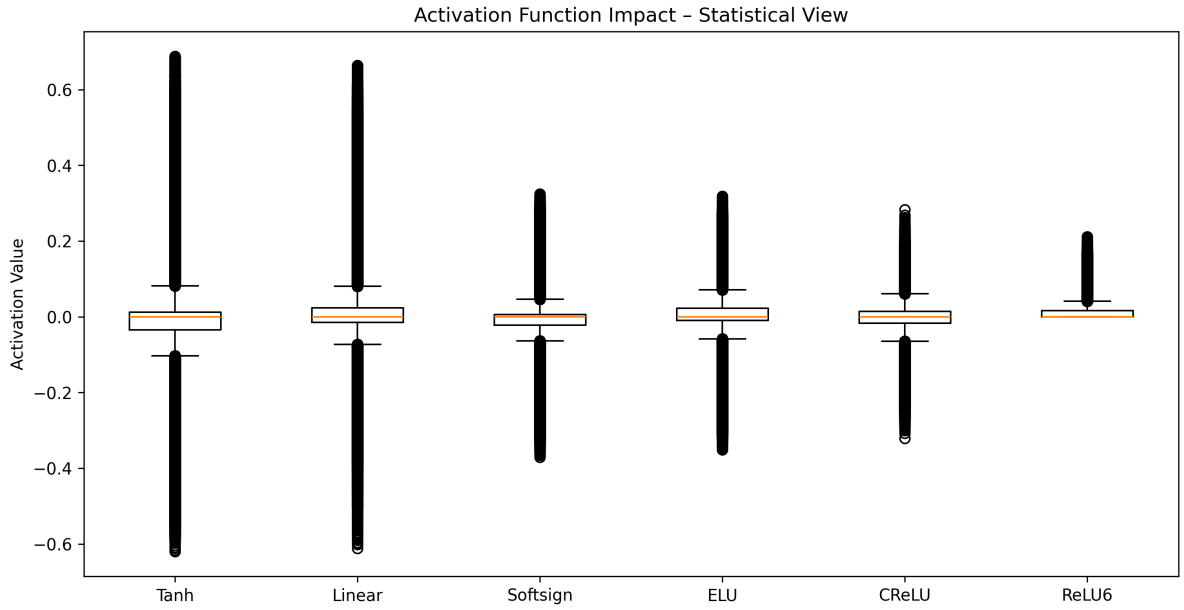


Fig. 3.7 Statistical Distribution of Activation Values Across Different Activation Functions

Fig 3.7 provides a comparative statistical analysis across multiple blocks to understand the effect of different AFs. The results highlight the ability of heterogeneous AFs to efficiently control the spread of activations and avoid both saturation and unnecessary spread, facilitating

the learning of stable features.

Table 3.2 Statistical Summary of Activation Distributions Across Convolutional Blocks

Convolutional Block	Mean Activation	Variance
Conv Block 1	-0.0023	0.0162
Conv Block 2	0.0089	0.0081
Conv Block 3	-0.0085	0.0040
Conv Block 4	0.0043	0.0028
Conv Block 5	-0.0027	0.0018
Conv Block 6	0.0120	0.0004

Table 3.2, which contains the mean and variance of activations in all the convolutional blocks, provides additional quantitative insights. The continuous decrease of the variance from 0.0162 in the first block to 0.0004 in the sixth block implies the improvement of the feature representations. At the same time, the mean values are kept small, around zero, implying constant gradient flow and balance in activation.

3.5.5 Feature Space Projection Analysis

After the block-wise feature representation analysis and the statistical analysis of the activation distributions, in this section it is examined how well the features extracted by the proposed CHAF architecture are organized in the final embedding space for class separability. The analyses above have been concerned with the nature and stability of the learned features, whereas feature space projection allows insights into the global discriminative structure of the features.

In this regard, the flatten layer's high dimensional feature vectors from the CHAF-CNN are embedded into a 2D space using three complementary dimensionality reduction methods: Principal Component Analysis (PCA), t-distributed Stochastic Neighbour Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP). The projections which are created are then gathered together in Fig 3.8 for a visual comparison of the linear and non-linear feature organization.

The 2D feature space projections of CHAF-CNN features by PCA, t-SNE and UMAP are presented in Fig 3.8. The PCA projection shows the distribution of features globally (as a linear view). Some class overlap is found, but obvious clustering patterns do exist, which means the features extracted under linear projection contain still meaningful discriminative

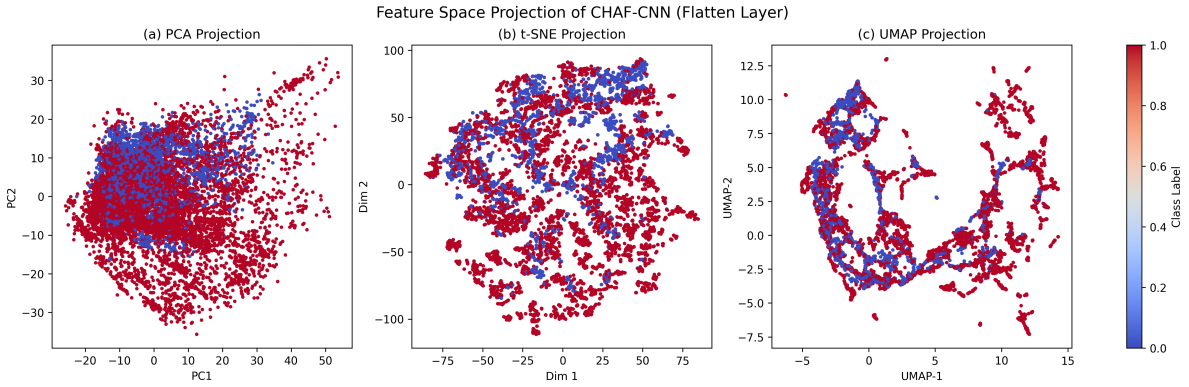


Fig. 3.8 Feature Space Visualization of CHAF-CNN Flatten Layer Using PCA, t-SNE, and UMAP Projections

information.

A t-SNE projection shows better clustering of local classes, more tightly grouped clusters and less variation within each class than PCA. This implies that the features extracted by CHAF maintain more neighborhood-level similarity relationship.

This is further backed up by the UMAP projection that yields cluster with better manifold continuity and tighter and more distinct separation. This indicates that the learned features are distributed in a structured low dimensional space, with samples belonging to the same class being well separated from other classes, but not too far apart.

In general, the feature projections show that the proposed CHAF architecture produces well-structured and discriminative embeddings. This is also a key reason why the good performance of the hybrid architecture, as the classification ability of the KNN classifier largely relies on the class separability in terms of distance. The results of the feature space projection analysis confirm that the performance improvement of CHAF is meaningful, which indicates that it is reasonable to represent the features in a meaningful organization, and its classification robustness and meaningfulness are enhanced.

3.5.6 Evaluation Metrics

Regardless of the type of next-generation ECG classification system (based on conventional ML or advanced DL architectures), they have to be evaluated with the help of objective evaluation metrics. These metrics give quantitative data about the success of a model at its job, and allow for comparisons of different models. Such evaluation is especially crucial in ECG classification due to the potential for severe clinical implications in the event of misclassification.

The measure of accuracy is often employed as an overall measure of performance, but in imbalanced medical database, it can be misleading. For instance, there are more serious consequences if an abnormal ECG is misclassified as normal than if a normal ECG is misclassified as abnormal. Thus, other measures like precision, recall and F1 are employed to achieve a more accurate and clinically relevant assessment. Furthermore, inference time is also taken into account because if the systems of practical diagnosis of an ECG are used in real time or in wearable form, they need to be accurate and also computationally efficient.

These are calculated based on a confusion matrix that is compiled by comparing the predicted labels and actual class labels. True positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) are its constituent parts. TP represents aberrant ECG signals correctly diagnosed as abnormal, TN represents normal ECG signals correctly classified as normal, FP represents normal ECG signals incorrectly classified as abnormal, and FN represents abnormal ECG signals incorrectly classified as normal.

The accuracy is the percentage of correctly classified ECG samples over the whole number of samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.5}$$

Precision indicates how many of the ECG signals predicted as abnormal are actually abnormal, thereby reflecting the model's ability to reduce false alarms.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.6}$$

Recall (or sensitivity) measures how effectively the model identifies all abnormal ECG cases and is especially important in medical diagnosis to avoid missed detections.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.7}$$

F1-score, which is the harmonic mean of precision and recall, provides a balanced evaluation and is particularly useful in imbalanced classification problems.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.8}$$

The proposed architecture is evaluated not only in terms of classification performance, but also computational efficiency through execution time analysis, where the total time required for feature extraction and classification is measured. The experiments were all conducted on

the AMD Ryzen 7 4800H 2.90 GHz processor with 16 GB RAM. All experiments are run on the same hardware and software platform, which provides a valid comparison of execution time between models.

3.5.7 Evaluating CNN Block Depth

CNNs are used as the base feature extractor due to their effectiveness in modeling ECG signals and their relatively low computational cost. However, the depth of the network, defined by the number of convolutional blocks, significantly affects both classification performance and computational efficiency. Increasing depth does not always guarantee better performance and may lead to overfitting and increased complexity.

To determine the optimal depth, a systematic evaluation was conducted using a baseline CNN with a homogeneous ReLU AF. All experiments were performed with a fixed 70:30 training–testing split for consistency. The results for different block configurations are summarized in Tables 3.3.

Table 3.3 Layer Details and Performance of CNN Architecture with ReLU AF

Block Number	**Layer Details**	**Time (s)**	**Accuracy (%)**
4	3rd block + Conv1D($f = 64$, $ks = (3,)$) MaxPool1D($ps = (3,)$, $s = 2$) Total Parameters: 106769	25.00	98.53
5	4th block + Conv1D($f = 128$, $ks = (3,)$) MaxPool1D($ps = (3,)$, $s = 2$) Total Parameters: 230033	32.68	99.07
6	5th block + Conv1D($f = 256$, $ks = (3,)$) MaxPool1D($ps = (3,)$, $s = 2$) Total Parameters: 525713	47.12	99.18
7	6th block + Conv1D($f = 512$, $ks = (3,)$) MaxPool1D($ps = (3,)$, $s = 2$) Total Parameters: 1575825	84.84	98.80
8	7th block + Conv1D($f = 1024$, $ks = (3,)$) MaxPool1D($ps = (3,)$, $s = 2$) Total Parameters: 4200337	390.109	98.69

As can be seen in the Tables 3.3, the number of convolutional blocks is increased, and the accuracy is improved at first, reaching the highest accuracy with 6 blocks (99.18%). If the depth is further increased, the accuracy will decrease, besides the number of model parameters

and the inference time will significantly increase. This suggests that deeper models will have decreasing returns with possible overfitting.

Given this analysis, the selected CNN architecture for this analysis is composed of 6 blocks, which is the best trade-off between accuracy and computational efficiency. This setup is thus fixed for the rest of the experiments conducted with the proposed CHAF architecture, therefore minimizing the effect of the growing depth of the models on the outcomes.

3.5.8 Comparative Analysis of Activation Functions in CNN

After identifying the optimal CNN depth as six convolutional blocks (Section 3.5.7), the next analysis examined the effect of different AFs on classification performance. In the baseline model, ReLU was used uniformly across all hidden layers, representing a homogeneous activation setup, while Sigmoid was used in the output layer for binary classification.

To isolate the influence of AFs, the same six-block CNN architecture was trained and evaluated by applying different AFs individually across all convolutional layers. This ensured that performance differences were due solely to the activation behavior rather than changes in architecture. The corresponding execution time and classification accuracy are presented in Tables 3.4.

Table 3.4 Performance of CNN for Different AFs with the Number of Convolutional Blocks Fixed at 6

S.No	Activation Function	Time (s)	Accuracy (%)
1	Swish	75.52	99.23
2	SoftMax	75.55	72.14
3	Exponential Linear Unit	72.36	99.33
4	Parametrized ReLU	77.77	99.16
5	Scaled Exponential Linear Unit	52.84	99.04
6	Concatenated ReLU	67.36	98.92
7	ReLU-6	66.19	99.16
8	Leaky ReLU	68.32	99.27
9	Tanh	46.91	99.46
10	Sigmoid	45.48	72.14
11	Gaussian Error Linear Unit	54.41	99.34
12	Linear Function	44.59	99.49
13	SoftSign	71.74	99.31
14	SoftPlus	71.45	96.94
15	Mish	60.26	99.55

As shown in Tables 3.4, most AFs achieved high classification accuracy, with several outperforming the baseline ReLU-based CNN. In particular, Tanh, ELU, GELU, Mish, and Softsign demonstrated consistently strong results, indicating their ability to learn rich and discriminative ECG features. In contrast, Sigmoid and Softmax showed substantially lower accuracy when used in hidden layers, confirming that they are less suitable for deep feature extraction.

Compared with the baseline ReLU configuration (99.18%), several AFs achieved comparable or improved performance, confirming that activation choice significantly influences CNN representational capability. These results suggest that using a single AF throughout the network may limit feature diversity and learning flexibility.

Although several AFs improved classification performance, the CNN still relies on dense classification layers, which increase parameter count and computational cost. This limitation motivates the transition toward a hybrid CNN–ML architecture, where the CNN is used only for feature extraction and lightweight ML classifiers are employed for final classification, as explored in the proposed CHAF architecture.

3.5.9 Baseline CNN–ML Model with ReLU Activation Function

The results shown in Sections 3.5.7 and 3.5.8 proved that a six-layer CNN model gives a good trade-off between efficiency and computational cost, and that AFs play a very important role in the process of feature learning. However, traditional CNNs still use fully connected layers that cause an increase in the number of parameters, memory consumption, and computational overheads.

In order to overcome this problem, we propose a baseline hybrid CNN-ML architecture in which the CNN model serves only as a feature extractor while the classification process is performed by ML algorithms. This approach will help reduce the computational costs, but still maintain the discrimination capabilities of deep features. The general architecture is depicted in Fig 3.9.

In this approach, the preprocessed ECG signals at the level of beats are fed through several 1D convolution and max-pooling layers with a homogeneous ReLU activation function. After feature maps extraction, the flattening procedure is applied to generate a 1D feature vector that is used as an input to ML classifiers such as RF, XGBoost, Histogram Gradient Boosting (HGB), KNN, and SVM.

Table 3.5 Performance Analysis of Different Machine Learning Algorithms Using Features Extracted from Different Convolution Blocks

Block	Layers Details and Total Parameters	ML Algorithm	Time (s)	Accuracy	Precision (0/1)	Recall (0/1)	F1 Score (0/1)
4	3rd block + Conv1D (f=64, ks=(3,)) MaxPool1D (ps=(3,), s=2) **Total Parameters: 106,769**	RF	4.95	98.78	99/99	97/100	98/99
		XGB	5.31	98.90	99/99	97/100	98/99
		HGB	16.07	98.87	99/99	97/100	98/99
		K-NN	0.006	98.00	95/99	97/98	96/99
		SVML	3.49	98.39	97/99	97/99	97/99
		SVMR	4.78	98.37	97/99	97/99	97/99
		SVMP	3.72	98.55	97/99	97/99	97/99
5	4th block + Conv1D (f=128, ks=(3,)) MaxPool1D (ps=(3,), s=2) **Total Parameters: 230,033**	RF	3.33	98.99	99/99	97/100	98/99
		XGB	4.11	99.28	99/99	98/100	99/100
		HGB	15.97	99.15	99/99	98/100	98/99
		K-NN	0.006	99.06	98/99	98/99	98/99
		SVML	1.45	99.03	98/99	98/99	98/99
		SVMR	4.10	99.12	99/99	98/99	99/99
		SVMP	4.35	99.08	99/99	98/99	98/99
6	5th block + Conv1D (f=256, ks=(3,)) MaxPool1D (ps=(3,), s=2) **Total Parameters: 525,713**	RF	3.05	99.10	99/99	98/100	98/99
		XGB	2.56	99.22	99/99	98/100	99/99
		HGB	11.15	99.08	99/99	98/100	98/99
		K-NN	0.006	99.31	99/99	99/100	99/100
		SVML	1.92	99.26	99/99	98/100	99/99
		SVMR	2.52	99.12	99/99	98/99	98/99
		SVMP	2.71	99.10	99/99	98/100	98/99
7	6th block + Conv1D (f=512, ks=(3,)) MaxPool1D (ps=(3,), s=2) **Total Parameters: 1,575,825**	RF	2.44	98.30	97/99	97/99	97/99
		XGB	3.84	98.55	97/99	98/99	97/99
		HGB	14.95	98.55	97/99	98/99	97/99
		K-NN	0.007	98.44	97/99	97/99	97/99
		SVML	5.41	98.64	97/99	97/99	98/99
		SVMR	6.11	98.52	98/99	97/99	97/99
		SVMP	7.48	98.14	97/99	96/99	97/99
8	7th block + Conv1D (f=1024, ks=(3,)) MaxPool1D (ps=(3,), s=2) **Total Parameters: 4,200,337**	RF	3.59	98.05	96/99	97/99	96/99
		XGB	6.50	98.02	96/99	97/99	96/99
		HGB	16.04	97.96	96/99	97/98	96/99
		K-NN	0.007	98.14	97/99	96/99	97/99
		SVML	5.81	97.98	96/99	97/99	96/99
		SVMR	5.74	98.02	96/99	97/99	96/99
		SVMP	22.41	97.25	93/99	98/97	95/98

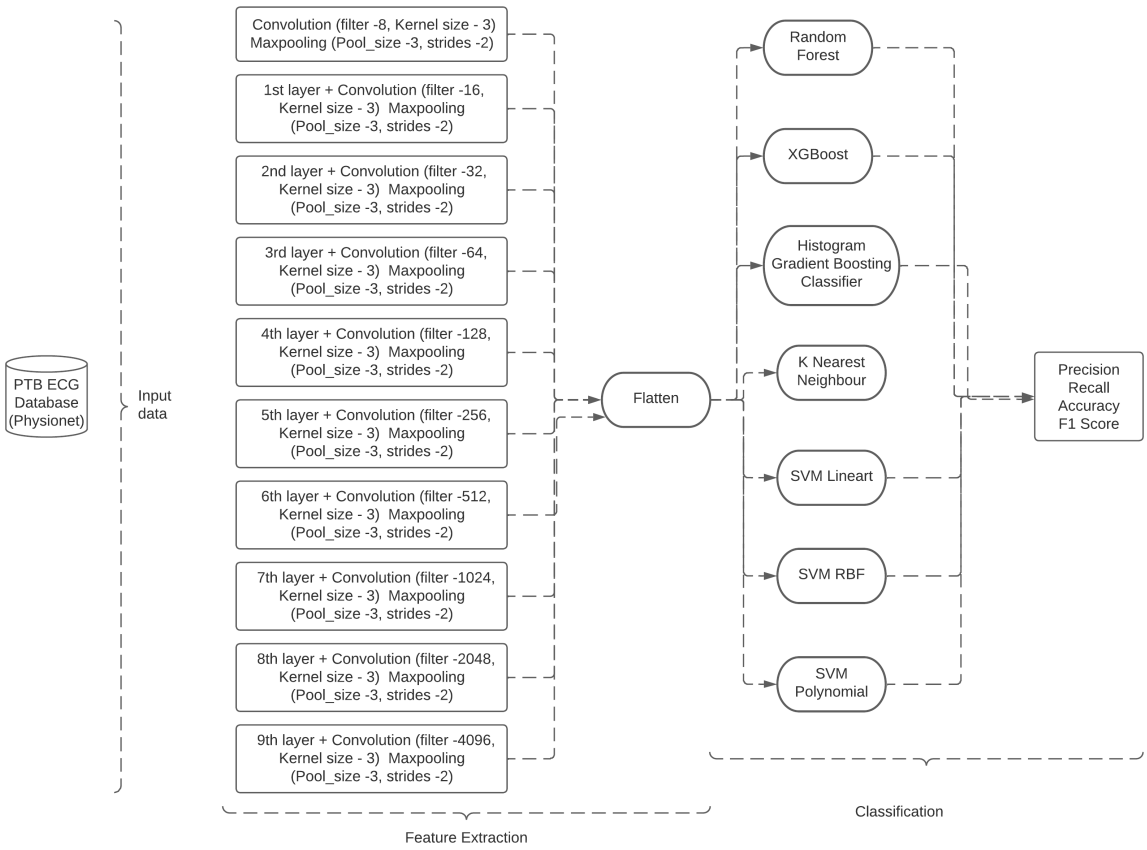


Fig. 3.9 Proposed CNN-Based Feature Extraction and Machine Learning Classification Architecture

The performance of this hybrid model across different CNN depths and classifiers is summarized in Table 3.5 . The results show that the six-block CNN–ML configuration provides the best trade-off between accuracy and inference time. Increasing depth beyond this point results in higher computational cost without significant performance improvement.

A key observation is that replacing dense layers with ML classifiers significantly reduces inference time while maintaining comparable accuracy, precision, recall, and F1-score. Among the classifiers, KNN consistently achieves strong performance, particularly with deeper feature representations. This supports earlier findings that CHAF-like feature spaces are compact and well-structured, making them suitable for distance-based classification.

Overall, the baseline CNN–ML architecture demonstrates that efficient classification can be achieved by combining deep feature extraction with lightweight classifiers. However, since this model still relies on a homogeneous activation strategy, its ability to capture diverse ECG characteristics remains limited. This limitation directly motivates the proposed CHAF architecture, which introduces activation heterogeneity to further enhance feature diversity and

robustness.

3.5.10 Effect of Activation Function Variations in Block-6 CNN–ML Architecture

After establishing the six-block CNN–ML architecture as the optimal baseline in Section 3.5.7, this experiment examines the effect of different AFs within the same architecture. The objective is to evaluate how the choice of AF, when applied uniformly across all convolutional blocks, influences both classification performance and computational efficiency, without altering the network depth or classifier design.

Accordingly, the six-block CNN feature extractor is retained, while different AFs are individually applied across all hidden layers. Several ML methods, including as RF, XGBoost, HGB, KNN, and SVM versions, are then used to classify the collected features. Table 3.6 provides a summary of the findings.

Table 3.6 Performance of CNN–ML Architecture for Different Activation Functions by Fixing the Number of Blocks as 6

S.No	Activation Function	ML Algorithm	Time (s)	Accuracy	Precision (0/1)	Recall (0/1)	F1 Score (0/1)
1	Swish	RF	10.58	98.80	99/99	97/99	98/99
		XGB	18.08	98.87	98/99	98/99	98/99
		HGB	22.78	98.96	98/99	98/99	98/99
		K-NN	0.004	98.85	98/99	97/99	98/99
		SVML	1.631	98.85	98/99	98/99	98/99
		SVMR	15.05	98.57	99/98	96/100	97/99
		SVMP	10.57	96.81	98/96	90/99	94/98
2	Softmax	RF	4.17	72.32	00/72	00/100	00/84
		XGB	6.36	80.18	67/84	56/89	61/87
		HGB	19.63	79.77	68/83	51/91	58/87
		K-NN	0.004	68.63	38/74	21/87	27/80
		SVML	23.97	72.32	00/72	00/100	00/84
		SVMR	23.59	72.32	00/72	00/100	00/84
		SVMP	19.47	27.67	28/00	100/00	43/00
3	Exponential Linear Unit	RF	9.29	98.94	99/99	98/99	98/99

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Table 3.6 Continued from previous page

S.No	Activation Function	ML Algorithm	Execution Time (s)	Accuracy	Precision (0/1)	Recall (0/1)	F1 Score (0/1)
		XGB	12.18	98.96	98/99	98/99	98/99
		HGB	20.18	99.10	99/99	98/99	98/99
		K-NN	0.008	99.22	99/99	98/100	99/99
		SVML	0.57	99.17	99/99	98/100	99/99
		SVMR	6.26	99.08	99/99	98/100	98/99
		SVMP	1.20	99.08	99/99	98/99	98/99
4	Parameterized ReLU	RF	10.18	98.76	98/99	97/99	98/99
		XGB	13.34	98.90	98/99	98/99	98/99
		HGB	15.51	98.80	99/99	97/99	98/99
		K-NN	0.007	98.90	98/99	98/99	98/99
		SVML	0.91	98.83	98/99	98/99	98/99
		SVMR	1.36	98.71	98/99	97/99	98/99
		SVMP	1.17	98.64	98/99	97/99	98/99
5	Scaled Exponential Linear Unit	RF	10.24	98.85	98/99	97/99	98/99
		XGB	12.85	98.99	99/99	98/99	98/99
		HGB	20.87	99.06	99/99	98/99	98/99
		K-NN	0.004	99.17	98/99	99/99	99/99
		SVML	0.77	99.06	98/99	98/99	98/99
		SVMR	42.99	97.18	99/96	90/100	95/98
		SVMP	2.28	99.22	99/99	99/99	99/99
6	Concatenated ReLU	RF	12.49	99.22	99/99	99/100	99/99
		XGB	19.53	99.19	99/99	99/100	99/99
		HGB	41.68	99.17	99/99	99/100	99/99
		K-NN	0.011	99.54	100/100	99/100	99/100
		SVML	1.03	99.26	99/99	99/100	99/99
		SVMR	40.56	98.41	99/98	97/100	97/99
		SVMP	1.42	99.22	99/99	99/100	99/99

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Table 3.6 Continued from previous page

S.No	Activation Function	ML Algorithm	Execution Time (s)	Accuracy	Precision (0/1)	Recall (0/1)	F1 Score (0/1)
7	ReLU6	RF	4.71	98.96	98/99	98/99	98/99
		XGB	8.47	99.06	99/99	98/99	98/99
		HGB	18.26	99.08	98/99	98/99	98/99
		K-NN	0.008	99.24	99/99	99/99	99/99
		SVML	1.17	98.94	98/99	98/99	98/99
		SVMR	2.51	99.28	99/99	99/100	99/100
		SVMP	1.52	99.06	98/99	98/99	98/99
8	Leaky ReLU	RF	10.90	99.03	99/99	98/100	98/99
		XGB	13.93	98.99	99/99	98/100	98/99
		HGB	19.25	98.94	99/99	98/99	98/99
		K-NN	0.004	98.94	98/99	98/99	98/99
		SVML	0.86	99.19	99/99	98/100	99/99
		SVMR	2.30	99.31	99/99	98/100	99/100
		SVMP	1.66	99.10	99/99	98/100	98/99
9	ReLU	RF	6.30	98.90	98/99	98/99	98/99
		XGB	8.80	98.92	98/99	98/99	98/99
		HGB	16.08	98.90	98/99	98/99	98/99
		K-NN	0.004	99.15	99/99	98/100	98/99
		SVML	0.89	99.03	98/99	98/99	98/99
		SVMR	2.64	99.19	99/99	98/99	99/99
		SVMP	1.52	99.08	98/99	98/99	98/99
10	Tanh	RF	9.02	99.06	99/99	98/100	98/99
		XGB	12.27	99.17	99/99	98/99	99/99
		HGB	20.92	99.03	98/99	98/99	98/99
		K-NN	0.004	99.45	99/99	99/100	99/100
		SVML	0.84	99.19	99/99	98/100	99/99
		SVMR	5.94	99.26	99/99	98/100	99/99

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Table 3.6 Continued from previous page

S.No	Activation Function	ML Algorithm	Execution Time (s)	Accuracy	Precision (0/1)	Recall (0/1)	F1 Score (0/1)
		SVMP	1.44	99.24	99/99	98/100	99/99
11	Sigmoid	RF	5.411	92.57	87/95	86/95	86/95
		XGB	15.47	92.87	88/95	87/95	87/95
		HGB	23.77	92.37	86/95	86/95	86/95
		K-NN	0.004	90.60	81/95	86/92	84/93
		SVML	45.66	72.32	00/72	00/100	00/84
		SVMR	43.51	72.32	00/72	00/100	00/84
		SVMP	52.35	72.32	00/72	00/100	00/84
12	Gaussian Error Linear Unit (GELU)	RF	13.80	98.71	99/99	97/99	98/99
		XGB	18.61	98.83	98/99	97/99	98/99
		HGB	27.03	99.06	99/99	98/100	98/99
		K-NN	0.007	98.71	98/99	97/99	98/99
		SVML	1.73	98.85	98/99	98/99	98/99
		SVMR	19.11	99.01	98/99	97/100	98/99
		SVMP	24.97	96.70	99/96	89/99	98/98
13	Linear	RF	13.53	99.01	99/99	98/100	98/99
		XGB	15.46	99.15	99/99	98/99	98/99
		HGB	28.55	99.12	99/99	98/99	99/99
		K-NN	0.008	99.40	99/99	99/99	99/100
		SVML	0.85	99.22	99/99	99/99	99/99
		SVMR	6.09	99.22	99/99	98/99	99/99
		SVMP	4.33	99.33	99/99	98/99	99/99

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Table 3.6 Continued from previous page

S.No	Activation Function	ML Algorithm	Execution Time (s)	Accuracy	Precision (0/1)	Recall (0/1)	F1 Score (0/1)
14	Softsign	RF	11.54	99.10	99/99	98/100	99/99
		XGB	13.71	99.15	99/99	98/99	98/99
		HGB	22.43	99.12	98/99	98/99	98/99
		K-NN	0.012	99.42	99/100	99/100	99/100
		SVML	1.35	99.31	99/99	98/100	99/100
		SVMR	1.35	99.49	99/100	99/100	99/100
		SVMP	3.76	99.45	99/99	99/100	99/100
15	Softplus	RF	8.75	97.13	95/98	95/98	95/98
		XGB	28.48	97.68	96/98	96/98	96/98
		HGB	23.70	97.50	96/98	95/98	95/98
		K-NN	0.004	97.83	95/99	96/98	95/98
		SVML	32.67	95.78	92/97	93/97	92/97
		SVMR	159.30	79.10	100/78	25/100	39/87
		SVMP	5.95	95.92	92/97	93/97	93/97
16	Mish	RF	8.16	98.80	99/99	97/100	98/99
		XGB	5.76	99.01	99/99	98/100	98/99
		HGB	15.29	98.87	96/98	95/98	95/98
		K-NN	0.006	99.08	99/99	98/99	98/99
		SVML	1.13	98.90	98/99	98/99	98/99
		SVMR	2.87	99.15	99/99	98/100	98/99
		SVMP	8.41	97.89	99/98	93/100	96/99

The results show a clear variation in performance across AFs. Functions such as Softmax, Sigmoid, and Softplus consistently yield inferior classification performance, indicating their limited suitability for hidden-layer feature learning in ECG-based CNN–ML models. In contrast, AFs such as ReLU6, CReLU, ELU, Softsign, Linear, and Tanh demonstrate strong and stable performance across accuracy, precision, recall, and F1-score, while maintaining relatively low inference time.

Another important observation is the consistently strong performance of the KNN classifier across a wide range of AFs. Compared to ensemble-based and kernel-based classifiers, KNN often achieves competitive or superior accuracy with very low inference time. This further supports the earlier analyses showing that the learned feature space is compact, structured, and well suited for distance-based classification.

Overall, this experiment confirms that AF selection remains critical even in CNN–ML hybrid models. The strong performance of several AF–classifier combinations, particularly those involving KNN, directly motivates the proposed CHAF architecture, where multiple AFs are strategically integrated across convolutional blocks to further improve feature diversity and classification robustness.

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and stable performance across accuracy, precision, recall, and F1-score, while maintaining relatively low inference time.

In general, this experiment verifies the importance of AF selection despite hybridization using CNN and ML. The good results achieved through the use of various AF and classifier pairs, especially the use of KNN, are what motivate the design of CHAF, in which multiple AFs are incorporated into convolutional layers.

3.6 Optimization and Evaluation of the CHAF architecture

The output of the meticulous analysis undertaken in the course of this chapter is CHAF architecture. Previous studies identified the optimal CNN depth, assessed the effect of each individual AF in a homogeneous setting, and proved that moving from dense layers to lightweight ML classifiers reduces computational complexity but preserves classification results. Inspired by those findings, the CHAF design employs heterogeneous AFs using convolutional blocks in a hybrid CNN–ML structure.

Unlike classic CNNs, in which the same AF is used at all layers, CHAF uses diverse AFs for different convolutional blocks. The fact that heterogenous AFs have complementary nonlinear properties and can thus more effectively detect different ECG morphological features serves as the motivation behind CHAF. Table 3.9 presents the comparison of various heterogeneous AF designs in combination with different ML classifiers.

As shown in Table 3.9, the evaluated CHAF configurations consistently achieve high classification performance, with several combinations exceeding 99.5% accuracy while maintaining very low inference time. Among the classifiers, KNN consistently delivers the most favorable performance across multiple activation combinations, particularly in terms of execution efficiency and balanced classification metrics. This is consistent with the earlier feature distribution and feature space analyses, which showed that the learned embedding are compact, structured, and well suited for distance-based classification.

For the PTB database, the six-block CNN backbone was retained, and six top-performing AFs were selected for heterogeneous assignment. Although this results in a theoretical search space of 720 possible activation permutations, only representative high-performing configurations are reported due to practical and space constraints. These selected combinations are

sufficient to demonstrate the effectiveness of the proposed heterogeneous activation strategy.

Table 3.7 Performance of CHAF Architecture with Different AF across Varying K Values

Model	K = 3			K = 5			K = 7			K = 9		
	Acc	**F1**	**IT**	**Acc**	**F1**	**IT**	**Acc**	**F1**	**IT**	**Acc**	**F1**	**IT**
CHAF-KNN tanh	99.3	99.5	0.008	99.2	99.5	0.006	99.4	99.6	0.006	99.0	99.3	0.006
CHAF-KNN linear	99.3	99.5	0.006	99.1	99.4	0.006	99.1	99.4	0.007	99.1	99.3	0.007
CHAF-KNN Softsign	99.2	99.4	0.007	99.2	99.4	0.011	99.4	99.5	0.008	99.3	99.5	0.012
CHAF-KNN elu	99.4	99.6	0.010	99.2	99.4	0.009	99.1	99.4	0.008	99.1	99.4	0.009
CHAF-KNN swish	99.1	99.4	0.021	99.2	99.5	0.013	99.4	99.6	0.021	99.0	99.3	0.022
CHAF-KNN relu	99.1	99.4	0.008	99.0	99.2	0.009	99.0	99.2	0.008	99.1	99.4	0.001
CHAF-KNN all AFs	99.1	99.3	0.006	**99.5**	**99.6**	**0.008**	99.4	99.6	0.007	99.2	99.4	0.009

Further analysis was carried out to optimize the KNN classifier, given its consistently strong performance. The results in Table 3.7 show that a neighborhood size of k=5 provides the most effective balance between bias and variance, yielding stable and high classification performance across multiple activation configurations. The Minkowski distance metric was found to be the most suitable for measuring similarity in the learned feature space.

Table 3.8 Performance Comparison of CHAF–ML Architectures

CHAF–ML Algorithm	Time (s)	Accuracy (%)	Precision	Recall	F1-Score
CHAF-RF	2.90	99.36	99/99	99/99	100/99
CHAF-XGB	6.77	99.25	99/99	99/99	100/99
CHAF-HGB	16.58	99.25	99/99	99/99	99/99
CHAF-KNN	**0.008**	**99.55**	**99/100**	**99/100**	**100/100**
CHAF-SVML	0.706	99.41	100/99	100/99	100/100
CHAF-SVMR	15.54	99.38	99/99	100/99	99/100
CHAF-SVMP	2.09	99.43	100/99	100/99	100/99

Table 3.8 summarizes the performance of the selected CHAF–ML architectures obtained after evaluating multiple heterogeneous activation combinations across different convolutional layers. In the optimal configuration, the convolutional blocks employed tanh, linear, softsign, elu, crelu, and relu6 AFs sequentially with progressively increasing filter dimensions from 8 to 256. Among the various activation ordering strategies investigated, this specific heterogeneous arrangement achieved the best trade-off between ACC, and computational efficiency for ECG signal classification.

Table 3.9 Comparison with the State-of-the-Art Approaches

S.No	Methodology	Database	Accuracy	Precision	Recall	F1 Score	Time (s)
1	CNN	PTB	99.18	98.62	99.03	99.01	47.12
2	CNN–XGBoost–ReLU7 (Thongsuwan *et al.*, 2021)	PTB	99.38	99.38	99.28	99.20	1.23
3	**Proposed CHAF–KNN + each layer with different AF (tanh, linear, softsign, elu, crelu, relu6)**	**PTB**	**99.55**	**99.64**	**99.55**	**100**	**0.008**
4	CNN	MIT–BIH	99.06	99.00	98.83	98.91	295.26
5	CNN–KNN	MIT–BIH	98.52	98.64	98.12	98.37	0.062
6	CNN–XGBoost	MIT–BIH	98.36	98.39	98.36	98.37	13.8
7	**Proposed CHAF–KNN + each layer with different AF (relu6, elu, softsign, crelu, tanh)**	**MIT–BIH**	**99.08**	**98.88**	**99.05**	**98.96**	**0.07**

Selection rationale and sensitivity. AFs were first evaluated individually (Tables 3.4 and Table 3.6). Sigmoid, Softmax, and Softplus were excluded due to poor hidden-layer performance, and six AFs with strong, stable accuracy and low execution time were retained. They exhibit complementary responses: Tanh, Linear, Softsign, and CReLU preserve response polarity (relevant to inverted T waves and ST deviation), ELU retains a non-zero negative response, and ReLU6 provides a bounded output suited to distance-based KNN. Performance depended strongly on excluding unsuitable AFs but varied little (≈ 99.0–99.5%) among suitable ones. The best ordering differed between PTB and MIT-BIH; therefore, no ordering is claimed to be universally optimal. The justification is an empirical inductive-bias argument rather than a formal proof of increased representational capacity. The best homogeneous configuration (CReLU–KNN, 99.54%) is close to CHAF–KNN (99.55%). Hence, the benefit of CHAF is consistent feature representation with very short execution time rather than a large accuracy gain. PReLU (Tables 3.4 and Table 3.6) was evaluated as a homogeneous option; a controlled comparison with CHAF is left for future work.

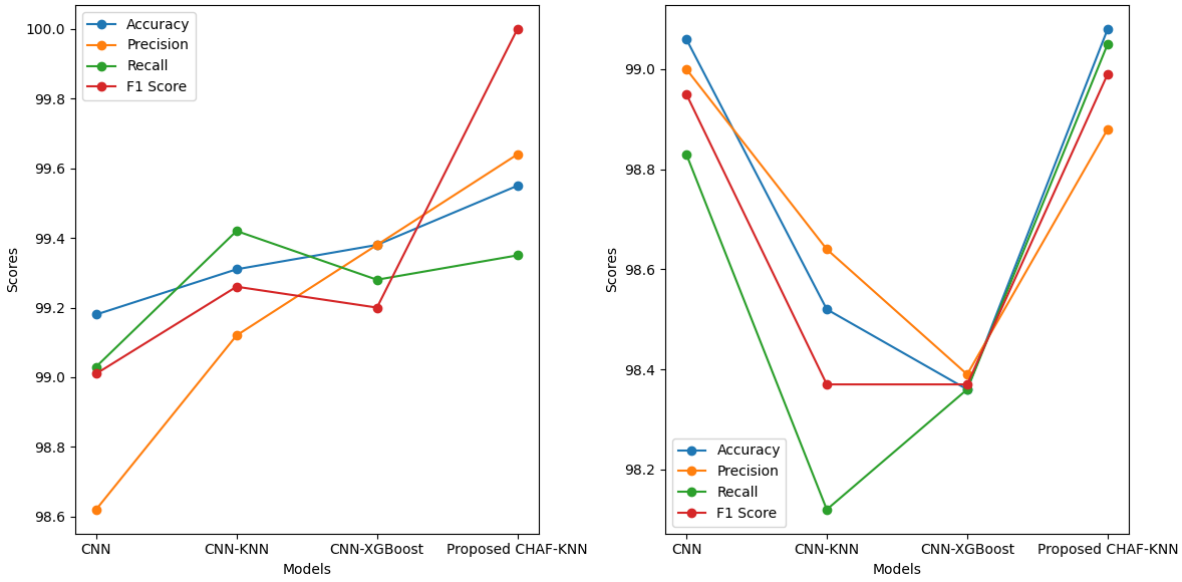


Fig. 3.10 Comparative Performance Analysis of CNN, CNN–KNN, CNN–XGBoost, and Proposed CHAF–KNN Models on PTB and MIT–BIH Databases using Accuracy, Precision, Recall, and F1 Score Metrics

3.7 Comparative Performance Analysis of the CHAF Architecture

The effectiveness of the proposed CHAF architecture is evaluated through a comparative analysis against baseline CNN and CNN–ML models on both the PTB and MIT-BIH databases. The comparison includes a standard CNN with homogeneous ReLU activation and dense layers, as well as hybrid models where CNN is used for feature extraction and ML classifiers perform classification. Since heterogeneous activation has not been previously explored in this context, the evaluation focuses on empirical performance validation.

The results summarized in Table 3.9 show that while the standard CNN achieves high accuracy, it incurs significantly higher computational cost. For the PTB database, the CNN achieves 99.18% accuracy with an inference time of 47.12 seconds. The CNN–XGBoost baseline improves efficiency, reducing inference time to 1.23 seconds with comparable accuracy. However, the proposed CHAF–KNN architecture further improves this trade-off, achieving 99.55% accuracy with a significantly reduced inference time of 0.008 seconds.

A similar trend is observed on the MIT-BIH database, where the standard CNN requires over 295 seconds, while hybrid models reduce computation at the cost of slight performance degradation. In contrast, CHAF–KNN achieves the highest accuracy of 99.08% with an inference time of only 0.07 seconds. For this database, a five-block configuration was adopted to

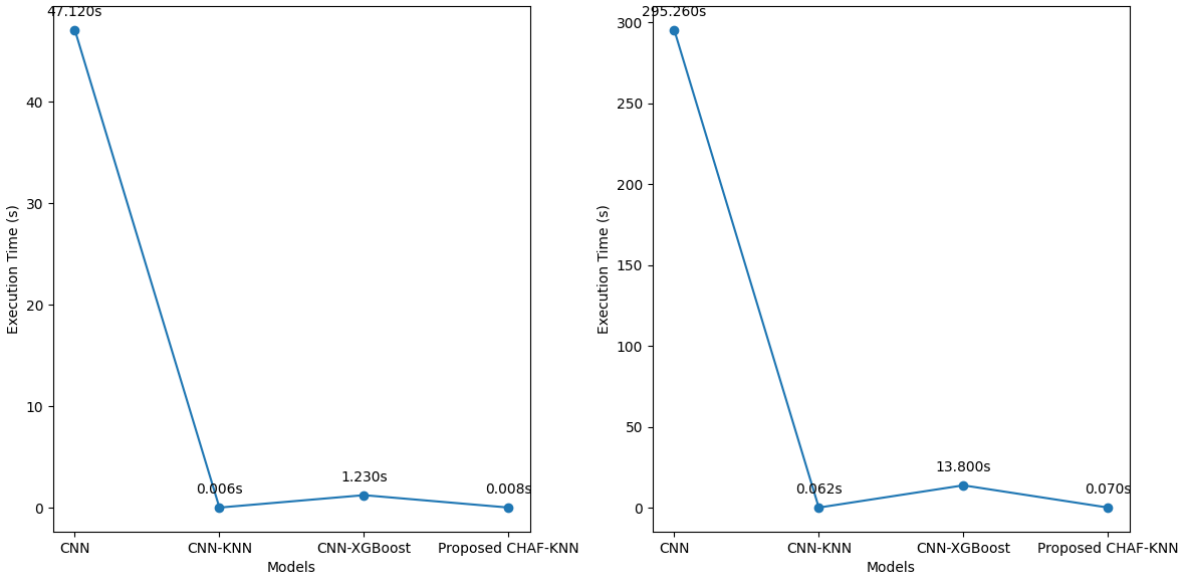


Fig. 3.11 Inference Time Comparison of CNN, CNN–KNN, CNN–XGBoost, and Proposed CHAF–KNN Models on PTB and MIT–BIH Databases

ensure stable training. The robustness and generalization potential of the suggested architecture are consistent performance across both datasets.

Fig 3.10 and Fig 3.11 provide more illustrations of the comparing results, showing that CHAF–KNN routinely delivers better or comparable classification metrics with low inference time. This improvement is attributed to the combination of heterogeneous activation-based feature extraction and the lightweight, distance-based KNN classifier, which together produce compact and discriminative feature representations. CHAF–KNN vs CReLU–KNN (99.55 vs 99.54%), and the dependence on dataset and ordering.

Overall, the findings show that compared to traditional CNN and CNN–ML models, the suggested CHAF architecture strikes a superior compromise between accuracy and computational efficiency. CHAF makes it possible to classify ECGs efficiently, robustly, and reliably by utilizing activation heterogeneity and removing bulky classification layers. This makes potentially suitable for resource-constrained applications.

3.8 Chapter Summary

In this chapter, CHAF was introduced as a heterogeneous activation-based feature learning framework for automated ECG classification which overcomes the weakness of CNN-based approaches due to the use of a unique AF within the network. With the ability to introduce activation diversity within sequential convolutional blocks, CHAF allows the network to ex-

tract different representations from 1D ECG data which help it learn more about the intricate nonlinear and morphological aspects of heart signals.

The effectiveness of the framework was validated through tests on two datasets for ECG classification: PTB Diagnostic ECG Database, where the task is binary classification, and MIT–BIH Arrhythmia Database, which includes multi-class classification. The extracted representations were used to classify the input data with ML algorithms; the CHAF-KNN configuration outperformed other configurations, with an accuracy rate of 99.55% and F1-score of 99.68% and an inference time of 0.008 s for PTB and an accuracy rate of 99.08% and F1-score of 98.99% and an inference time of 0.07 s for MIT–BIH.

In addition to its numerical metrics, feature visualization and activation distribution analyses showed that CHAF learns the representation of ECG signals in a hierarchical and structural way, where the learning progresses from local signal properties to discriminative cardiac features. The integration of heterogenous feature extraction and lightweight machine learning classification makes an efficient and potentially clinically useful for ECG analysis.

Whereas CHAF is capable of extracting local waveform morphology information from the ECG signals efficiently, the convolutional architecture of CHAF is somewhat limited in its capability of modeling long-range temporal correlations. Inspired by the limitation, the next chapter will introduce ECG-ResViT, a hybrid framework of CNNs and Transformers for joint learning of local waveform morphology and global temporal correlation.

Chapter 4

ECG-ResViT: A HYBRID CNN-VIT MODEL FOR ECG CLASSIFICATION

4.1 Chapter Overview

CHAF framework (Premanand and Narayanan, 2023), the first contribution of this dissertation, is discussed in the previous chapter. Unlike the conventional approach of using an AF in all convolutional layers of a CNN model, CHAF framework uses activation diversity between consecutive convolutional layers and hence achieves complementary ECG feature representations. With the help of activation diversity using heterogeneous AFs and ML classifiers such as KNN classifier, the proposed framework provides high-level ECG classification on both PTB Diagnostic ECG Database and MIT–BIH Arrhythmia Database. This shows that the inclusion of activation diversity enhances the ability of feature representation of a CNN network for ECG signals and hence provides efficient classification of heart arrhythmias.

Success of CHAF framework paved way for the realization of the need for learning rich morphological representations of ECG signals from their raw data format. With the help of heterogeneous feature extraction through allowing convolutional layers to learn specific nonlinear transformation of signals, CHAF framework was able to achieve a larger spectrum of waveform characteristics. This forms the base of this dissertation work.

Although there are many merits for CHAF framework, its feature learning process is still convolution-based (Ansari *et al.*, 2023). Although convolutions can effectively extract waveform morphology information, their receptive fields are localized and concentrate only on short-term signal dependencies (Kim *et al.*, 2025). Thus, the learned features are heavily biased toward morphological information in limited temporal neighbors. On the other hand, in clinical ECG diagnosis, diagnostic information usually involves longer time intervals than single cardiac cycle and is reflected in long-term temporal dependencies of rhythms and beat-to-beat variations of the heart (Wang *et al.*, 2026).

In addition, the machine learning classifiers were used as the decision-making module in the original CHAF framework. Even though this approach enabled efficient inference process and good classification results, it limited the capacity of the framework to learn deeper feature representation interactions in an end-to-end way. These points show that the improvement of the model can be reached through introducing global temporal dependency modeling methods along with local morphological learning.

Due to this deficiency, the second contribution of this thesis examines a radically different approach to representation learning. Instead of paying attention only to local waveform attributes, the proposed ECG-ResViT model fuses residual-dilated convolutional learning with transformers to learn both detailed morphological features and global cardiac dependencies in the time series. The proposed local-global learning approach is meant to offer a holistic view of ECG behavior and provide a unifying framework for reliable automated cardiac diagnosis.

Approximately 17.9 million deaths annually, or 32% of total fatalities globally, are attributable to cardiovascular diseases (CVDs), which remain the leading cause of mortality (World Health Organization, 2023). Some cardiac conditions include arrhythmia (ARR), congestive heart failure (CHF), and NSR anomalies, and early diagnosis of such conditions is crucial since a delay can significantly increase the risk of adverse health outcomes. An electrocardiogram (ECG) remains the most widely used noninvasive technique for monitoring heart electrical activity and identifying such disorders. Therefore, developing accurate automated ECG analysis systems has become an important research problem in modern medicine.

Automated ECG classification approaches have made substantial progress recently, thanks to DL advancements. In particular, the ability of CNN-based models to identify distinctive morphological features of the data makes them very effective in eliminating the necessity of manual feature engineering (Wang *et al.*, 2021; Śmigiel *et al.*, 2021). There is, however, one drawback of using CNNs related to their incapacity to capture long-range temporal dependencies among different cardiac cycles since the primary objective of CNNs is identification of local features of the waveform. Many illnesses are characterized by rhythm disorders, inter-beat variability, and evolving temporal dependencies, and not by individual waveforms, thus making temporal dependencies important in the field of medicine.

In order to solve the problem of temporal dependency modeling, various architectures

based on recurrent neural networks such as RNNs, LSTMs, and GRUs are widely used (Sun, 2023; Alamatsaz *et al.*, 2024; Clement Virgeniya and Ramaraj, 2021). Even though these architectures can model dependencies, their iterative nature makes them inefficient and unstable in terms of learning long-term dependencies.

On the other hand, there has been increasing interest in transformer-based networks as an effective framework for analyzing sequential data by using self-attention operations that allow the model to exploit global temporal relations in parallel (Vaid *et al.*, 2023; Halder *et al.*, 2024). ViT model has gone even further and has demonstrated remarkable ability in the analysis of structured data with potential applications in ECG classification problems (Liu *et al.*, 2022; Alkegany and Ali, 2024). Nonetheless, models based on transformers are not efficient at capturing the local information from waveforms and might need much training data to perform well.

After extracting the knowledge on feature extraction from CHAF, Chapter 4 introduces ECG-ResViT as the second significant contribution of the thesis. In this newly proposed network, the combination of residual dilated CNN and transformer has been used for the extraction of local morphological features and long-range dependence between cardiac cycles in the raw 1D ECG data. Even though the ViT encoder uses the mechanism of positional encoding and MHSA for learning global dependency between cardiac cycles in the temporal domain, the residual dilated CNN helps to increase the receptive field while retaining the wave features.

The integration of convolutional learning along with attention-based learning in a single framework in the form of ECG-ResViT allows addressing the main drawback of CHAF architecture, which is the inability to model global temporal dependencies explicitly. Different from the methods that convert ECG signals to images, the suggested framework uses the intrinsic one-dimensional nature of the signal throughout the learning procedure, making it possible to analyze its morphological and rhythmic features. The efficacy of the proposed framework is confirmed by the experiment carried out on the PhysioNet databases for ARR, CHF, and NSR signals, with the accuracy and ROC–AUC scores equal to 99.13% and 99.94%, respectively.

The rest of this chapter will be structured as follows. Section 4.2 will cover the literature review for CNN-based, recurrent, transformer-based, and hybrid models used for ECG classi-

fication. In Section 4.3, the new ECG-ResViT model is described, together with its motivation, formulation, and computation. The experimental setup, data sets, pre-processing methods, and evaluation criteria are covered in Section 4.4. In Section 4.5, the implementation, ablation study, and performance comparisons are discussed. Lastly, Section 4.6 concludes this chapter and suggests possible future work.

4.2 Need for Hybrid Architecture

The theoretical underpinnings of the suggested ECG-ResViT architecture are presented in this section. The current section concentrates on the fundamental DL processes pertinent to ECG analysis, along with their advantages and disadvantages, whereas Chapter 2 offered a thorough assessment of the body of existing work. In particular, it describes how CNNs, residual and dilated convolutional blocks, and transformer-based attention mechanisms are used to represent ECG signals and emphasises how their complementing features drive the creation of a hybrid CNN–Transformer architecture.

4.2.1 Convolution Neural Network for Local ECG Features Extraction

CNNs can automatically learn informative feature representations directly from raw signals, they have become one of the most popular architectures for ECG signal analysis (Abuzied *et al.*, 2025; Khan *et al.*, 2023; Yu, 2023). CNNs are very good at extracting short-term temporal patterns and local morphological features from one-dimensional time-series data in ECG applications.

The convolution procedure, which applies learnable kernels throughout the ECG signal to extract pertinent characteristics, is the core function of a CNN. One way to express a 1D convolution operation is as

$$y(t) = \sum_{i=1}^{K} x(t+i)\, w(i) \tag{4.1}$$

where $y(t)$ is the final feature map, $w(i)$ is a learnable kernel of size K, and $x(t)$ represents the input ECG signal. By combining data from nearby temporal sections of the signal, the network learns localised patterns through this process.

Convolutional filters gradually identify clinically significant waveform characteristics during training, such as P-waves, QRS complexes, T-waves, and the morphological changes that

go along with them. These learnt representations capture important signal characteristics like amplitude fluctuations, slope changes, waveform symmetry, and transition dynamics that are necessary for a precise cardiac diagnosis.

Feature learning becomes hierarchical with increasing network depth. While deeper layers integrate these representations to acquire increasingly abstract and task-specific cardiac characteristics, early convolutional layers concentrate on low-level waveform structures. Without the requirement for explicit feature design, CNNs have demonstrated great performance in ECG classification and arrhythmia detection tasks thanks to this hierarchical learning capabilities (Abuzied *et al.*, 2025; Khan *et al.*, 2023).

Despite these benefits, CNNs mainly concentrate on short-range signal dependencies and naturally function within localised receptive fields. Deeper structures do not clearly reflect long-range temporal interactions across numerous cardiac cycles, despite the fact that they can partially enlarge the receptive field. As a result, traditional CNN-based architectures might not adequately capture rhythm-level features and more general temporal dependencies that are frequently crucial for clinical diagnosis (Allam *et al.*, 2024; Wang *et al.*, 2026; Huang *et al.*, 2024; Sun, 2023).

4.2.2 Residual and Dilated Convolutions for Deep Feature Enrichment

Conventional CNNs are quite good at learning local ECG morphology, as was covered in the preceding section, but they are still not very good at capturing larger temporal correlations that span several cardiac cycles. Modern ECG classification frameworks frequently use residual connections and dilated convolutions to get over this restriction and improve feature representation. This allows for deeper feature learning and better temporal context modelling (Khan *et al.*, 2023).

Deeper networks can be taught more successfully thanks to residual learning, which creates shortcut connections that immediately transfer information across layers. The expression for a residual block is

$$y = F(x) + x \tag{4.2}$$

where y is the final output, x is the input feature map, and $F(\cdot)$ is the nonlinear transformation learned within the block. Residual connections allow the network to gradually improve

both higher-level cardiac representations and low-level waveform features by maintaining information from previous layers (Khan *et al.*, 2023; Ahmed, Abbas, Daadaa, Qureshi, Perumal and Ibrahim, 2023). For ECG analysis, where diagnostically useful information is present at several levels of abstraction, such hierarchical feature learning is very advantageous.

While residual connections improve feature propagation, dilated convolutions enhance the temporal coverage of the network without increasing the number of learnable parameters (Ma *et al.*, 2021). A 1D dilated convolution is defined as

$$y(t) = \sum_{i=1}^{K} x(t + d \cdot i)\, w(i) \tag{4.3}$$

where the dilation rate is indicated by d. Dilated convolutions increase the receptive field and allow the network to capture data over larger temporal regions while maintaining sensitivity to local waveform features by introducing regulated space between kernel elements.

This approach becomes particularly helpful in ECG analysis, where long-term temporal correlations spanning several heart beats along with subtle morphology changes of a single heartbeat can give valuable insights about the patient's state. The combination of dilated convolutions and residual learning helps the network learn feature representations on multiple temporal scales ranging from local wave morphology to more general rhythmic patterns.

Thus, the introduction of residual connections along with dilated convolutions in modern deep learning-based ECG models is a good strategy (Xia *et al.*, 2024). In combination, these techniques bring stable optimization, multiscale feature learning, and better temporal representations, which helps to improve results for such tasks as arrhythmia detection and ECG classification (López-Espejo, 2019; Kuila *et al.*, 2022).

4.2.3 Transformers and Self-Attention for Global Temporal Modeling

The transformer model architecture has proved to be a promising solution to sequence modeling thanks to its ability to establish interactions between all positions in an input sequence via the self-attention module (Shah *et al.*, 2025; Zhao, 2023; Ikram *et al.*, 2025). This property is particularly advantageous for the problem of ECG classification since diagnostic information can be spread out over several cardiac cycles in terms of temporal dependencies, rhythms, and inter-beat interactions.

In contrast to CNNs which mostly rely on the extraction of local features and recurrent

networks which sequentially process input signals, Transformers successfully handle global temporal dependencies and enable parallel computation (Zhao, 2023; Alghieth, 2025). The essential self-attention function is defined as follows

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4.4}$$

where Q, K, and V stand for the query, key, and value matrices, and d_k is the dimensionality of key vectors. The above-mentioned matrices can be obtained from the input sequence X by means of projections with learnable weight matrices:

Due to the learnt dependencies between different regions of the ECG signal in time, the self-attention mechanism allows the model to concentrate on the relevant cardiac events for diagnosis as well as learn the distant patterns of rhythms, which cannot be captured by traditional methods like convolutions or RNNs. Additionally, the learnt attention weights will enhance the interpretability of the model by emphasizing the time segments which contribute most towards the decision made by the model.

4.2.4 Vision Transformer Principles Adapted to 1D ECG Signals

Initially developed for image processing purposes, ViT is now considered an effective architecture for modelling long-range dependencies in sequential biomedical signals such as ECG. The use of ViT is particularly valuable for identifying rhythm-related dependencies and long-range dependencies of the heart since, unlike convolution-based methods, the transformer uses self-attention techniques in order to create relationships between distant temporal portions of a signal.

A continuous 1D signal is first split into the sequence of temporal fragments, referred to as tokens, in case of ECG data. A learnable linear mapping is performed in order to map each token, which corresponds to the localised portion of the ECG waveform, to the higher dimensional embedding space:

$$z_i = x_i W_e + b_e \tag{4.5}$$

where x_i denotes the i-th ECG segment, W_e and b_e are learnable embedding parameters, and z_i represents the corresponding token embedding.

Since transformer architectures do not inherently preserve temporal order, positional information is incorporated into the token embeddings using sinusoidal positional encoding:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d}}}\right) \tag{4.6}$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d}}}\right) \tag{4.7}$$

where pos denotes the token position and d represents the embedding dimension. These encodings preserve the temporal structure of the ECG sequence and enable the model to distinguish the relative ordering of cardiac events.

Transformer encoder blocks, consisting of feedforward and MHSA layers, are used in order to transform generated token embeddings. Tokens within the sequence have the ability to communicate with one another through self-attention, which means that the model is capable of learning not only global temporal dependencies covering several cardiac cycles but also local features of the waveform. This quality is especially useful when trying to detect cardiac diseases where the diagnosis indicators are spread over large time frames.

Architectures based on ViT provide a robust framework for analysing ECG signals due to their ability to model long-range temporal dependencies. They have been successful in the detection of arrhythmias, conduction disorders, and irregular rhythms that cannot be identified through localized feature extraction alone (Dong *et al.*, 2023; Vaid *et al.*, 2023)

4.2.5 Hybrid CNN–Transformer Architectures

Hybrid CNN-transformer architectures have been found to be an effective method for ECG signal analysis due to their ability to take advantage of the complementary benefits provided by the two approaches of convolutional and attention-based neural networks (Zhou *et al.*, 2023; Kim *et al.*, 2025). In such hybrid systems, CNN models are used to identify the local morphological features of ECG signals, whereas transformers capture the global temporal relationships among the learned features.

By performing feature extraction and temporal compression prior to transformer processing, CNN modules reduce the sequence length presented to the attention mechanism. This not only improves the efficiency of self-attention but also allows the transformer to focus on learning higher-level temporal dependencies and rhythm-level patterns. Consequently, hybrid

architectures have demonstrated promising performance in a variety of ECG classification tasks.

Despite these advantages, several challenges remain. Many existing frameworks rely on conventional convolutional backbones that may not fully capture multi-scale temporal characteristics of ECG signals. Furthermore, rather than being created especially for one-dimensional biomedical signals, tokenisation techniques are frequently modified from generic transformer topologies. Therefore, it is still an open research challenge to properly balance global temporal modelling, local feature learning, and practical deployment.

Improved CNN backbones, MHSA mechanisms, adaptive positional encoding, and sophisticated preprocessing approaches have been shown to yield significant gains in recent studies (Kim *et al.*, 2025; Arabi *et al.*, 2025; Krishna *et al.*, 2024).

Motivated by these observations, the proposed ECG-ResViT architecture integrates residual–dilated CNN-based feature extraction with a ViT encoder tailored for 1D ECG analysis. By combining multi-scale convolutional learning with transformer-based temporal modeling, ECG-ResViT aims to develop richer and more discriminative representations of cardiac activity for automated ECG classification.

4.3 Proposed ECG-ResViT Architecture

Building upon the limitations identified in Section 4.2, an effective ECG classification framework should be capable of learning both detailed waveform morphology and long-range temporal dependencies from cardiac signals. Motivated by this requirement, the proposed ECG-ResViT architecture integrates residual–dilated convolutional learning with transformer-based temporal modeling to provide a unified framework for comprehensive ECG analysis.

The framework ECG-ResViT is designed based on a hybrid architecture where a residual-dilated CNN is utilized as the frontend feature extractor, extracting the representative local and multi-scale features from the raw ECG signals. After that, the extracted features are converted into the tokenized form and fed into the encoder of ViT to learn the global temporal relations as well as the characteristics at the rhythm level.

As the input ECG signals retain their native 1D form during the learning process, ECG-ResViT keeps the physiological information of the input data intact while learning the repre-

sentative features from the data. Hence, the proposed framework presents a balanced approach towards the automatic classification of ECG.

4.3.1 Limitation of Hybrid architecture

There is much advancement made regarding DL techniques for ECG classification based on the review of available literature. Nonetheless, there remain many challenges to face in order to effectively learn not only the local but also the global characteristics of cardiac signals. P-waves, QRS waves, and T-waves are some examples of local morphological characteristics that CNN-based approaches have shown great success in extracting. Nonetheless, convolution processes by nature are local and hence cannot explicitly represent rhythm level patterns and long range temporal connections which often extend beyond a few heartbeats.

To overcome this limitation, researches have applied recurrent networks like RNN, LSTM, and GRU for ECG analysis. Though these models can capture temporal relationships, their sequential nature makes learning complicated and computationally inefficient, especially when dealing with long ECG signals.

On the other hand, recent works based on transformer models and hybrid CNN-transformer models have shown promise by taking advantage of locality through feature extraction as well as temporal dependencies through self-attention. Despite showing promising results, these works have been either overly complex or have not been explicitly tailored to learn hierarchical representations of raw 1D ECG sequences. Therefore, finding an effective solution that incorporates both local morphology as well as global temporal modeling is an open problem.

Inspired by these findings, we present ECG-ResViT, which combines residual-dilated CNN modules with a Vision Transformer encoder. Residual-dilated CNN modules capture local characteristics of the waveforms while also capturing hierarchical patterns of the waveforms using an extended receptive field. The ViT encoder takes advantage of temporal relationships through self-attention mechanism to learn the information at the morphological as well as rhythm levels of the waveforms in one architecture. Through an integration of both local and global representation learning methods, ECG-ResViT gives a better insight into the dynamics of the ECG and provides an effective solution for automated cardiac diagnosis.

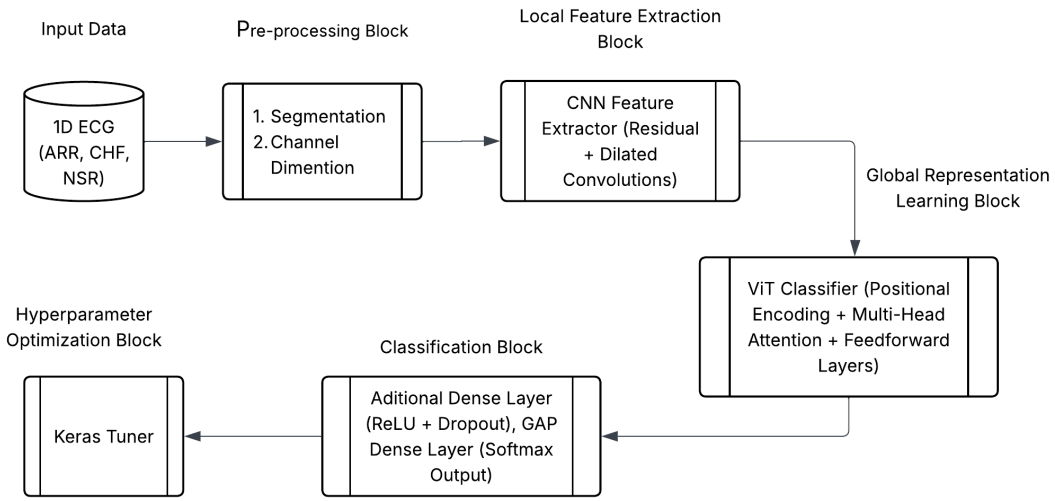


Fig. 4.1 Proposed Hybrid CNN–ViT Architecture for ECG Signal Classification

4.3.2 Overall Architecture Overview

As shown in Fig. 4.1, the overall architecture of ECG-ResViT proposes a novel solution for extracting both local morphological features and long-distance temporal dependencies in ECG signal data. First of all, the ECG signal is segmented into a number of fixed-length segments, ensuring that the temporal nature of heart beats is maintained. The extracted segments are next fed into the residual-dilated CNN block to learn the rich features corresponding to the waveform morphology and short-term temporal variations.

The convolutional features are aggregated and embedded as feature tokens, which will be used as the inputs to the transformer block. In the transformer block, the positional encoding and MHSA are leveraged to learn temporal relationships across multiple cardiac cycles. The interactions among all the token representations can effectively capture the global information in terms of cardiac rhythm, which cannot be learned via convolutional layers. Finally, the transformer-encoded features are further aggregated and fed into a lightweight classifier to output the probability of each cardiac disease.

One of the main strengths of the ECG-ResViT model is that it allows integrating both convolutional and transformer learning methods into a single structure. The task of learning the features of a local waveform is solved using the residual–dilated CNN block; at the same time, the ViT encoder learns global temporal dependencies and connections between elements.

The design of the proposed model was based on four main ideas:

1. Joint learning of local morphological features and global temporal dependencies through the integration of CNN and transformer architectures,
2. Compact and efficient feature representation through aggregation and tokenization,
3. Preservation of the native one-dimensional temporal structure of ECG signals throughout the learning process, and
4. Enhanced interpretability through attention-based modeling of temporal relationships.

4.3.3 Local Feature Extraction Using Residual–Dilated CNNs

The initial phase of the suggested ECG-ResViT architecture is in charge of using raw ECG data to directly learn useful local and multi-scale representations. In order to do this, residual blocks and dilated convolutions are used in the design of a convolutional feature extraction module, which allows the network to capture both the more general temporal patterns and fine-grained waveform features found in cardiac activity.

Let the input ECG segment be represented as

$$x \in \mathbb{R}^{L\times 1} \tag{4.8}$$

where L represents the input ECG segment's length. Convolutional layers are used to treat the signal initially, gradually converting the raw waveform into higher-level feature representations.

A standard 1D convolution operation is defined as

$$y(t) = \sum_{i=0}^{K-1} w(i)\, x(t-i) \tag{4.9}$$

where $w(i)$ represents a learnable kernel of size K. While effective in capturing local morphological structures such as P waves, QRS complexes, and T waves, standard convolutions are inherently limited in their ability to model long-range temporal relationships.

To overcome this limitation, dilated convolutions are incorporated within the convolutional blocks. A dilated convolution is expressed as

$$y(t) = \sum_{i=0}^{K-1} w(i)\, x(t-d\cdot i) \tag{4.10}$$

where d represents the dilation rate. Using dilated convolutions, control of the space between kernel elements provides an expanded receptive field without increasing the kernel size

or the number of parameters. In the proposed architecture, an increasing sequence of dilation rates

$$d \in \{2, 4, 8, 16, 32\} \tag{4.11}$$

allows the network to learn features of the ECG signal at multiple temporal scales, including extended cardiac rhythms to localized waveform characteristics.

To facilitate effective feature learning, residual blocks are included within the architecture. Residual learning consists of the following mapping:

$$y = \mathcal{F}(x) + x \tag{4.12}$$

where $\mathcal{F}(\cdot)$ is the non-linear mapping that is learned within the block. With the use of skip connections, the network is able to learn more efficient feature representations at deeper layers and allows efficient backpropagation of gradients.

With the combination of dilated convolutions and residual learning, feature extraction becomes hierarchical, where deeper layers learn wide temporal relations and shallower layers learn local waveform properties. The feature map of the last residual-dilated convolutional block is expressed as follows:

$$H \in \mathbb{R}^{L' \times C} \tag{4.13}$$

where L' denotes the resulting temporal resolution and C represents the number of learned feature channels.

To obtain a compact representation suitable for subsequent transformer processing, GAP is applied along the temporal dimension:

$$g_c = \frac{1}{L'} \sum_{t=1}^{L'} h_c(t) \tag{4.14}$$

where $h_c(t)$ denotes the activation of the c-th feature channel at time step t. This operation generates a pooled feature vector

$$z \in \mathbb{R}^C \tag{4.15}$$

that summarizes the most informative local and multi-scale characteristics learned from the ECG signal. By combining residual learning, multi-scale temporal modeling, and global feature aggregation, this module produces rich and discriminative feature representations that serve as the foundation for subsequent transformer-based global dependency modeling.

4.3.4 Feature Tokenization and Embedding Strategy

Following local feature extraction through the residual–dilated CNN module, the learned feature representations are transformed into a token-based sequence suitable for transformer processing. Since ViT operate on sequences of embeddings rather than continuous feature maps, a feature tokenization strategy is employed to establish a seamless transition between convolutional feature learning and global temporal modeling.

Let the output of the CNN module be represented as

$$H \in \mathbb{R}^{L' \times C} \tag{4.16}$$

where L' denotes the temporal resolution of the learned feature map and C represents the number of feature channels. This representation captures rich local and multi-scale ECG characteristics but is not directly compatible with transformer-based processing.

To obtain a compact and structured representation, the aggregated CNN feature vector

$$z \in \mathbb{R}^{C} \tag{4.17}$$

is projected into a fixed-length sequence of token embeddings through a learnable transformation:

$$E = \text{reshape}(zW_e + b_e) \tag{4.18}$$

where $W_e \in \mathbb{R}^{C \times (T \cdot d)}$ and b_e are trainable parameters, T denotes the number of tokens, and d represents the embedding dimension. The resulting token embedding matrix is given by

$$E \in \mathbb{R}^{T \times d} \tag{4.19}$$

Each token combines the morphological and temporal information that the convolutional layers have acquired to create an abstract representation of the ECG signal. The transformer

may concentrate on significant temporal associations instead of low-level waveform details because these tokens, in contrast to raw signal segments, reflect higher-level feature abstractions.

To preserve temporal ordering, positional information is incorporated into the token embeddings:

$$Z_0 = E + P \tag{4.20}$$

where $P \in \mathbb{R}^{T \times d}$ denotes the positional encoding matrix. A sinusoidal positional encoding scheme is adopted:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d}}}\right) \tag{4.21}$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d}}}\right) \tag{4.22}$$

where pos denotes the token position and i indexes the embedding dimension. These encodings enable the model to retain temporal ordering information while learning relationships across different regions of the ECG signal.

The resulting position-aware token sequence

$$Z_0 \in \mathbb{R}^{T \times d} \tag{4.23}$$

acts as the ViT encoder's input. The suggested tokenisation system, which is a crucial part of the ECG-ResViT architecture, successfully connects local morphological feature extraction and global temporal dependency modelling by transforming convolutional feature maps into compact information tokens.

4.3.5 Vision Transformer Encoder for Global Temporal Modeling

The ECG signal is portrayed as a compact series of position-aware tokens appropriate for transformer-based processing after feature aggregation and tokenisation. The ViT encoder makes it possible to explicitly describe long-range temporal connections, which enables the network to learn rhythm-level features and interactions spanning several cardiac cycles, in contrast to convolutional processes that mainly capture local patterns.

Let the input to the transformer encoder be denoted as

$$Z_0 \in \mathbb{R}^{T \times d} \tag{4.24}$$

where T denotes the number of tokens and d represents the embedding dimension. Each token encodes locally enriched ECG features extracted by the residual–dilated CNN module.

The core operation of the transformer encoder is the scaled dot-product self-attention mechanism. Given an input token sequence Z, the query, key, and value representations are computed as

$$Q = ZW_Q, \quad K = ZW_K, \quad V = ZW_V \tag{4.25}$$

where W_Q, W_K, and W_V are learnable projection matrices. The self-attention operation is defined as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4.26}$$

This method allows each token to include information from other tokens, thus helping the model learn global time-based relations which cannot be learned by using only convolution.

MHSA is used in order to improve the representation power of the model. Multiple attention heads help the model focus on various temporal relations between the ECG signal:

$$\textit{MHSA}(Z) = \textit{Concat}(head_1, \ldots, head_H)W_O \tag{4.27}$$

where

$$head_h = \textit{Attention}\left(ZW_Q^h, ZW_K^h, ZW_V^h\right) \tag{4.28}$$

and W_O denotes the output projection matrix.

A position-wise feedforward network (FFN) combined with residual connections and layer normalization comes after MHSA in each transformer encoder block:

$$Z' = \textit{LayerNorm}\left(Z + \textit{MHSA}(Z)\right) \tag{4.29}$$

$$Z_{out} = \textit{LayerNorm}\left(Z' + FFN(Z')\right) \tag{4.30}$$

where

$$FFN(x) = \max(0,\, xW_1 + b_1)\, W_2 + b_2 \tag{4.31}$$

While the residual pathway retains the previous representation, the feedforward pathway and attention mechanism further enhance the temporal features. The ViT encoder operates on feature tokens derived from the residual-dilated CNN, thus augmenting local morphology-based representation learning with the global temporal feature learning approach. This is how the local and global feature learning approach helps ECG-ResViT capture the nuances of ECG waves.

Choice of ViT: An encoder-only ViT was chosen because one label is required per segment, and it operates on compact CNN-derived tokens rather than raw samples. A dedicated temporal transformer is a valid alternative; however, no direct controlled comparison was performed. Therefore, the claim is limited to the evaluated experimental conditions.

4.3.6 Classification Layer and Output Representation

After the process of global temporal modeling via the ViT, the produced token representations carry information on both local morphological features and long-range temporal dependencies of the ECG signal. The produced enriched token representations are then used to generate diagnostic predictions using a lightweight classification layer.

The output of our transformer encoder can be expressed as

$$Z_L \in \mathbb{R}^{T \times d} \tag{4.32}$$

where d is the size of the embeddings and T is the number of tokens. Each token embedding contains contextual information using self-attention mechanism of the appropriate ECG segment.

In order to get a global compact representation of the signal, GAP operator is performed over the token dimension:

$$g = \frac{1}{T} \sum_{t=1}^{T} z_t \tag{4.33}$$

where $z_t \in \mathbb{R}^d$ represents the token embedding of the t-th token. This technique produces

one feature vector containing information from all tokens.

$$g \in \mathbb{R}^d \tag{4.34}$$

thereby providing a holistic representation of the ECG signal while reducing sensitivity to local temporal variations.

The aggregated feature vector is then projected into a more discriminative feature space using a fully connected layer:

$$h = \phi(W_c g + b_c) \tag{4.35}$$

where $\phi(\cdot)$ represents a non-linear AF and W_c and b_c are learnable parameters. Through this transformation, the globally aggregated representation can be used to teach the model class-specific feature interactions.

Finally, the transformed features are mapped to the target diagnostic classes through

$$o = W_o h + b_o \tag{4.36}$$

where W_o and b_o denote the trainable parameters of the output layer. The softmax function is used to turn the resultant logits into class probabilities:

$$y_i = \frac{\exp(o_i)}{\sum_{j=1}^{C} \exp(o_j)} \tag{4.37}$$

where y_i is the expected probability of class i and C is the number of diagnostic classes.

ECG-ResViT is an accuracy-oriented hybrid rather than a lightweight model. The number of parameters increases from 973,123 (Base-CNNViT) to 2,203,971 and 6,611,915 (Table 4.1), alongside accuracy gains from 96.30% to 99.13%. FLOPs, inference latency, and memory consumption were not measured. Therefore, a fair comparison with lightweight 1D CNNs under identical hardware conditions is left for future work.

In order to dedicate most of the model capacity to feature extraction and temporal representation learning, the classification module is purposefully kept small. The end-to-end classification system is completed by ECG-ResViT, which efficiently converts the learnt local

and global ECG characteristics into trustworthy diagnostic predictions by combining global token aggregation with a lightweight classifier.

4.3.7 Computational Complexity Analysis

Computational efficiency is an important consideration in ECG classification, particularly when analyzing long-duration recordings. This section examines the computational characteristics of the proposed ECG-ResViT architecture and highlights how its hybrid design balances local feature learning and global dependency modeling.

For a 1D convolutional layer with kernel size K, input length L, input channels C_{in}, and output channels C_{out}, the computational complexity is given by

$$\mathcal{O}\left(L \times K \times C_{in} \times C_{out}\right) \tag{4.38}$$

The proposed architecture utilizes residual-dilated convolutional blocks for local feature extraction. Dilated convolutions increase the size of the receptive field and help extract multi-temporal features from cardiac signals without requiring additional learnable parameters. Residuals make it possible to deepen the network with virtually no extra computation cost. Therefore, the complexity of the convolutional stage grows linearly with respect to input size.

Self-attention, which is used in the transformer block, has computational complexity equal to

$$\mathcal{O}(N^2 d) \tag{4.39}$$

where N is the sequence length and d is the embedding dimension. Because of its quadratic dependence on sequence length, straightforward usage of self-attention for processing of long ECG sequences can become rather costly from the computational perspective.

For this reason, ECG-ResViT implements tokenization and feature aggregation before the transformer stage. Let T be the number of created tokens and assume $T \ll L$. Then the complexity of the attention becomes

$$\mathcal{O}(T^2 d) \tag{4.40}$$

Thus, by using self-attention in the compact token sequence instead of in the whole input sequence, the network is capable of capturing long-term dependencies while remaining computationally efficient.

Overall complexity of ECG-ResViT equals

$$\mathcal{O}\left(LKC_{in}C_{out} + T^2d\right) \tag{4.41}$$

In this way, it can be observed how this design underlines the complementary feature of the architectural design. This is because the residual-dilated CNN blocks allow learning of local morphological features whereas the transformer network allows modeling of global temporal dependencies using a smaller token. When compared to CNNs that focus mainly on local features and transformers that have a quadratic complexity depending on the length of the sequence, ECG-ResViT becomes a good compromise.

4.4 Experimental Setup

This section describes the experimental setup that was employed for evaluating the proposed ECG-ResViT model. In order to provide an accurate and reproducible comparison between our approach and state-of-the-art approaches, this section provides information on the databases employed, the preprocessing steps, the experimental protocol, and the evaluation metrics. Training optimization parameters are among implementation-related details discussed in Section 4.5.

4.4.1 Dateset Description

Benchmarked ECG data sets from open access PhysioNet resources that have been widely recognized in cardiac signal analysis research are utilized in order to evaluate the proposed ECG-ResViT model. There are three relevant clinical states of the heart considered in this study including normal sinus rhythm (NSR) (Goldberger *et al.*, 2000), congestive heart failure (CHF) (Baim *et al.*, 1986; Goldberger *et al.*, 2000), and arrhythmia (ARR) (Moody and Mark, 2001; Goldberger *et al.*, 2000). The databases contain the ECG records of people suffering from various heart diseases and annotated by experts and thus they provide an appropriate foundation for assessment of the classification skills in realistic clinical conditions.

To make the assessment fair and representative, the same amount of recordings was chosen

from each group.

4.4.2 Data Preprocessing and Segmentation

All ECG records must undergo a standardized preprocessing step before training the model in order to achieve consistency and better quality of input data. To enable training on consistent data with a specific structure, all ECG records are split into temporal non-overlapping windows with fixed duration.

This method of splitting retains morphological properties of individual cardiac cycles, brings standardisation of input dimensionality for the DL model, and makes training and model evaluation possible. The DL model operates consistently without computational instability while capturing important cardiac features by modeling the entire record as a set of consistent temporal slices.

In order to make the model evaluation unbiased, splitting into temporal slices is done prior to dataset splitting and is performed at the record level. Thus, there is no possibility of using examples from one ECG record in both datasets (training and testing).

4.4.3 Experimental Protocol

ECG-ResViT is exhaustively validated with respect to a common experimentation methodology. To ensure that there is no occurrence of having any ECG signal of a subject in more than one of the train, validation, and test sets, the dataset is split in subject-independent ways for creating training, validation, and test datasets.

To reduce bias due to a single split of data and make our statistics more reliable, we adopt a cross-validation methodology. The dataset is divided in an identical manner into training, validation, and test datasets for all baselines and our proposed ECG-ResViT. Moreover, identical pre-processing steps are also applied to each dataset for all models. Model selection and hyper-parameter tuning are done only on the training and validation sets, whereas the test set is untouched. Hyper-parameter tuning used only the training and validation sets.

4.4.4 Evaluation Metrics

The standard criteria for ECG classification that are often used in ECG analysis were adopted to determine the performance of the proposed ECG-ResViT model. They are Accuracy, Precision, Recall (Sensitivity), and F1 score as discussed in section 3.5.6. In order to ensure

uniformity and fair comparison, the same evaluation criteria and metrics were used throughout the chapter.

4.5 Results and Discussion

The suggested ECG-ResViT architecture was tested on ECG data pertaining to three clinical cases: ARR, CHF, and NSR. The dataset comprised data that has been used before in the PhysioNet databases, namely, MIT-BIH Arrhythmia Database, MIT-BIH Normal Sinus Rhythm Database, and BIDMC Congestive Heart Failure Database as mentioned in Section 4.4.

A similar proportion of recordings were chosen from each category to guarantee a fair and representative analysis. The model was able to learn both local waveform characteristics and more general temporal patterns from a standardised input format since each ECG recording was divided into 10 non-overlapping segments of 500 samples. One-dimensional convolutional processing was then made easier by the addition of a channel dimension.

AdamW was employed for optimizing the neural network model, while Keras Tuner was employed for optimizing the vital hyper-parameters systematically. The optimized neural network model was trained using 40 epochs, with batch size, learning rate, dropout rate, and weight decay being set to 16, 0.001, 0.3, and 1×10^{-6} respectively. During the entire experimental investigation, the cross-validation approach was implemented to eliminate partition-based bias and ensure reliable evaluation of the performance of the model. The experiments were implemented using the TensorFlow/Keras platform on a system equipped with GPU.

4.5.1 Ablation Study on Model Components

Using controlled architectural modifications, a systematic ablation research was carried out to measure the contribution of different components within the proposed ECG-ResViT architecture. The goal was to assess how each design element affected model complexity and classification performance. The configurations listed below were taken into consideration:

1. **Base-CNNViT:** A baseline hybrid model that consists of a ViT with Linformer attention after a conventional CNN feature extractor.
2. **ResDilate-CNNViT + LS:** An improved hybrid model that includes label smoothing (LS) and residual and dilated convolutional blocks in the CNN stage.
3. **Tuned-ResDilate-CNNViT + LS:** The proposed ECG-ResViT model, which includes

label smoothing, ViT encoding, residual-dilated CNN blocks, and hyper-parameter tuning.

Table 4.1 presents the comparison results. The Base-CNNViT model achieved satisfactory performance, indicating that the combination of convolutional feature extraction and transformer-based temporal modeling provides a strong baseline for ECG classification. However, its performance remains limited by the relatively basic convolutional backbone.

Table 4.1 Ablation Study on Different Architectures for ECG Classification

S.No	Architecture	Parameters	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	Base-CNNViT	973,123	96.30	96.40	96.30	96.29
2	ResDilate-CNNViT + LS	2,203,971	97.78	97.92	97.78	97.79
3	**Tuned-ResDilate-CNNViT + LS**	**6,611,915**	**99.13**	**99.15**	**99.13**	**99.13**

Introducing residual and dilated convolutions in the ResDilate-CNNViT + LS configuration resulted in a clear improvement across all evaluation metrics, demonstrating the effectiveness of enhanced local and multi-scale feature extraction. The addition of LS further contributed to improved generalization.

The Tuned-ResDilate-CNNViT + LS configuration achieved the best overall performance, with an accuracy of 99.13%, confirming the benefit of combining residual–dilated convolutional feature extraction, efficient transformer-based global temporal modeling, regularization, and optimized hyper-parameters.

Overall, the ablation study confirms that each architectural component contributes meaningfully to the final model performance. In particular, the combination of enhanced CNN-based feature extraction and transformer-based temporal modeling is critical for achieving high accuracy ECG classification.

4.5.2 Comparison with State-of-the-Art Methods

For further analysis of the effectiveness of the proposed ECG-ResViT framework, the performance of the proposed framework was compared with the recent state-of-the-art method for ECG classification in ARR, CHF and NSR classes. The comparison results are summarized in Table 4.2.

From Table 4.2 it is seen that most existing works are based on deep CNN architectures, transfer learning or 1D ECG signals being converted to 2D time–frequency representations.

Table 4.2 SOTA Performance of DL Models on ECG Classification (ARR, CHF, NSR)

S.No	Study / Paper	Methodology	Model Used	Remarks	Accuracy / Metrics
1	Cinar et al. (2021) (Çınar and Tuncer, 2021)	1D + Hybrid AlexNet + SVM	AlexNet for feature extraction, SVM for classification	Outperforms traditional ML classifiers like SVM (68.75%) and KNN (65.63%)	96.77% (Acc)
2	Sabeenian et al. (2023) (Sabeenian and Sree Janani, 2023)	Transfer Learning (TL) on CNNs	ResNet18 and DenseNet	Used pretrained CNNs with fine-tuning for ECG classification	ResNet18: Raw ECG – 98.81%, Wavelet Data – 97.05%
3	Nitish Katal et al. (2023) (Katal *et al.*, 2023)	CNN vs GoogLeNet	CNN and GoogLeNet	Smaller CNN outperformed deeper GoogLeNet in 1D ECG signal setting	91.20%
4	Malleswari et al. (2024) (Malleswari *et al.*, 2024)	CWT + 2D Spectrograms + SqueezeNet	SqueezeNet CNN	Achieved highest performance using image-based transformation	98.7% (Acc)
5	Manjur Kolhar et al. (2024) (**?**)	CQ-NSGT-based time–frequency representation	CQ-NSGT + AlexNet	Achieved high validation and test accuracy with proper preprocessing	98.82% (Acc)
6	**Proposed**	**Hybrid CNN–ViT with Residual and Dilated Convolutions, Positional Encoding, and MHSA**	**CNN (local features) + ViT (global dependencies)**	**Combines spatial and temporal learning in 1D signals; Combines local and global learning on raw 1D ECG**	**99.13% (Acc) 99.94% (ROC–AUC)**

These techniques have good performance but need extra pre-processing and sometimes do not completely retain the temporal properties of ECG signals.

On the other hand, ECG-ResViT models the temporal information using a transformer while incorporating residual dilated convolutional learning and trains directly from 1D raw ECG signals. Long-range cardiac interdependence and local morphological patterns can be combined into a single design because to this combination.

The proposed model performed better than the compared models, under the reported settings; the compared studies use different protocols with an accuracy of 99.13% and a ROC–AUC of 99.94%. The results show that the joint learning of local and global ECG characteristics is effective and the ECG-ResViT can serve as a strong framework for automated diagnosis of the heart. Contribution of each component from Table 4.1, and a note that no repeated-seed variability is reported.

4.6 Chapter Summary

This chapter presented the ECG-ResViT that combines two DL models to learn two complementary features directly from raw 1D ECG signals instead of those frameworks that are mostly focused on either local waveform shape or a global temporal relationship. The proposed approach applies feature learning in the 1D form to the original signal, and maintains

the original temporal structure of the signal, so that clinically relevant cardiac information is not lost during the analytic process, or converted to image-based depiction of ECG signal.

The proposed architecture improves the convolutional feature extraction by incorporating residual connections and dilated convolutions. Residual learning allows for better optimization of the network and allows for the information to flow appropriately between layers, while dilated convolutions increase the field of view to account for cardiac patterns of different temporal scales without sacrificing the resolution of the information. These changes enable the network to capture more complex and discriminative local representations in ECG signals, including subtle differences in T-wave morphology, QRS morphology, and P waves.

The recovered representations are then further processed with a ViT adapted for 1D ECG processing to augment the local feature learning. It can model long-range temporal interactions over multiple cardiac cycles using positional encodings and MHSA techniques. This local–global representation learning approach allows the model to simultaneously learn both long-range rhythm-level dependencies and short-range morphological details that are often overlooked by the traditional CNN-based approach.

The experiments were conducted using benchmark ECG data of ARR, CHF and NSR datasets, and the efficiency of the proposed model is proven with 99.13% accuracy and ROC–AUC value of 99.94%. The findings validate the significance of applying deep convolutional learning, in conjunction with transformers for temporal modeling to interpret the ECG signal reliably. The chapter presents ECG-ResViT as a significant development from morphological models, presenting a general framework to identify cardiac relation and the local characteristics of the ECG waveform in a 1D signal.

This is in contrast to the traditional ECG classification systems that use only one representation of ECG or fixed fusion methods, the next chapter proposes BlendNet, a representation-aware feature fusion system which is able to fully utilize the complementary diagnostic information carried by multiple ECG representations. The motivation of the proposed approach lies in the fact that various representations reflect different aspects of cardiac activity and are not equally contributing to the classification process. BlendNet is a method that allows adaptive interaction between complementary feature spaces, leading to a more informative and discriminative representation for ECG signals in automated cardiac diagnosis.

Chapter 5

BLENDING-BASED DEEP LEARNING ARCHITECTURE FOR ECG SIGNAL CLASSIFICATION

5.1 Chapter Overview

As for the previous chapters, we aimed to improve the feature extraction and modeling of ECG signals using the DL architectures. Due to the complex structure of ECG signals, the process of learning representation becomes quite difficult and inefficient for regular CNN architectures, pushing to investigate new expressive DL architectures (Talukder *et al.*, 2025). Unlike homogeneous activation methods, the CHAF architecture introduced in Chapter 3 enriches classical CNN models by heterogeneous AFs within convolutional blocks allowing to learn a wider range of nonlinear ECG features.

In turn, the capability of this type of architecture is further improved in Chapter 4 by ECG-ResViT, which is a hybrid CNN-Transformer architecture that uses self-attention mechanism to model long-range temporal dependencies in ECG signals along with the development of convolution-based feature extraction by residual connections and dilated convolutions. These types of hybrid and attention-based architectures are consistent with the current trends in ECG arrhythmia detection that include the use of CNN-ViT models that show a better ability to capture complicated temporal patterns (Ansari *et al.*, 2023). Overall, CHAF and ECG-ResViT architectures considerably enhance ECG feature learning.

Recent advances in the field of ECG analysis suggest a paradigm shift from dependence on only one representation of ECG data towards application of a number of different complementary representations of cardiac activity, providing a more expressive set of features for discrimination (Zhou and Fang, 2024). Initially, most DL methods represented ECGs as 1D time series and used this representation to analyze waveform morphology and rhythm patterns. Nevertheless, this type of representation gives an insufficient amount of information about the spectral and structural dynamics associated with ECG data, especially concerning frequency

domain and phase based characteristics (Scarpiniti, 2024; Aarotale and Rattani, 2023).

In order to avoid these drawbacks, several research works have considered signal-to-image transformations converting 1D ECGs into 2D images and exploiting convolutional neural networks to take advantage of spatial learning. Image representations like Recurrence Plot (RP), scalograms, spectrograms, angular fields, and Markov Transition Field (MTF) provide diagnostic information that is difficult to obtain using waveform representation (Zhou and Fang, 2024; Scarpiniti, 2024; Aarotale and Rattani, 2023; Gutierrez-Ojeda *et al.*, 2023).

In practice, it has become increasingly evident that relying on a single ECG representation—either signal-based or image-based—is insufficient to fully characterize the complex, multi-scale nature of cardiac activity (Ansari *et al.*, 2023; Zhou and Fang, 2024). While signal-based models have demonstrated strong capability in learning temporal morphology and sequential dependencies, image-based representations provide enhanced sensitivity to frequency patterns, spatial textures, and recurrent dynamics embedded within ECG signals (Aarotale and Rattani, 2023; Narotamo *et al.*, 2024). This complementary behavior underscores the growing necessity of combining multiple representations to achieve more robust and generalizable ECG analysis (Zhou and Fang, 2024; Narotamo *et al.*, 2024).

For example, RP architecture combined with residual CNN have been reported to perform better for arrhythmia detection even in the context of resource-limited or imbalanced data scenario (Gutierrez-Ojeda *et al.*, 2023). Likewise, the methods based on learning using spectrogram, together with MTF and Gramian Angular Field (GAF), highlight the potential of transforming the time correlations of ECG signal into texture representation to improve robustness for a wide range of cardiac diseases (Zhou and Fang, 2024; Aarotale and Rattani, 2023).

From this background, multi-modal and fusion-based learning where ECG representations from various modalities are integrated rather than processing each modality in isolation has received increasing research attention (Oladunni and Aneni, 2025). Early, feature, and decision level fusion are some common methods used for fusion-based learning (Oladunni and Aneni, 2025). Amongst all these, feature-level fusion is of utmost interest because of its good trade-off between efficiency and effectiveness to integrate temporal, structural, and spectral representations through one single learning paradigm (Oladunni and Aneni, 2025).

In parallel, multi-modal architectures that combine heterogeneous ECG representations—such as RPs, GAF, and MTF—within a shared embedding space demonstrate that performance improvements increasingly arise from representation integration rather than merely increasing network depth (Zhou and Fang, 2024).

Motivated by this design gap in existing multi-modal ECG learning architectures—where performance gains are often achieved at the cost of increased architectural complexity and reduced interpretability—this chapter adopts a representation-blending perspective for ECG analysis (Oladunni and Aneni, 2025). Rather than deepening the network or introducing specialized fusion layers, the proposed approach employs an alpha-controlled feature blending strategy to integrate complementary ECG representations in a controlled and interpretable manner (Oladunni and Aneni, 2025).

Time-frequency properties from the CWT, on the other hand, provide information about frequency-dependent changes in the dynamics of the heart. While binary structures focus on edge-based information and morphology of the signal. The proposed framework learns complementary information from both perspectives before applying classification in order to enrich ECG representation without the need for complex and costly fusion blocks or attention mechanisms.

The proposed method highlights a simple and interpretable representation-blending technique in comparison to fusion-based approaches which depend on decision-level aggregation or multi-branch networks (Scarpiniti, 2024; Oladunni and Aneni, 2025). The proposed technique combines complementary ECG representations at a feature-level mechanism in a manner that ensures interpretability as well as low complexity, an essential requirement in the real-time clinical environment (Aarotale and Rattani, 2023; Oladunni and Aneni, 2025).

It is worth mentioning here that representation blending becomes the third main technical innovation of this thesis after CHAF and ECG-ResViT. This work also provides a conceptual shift from improving ECG feature learning via network architectures to integrating complementary representations before classification.

The rest of this chapter is organized as follows. The background and literature review concerning multi-modal fusion methods and representations of the ECG signal is presented in Section 5.2. The mathematical description of the alpha-controlled fusion process and the

proposed blending approach is provided in Section 5.3. The experiment configuration, including the database description and implementation details, is presented in Section 5.4. The performance assessment and comparative results are addressed in Section 5.5, followed by the impact of the blending process in different configurations. The chapter is concluded with Section 5.6.

5.2 Foundations of Multi-modal ECG Representation Learning

ECG signals have been extensively employed for cardiac signal analysis because of their non-invasive nature and high clinical relevance (Ansari *et al.*, 2023). Nevertheless, automatic interpretation of ECG signals is still a hard task since ECG signals are highly non-stationary and may be corrupted by noise and other types of artifacts like baseline wander and motion artifacts. Therefore, advanced learning systems are needed in order to properly process such signals. The thing is that a number of important anomalies can be revealed only through morphology changes and long-term rhythms over time.

5.2.1 Signal-Based (1D) ECG Modeling

In particular, most of the work done in the DL area of ECG analysis has concerned signal-based modeling, which involves processing the raw ECG signal in the form of a 1D time-series. Due to the fact that the 1D CNNs (Khan *et al.*, 2023; Khondowe *et al.*, 2023) are capable of capturing discriminative features of morphological patterns such as P-wave dynamics, QRS shape, T-wave dynamics, etc., their application in this area is popular. The potential of recurrent architectures such as GRU- and LSTM-based models (Ansari *et al.*, 2023; Gao *et al.*, 2019), besides 1D CNNs, which remain a basis for ECG features learning, was explored in order to learn sequential dependencies and rhythms dynamics when longer segments of ECG were analyzed. Transformer architectures (Ikram *et al.*, 2025; Rahman *et al.*, 2024) have become particularly interesting due to the ability of self-attention mechanism to explicitly model the long-range temporal dependencies between different cardiac cycles that are difficult to achieve using only convolutional or recurrent networks.

While the described approaches proved to be efficient in waveform learning, they do not allow for obtaining sufficient information about the frequency-domain dynamics and structure of the signal, which is crucial for detecting the subtle abnormalities and transient events.

5.2.2 Image-Based (2D) ECG Representations

To remedy the issues with 1D signal modelling and to enhance feature extraction process, most modern ECG analysis methods have adopted ECG waveform representation as 2D images, which are more suitable for convolutional feature learning. The transformation of signals to images allows the use of the superior spatial feature extraction abilities of 2D CNNs which are known to work well in the image classification field.

One of the types of ECG image representation includes the time–frequency domain image. CWT (Qiu, Li, Qi and Li, 2024*b*; Wang *et al.*, 2021) is often utilized in producing a scalogram that helps in identifying the evolution of spectral energy over time. Likewise, STFT and Stockwell transform-based mapping methods (Kim *et al.*, 2025; Jeong and Lim, 2021) generate spectrogram-type representations that highlight the localised frequencies associated with rhythm disorders and abnormal beats.

Apart from time-frequency representations, there are other techniques that follow structural and recurrence-based encoding schemes which map ECG signals into texture-like images. Recurrence plots (RP) (Zhang *et al.*, 2021; Mathunjwa *et al.*, 2022) show the periodicity and recurrent state transitions of cardiac activities and were demonstrated to be useful for effective arrhythmia classification when used with residual CNNs' feature learning. Generalized average frequency (GAF) and multidimensional transfer function (MTF) (Zhang *et al.*, 2021) are other types of structural encoding that represent temporal correlations and transition dynamics with image textures and are robust when used with deep features.

Thus, the 2D representations of ECG signals allow for learning spectral, structural, and temporal features in an image learning pipeline.

5.2.3 Hybrid Deep Learning Architectures

While image representations enhance feature visibility, effective ECG learning often requires combining multiple learning mechanisms and modeling paradigms. As a result, hybrid architectures (Xiong *et al.*, 2025; Krishna *et al.*, 2024) have become increasingly popular, where convolutional networks are used to learn local morphological patterns, while complementary components—such as recurrent models, attention mechanisms, transformers, or even classical ML classifiers—are employed to capture higher-level dependencies and decision boundaries. Recent research highlights that performance improvements do not always arise from increas-

ing network depth alone, but from integrating complementary learning components such as residual, attention mechanisms, multi-branch feature extraction, and DL–ML hybridization (Xiong *et al.*, 2025).

Such architectural enhancements were investigated in this thesis using CHAF, which improves CNN representation learning by integrating heterogeneous AFs across layers, and ECG-ResViT, which combines transformer-based self-attention with residual and dilated convolutional learning to jointly model long-range temporal dependencies and local morphology. These advancements are consistent with the larger trend in ECG analysis toward hybrid DL architectures, which take into account deployment flexibility, representational richness, and architectural efficiency.

5.2.4 Multi-modal Fusion and Representation Integration

As ECG learning matured, research increasingly shifted from selecting a single optimal representation toward combining multiple representations to exploit their complementary strengths. Multi-modal ECG architectures typically adopt one of three fusion strategies:

1. Early fusion, which involves fusing multiple representations at the input level,
2. Feature-level fusion, where learned embeddings are fused prior to classification, and
3. Decision-level fusion, where outputs from multiple models are aggregated.

The above approaches can be implemented in real systems by employing various forms of fusion (Khoshnevis and Sankar, 2022; Liu, Xiao and Hsieh, 2025), such as channel-wise fusion (Khoshnevis and Sankar, 2022; Qiu, Guo, Wang *et al.*, 2024), where representations are fused at the channel dimension of feature maps to enable cross-representations interaction. Attention-based fusion deceptively assigns weights to representation streams according to their relevance to the task, whereas hybrid residual fusion keeps the original feature streams intact and gradually fuses them with residual connections. Among all these fusion schemes, feature-level fusion is widely regarded as the most appropriate choice due to its trade-off between rich representation and high computational cost.

Even with such advancements, however, there are numerous ECG systems based on fusions (Liu *et al.*, 2026; Xiong *et al.*, 2025) that still depend on attention-heavy fusion mechanisms or deep multi-branch architectures that lead to increased complexity and reduced in-

terpretability of the model. Such an approach demonstrates a discrepancy between the high performance of the research models and practical diagnostic tools and poses challenges for the real-time application in resource-limited environments.

From a systems point of view, such a discrepancy requires a lightweight, transparent and controllable fusion mechanism capable of effectively fusing complementary ECG features without introducing any significant architectural complexity. In this regard, the following section describes the proposed blending-based DL framework, in which time-frequency scalogram features and binary structural representations are blended using the alpha parameter before classification.

5.3 Proposed Blending-Based ECG Classification Architecture

In this section, we introduce the proposed model called BlendNet, a representation-based neural network for ECG signal classification. The primary goal of introducing such a framework is to successfully leverage complementary information about ECG signals by means of their multiple representations in a simple, interpretable, and scalable model. As opposed to the existing approaches of feature concatenation, complicated multi-branch networks, and attention-driven fusion, BlendNet suggests using a representation-based approach to feature fusion that allows controlling the influence of each representation on the learning process.

Two different views of the same ECG signal are used in the proposed framework: the time-frequency scalogram view and the binary structural view of the signal. While the former represents the variation in the signal spectrum, the latter, extracted from the scalogram, focuses on the waveforms' morphology and structure. Thus, the combination of these two views provides a more detailed description of ECG dynamics than any of the two representations considered separately.

For learning discriminative features from the complementary representations, a two-stream CNN model is used, in which each stream learns the unique domain-specific feature embeddings. The learned representation is then blended using an alpha controlled blending approach to enable explicit tuning of the weightage of the contributions of spectral and structural features during feature fusion. Such a representation-based approach facilitates interactions between complementary features of the ECG signals without making use of the complex archi-

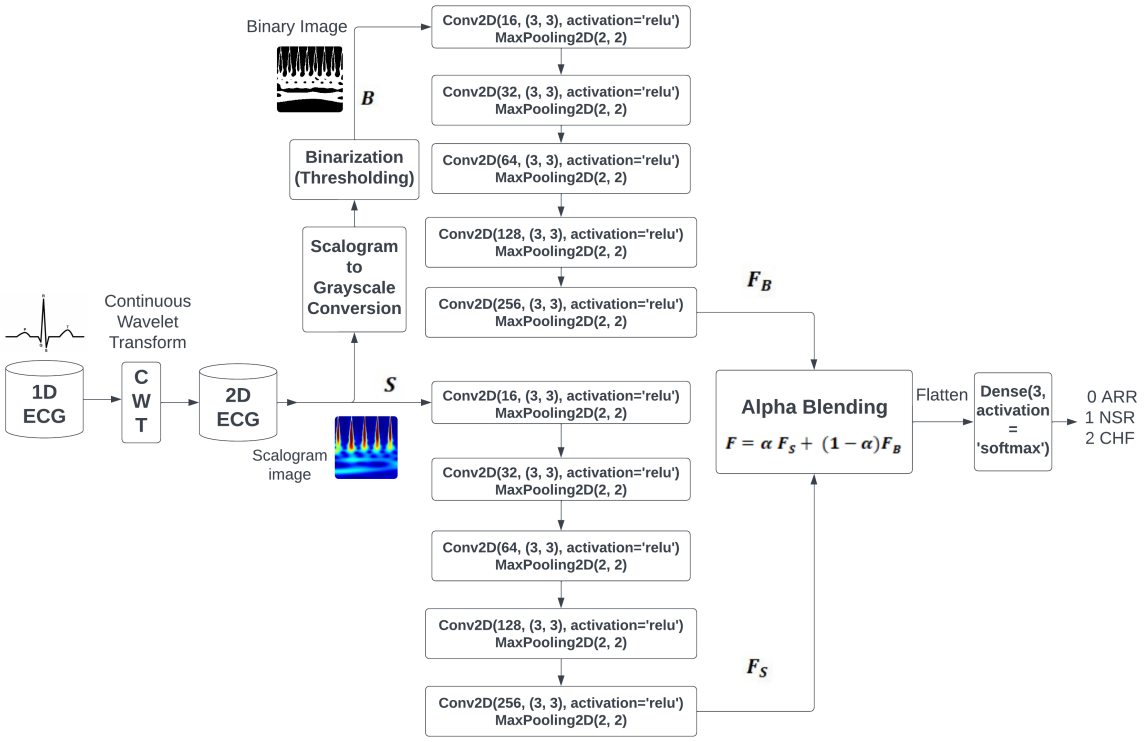


Fig. 5.1 Architecture of the Proposed BlendNet Architecture

tecture required by traditional fusion approaches.

The overall pipeline of the proposed BlendNet architecture is shown in Fig 5.1. It shows the entire process starting from representation generation to feature extraction, alpha controlled blending of features, and finally classification using complementary information of ECG signals.

5.3.1 Overall Workflow

The introduced BlendNet architecture starts with the segmentation of the input ECG signal into a number of fixed-length windows that provide a common representation for further processing. Each segmented window is represented as a time-frequency scalogram that provides a description of the spectral properties and frequency-dependent changes of heart activity. In order to enhance this representation, a binary structural map of the scalogram is created based on the thresholding procedure, where boundaries, waveform transitions, and other structural features are marked out of the ECG signal.

Both representations are fed to parallel CNN streams that allow for independent feature embedding learning for both types of the representation.

The learnt feature embeddings are later combined using the proposed alpha blending al-

gorithm, where the contributions of each of the representations are dynamically controlled while merging the features at the feature level. The explicit incorporation of both spectral and structural features during blending allows us to obtain a richer feature embedding compared to either of the modalities.

The blended feature embedding is ultimately fed into the classification stage, where the diagnostic decisions are made by means of a lightweight classification architecture. The presented approach enables efficient learning of both spectral and structural features of ECG.

5.3.2 ECG Signal-to-Image Transformation

In order to achieve efficient feature learning, the 1D ECG signal is mapped into two different image-based features that describe different facets of cardiac behavior (Scarpiniti, 2024; Qiu, Li, Qi and Li, 2024*a*). This process involves generating a time-frequency scalogram to capture the spectral information through time, and generating a binary structural map to highlight structural edges and transitions. These two features provide complementary spectral and structural information to be used in dual-stream feature learning and alpha-controlled fusion.

(a) Time–Frequency Scalogram Representation

ECG signals exhibit variations in time at different frequencies, thereby being inherently non-stationary. The CWT method is applied to model these variations as:

$$\mathrm{CWT}(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t), \psi^* \left(\frac{t-b}{a} \right) dt \tag{5.1}$$

Here, $x(t)$ represents the ECG signal, $\psi(\cdot)$ is the mother wavelet, while a and b stand for scale and translation variables, respectively. The coefficients produced by applying the CWT method are displayed in the form of a scalogram, which provides a time-frequency plot of the signal. Because of its multi-resolution analysis properties, the scalogram succeeds in modeling rapid features, like QRS complex, as well as slower features, like P and T waves.

(b) Binary Structural Representation

Whereas the scalogram provides detailed spectral data, it may also include intensity changes that do not contribute to classification in all cases. The scalogram is thus binarized to emphasize structure by applying thresholding:

$$B(i,j) = \begin{cases} 1, & \text{if } S(i,j) \geq \tau \\ 0, & \text{otherwise} \end{cases} \tag{5.2}$$

where $S(i,j)$ denotes the intensity of the scalogram at the point (i,j) and τ is the threshold value. Minor intensity changes are removed whereas critical points of waveforms' shapes and structural changes are emphasized. As a result, the network can focus on those morphological aspects which would otherwise remain unnoticed due to the original scalogram.

With the help of both scalogram and structural representation in binary form, the suggested approach includes spectral dynamics and morphological aspects in the final representation of the ECG signal.

5.3.3 Dual-Stream CNN Feature Extraction

The proposed BlendNet architecture employs two parallel CNN streams to learn complementary information from the generated ECG representations. The first stream processes the scalogram representation to capture spectral and time–frequency characteristics, while the second stream operates on the binary structural representation to learn morphological and edge-level patterns.

Let $\mathrm{CNN} * s(\cdot)$ and $\mathrm{CNN} * b(\cdot)$ denote the feature extractors for the scalogram and binary representations, respectively. The resulting feature embeddings are given by

$$F_s = \mathrm{CNN}_s(S) \tag{5.3}$$

$$F_b = \mathrm{CNN}_b(B) \tag{5.4}$$

where F_s and F_b are the spectral and structural feature embeddings obtained from learning.

The network can learn complimentary information from both the spectral and structural domains without compromising the distinct characteristics of each modality by independently learning their respective representations. The binary stream focuses on highlighting waveform discontinuity and shape properties, whereas the scalogram stream focuses on changes frequency-wise and rhythm-based information. By using this approach of two separate streams for learning, the model is able to build a more discriminative feature space.

Prior researches have demonstrated that fusion of complementary ECG representations is

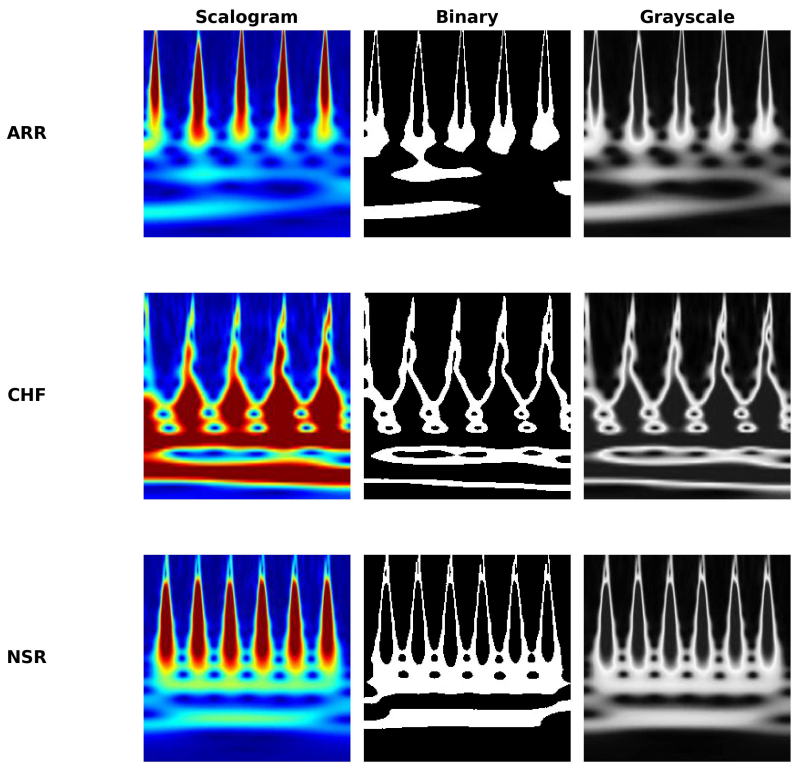


Fig. 5.2 Comparison of ECG Representations for ARR, CHF, and NSR Classes

capable of enhancing feature extraction and classification due to the utilization of information which is not sufficiently extracted in one domain (Yoon and Kang, 2023; Thapa *et al.*, 2025; Scarpiniti, 2024). In line with previous observations, Fig 5.2 demonstrates scalograms and binary structures of the ARR, CHF, and NSR categories. Despite the ability of scalograms to show different energy spectra, binary structures focus on structural borders and morphologies. These two complementary properties constitute the background for the design of the proposed approach.

5.3.4 Alpha-Controlled Feature Blending

The critical contribution of the proposed architecture BlendNet lies in the feature blending approach which incorporates a control parameter α that makes possible feature-aware fusion of spectral and structural ECG representations by means of a simple and understandable weight combination strategy. Let F_s and F_b be the feature embeddings obtained from the scalogram and binary streams, correspondingly. Then the representation fusion is performed according to the following formula:

$$F = \alpha F_s + (1 - \alpha) F_b \tag{5.5}$$

where $\alpha \in [0, 1]$ is a parameter regulating the ratio between spectral and structural features. Increasing values of α put more focus on the time-frequency components, while smaller ones highlight the morphological features. Intermediate values allow for the incorporation of both types of features into the model, providing the opportunity for learning more discriminative ECG representations.

The blending mechanism proposed above allows performing the feature-level fusion by means of a simple element-wise operation, in contrast to conventional approaches where such fusion is carried out either via concatenation, attention mechanisms, or complex multi-branch approaches. Thus, the proposed representation-aware fusion technique enables meaningful interactions between complementary ECG representations while preserving interpretability. Therefore, BlendNet offers an efficient framework for automatic ECG classification based on multimodal feature learning with minimal architectural complexity.

Fixed α is simple, parameter-free, and reproducible, but it applies the same weight to every sample and requires repeated training to select the optimal value. The best α also depended on the classifier (0.7 for deep backbones and lower values for classical classifiers, as shown in Table 5.4). Adaptive attention-based or gated fusion could learn sample-dependent weights, but would introduce additional parameters and may overfit on a dataset of only 162 recordings. A learnable scalar or a small gating mechanism would be a reasonable first step toward adaptive fusion; however, this approach has not been validated in the present study.

5.3.5 Classification Head

This blended feature vector F is then fed into a lightweight classification head to predict the probability distribution of the corresponding ECG class as follows:

$$\hat{y} = \mathrm{Softmax}(WF + b) \tag{5.6}$$

In this equation, the matrix W and vector b denote the learnable weight and bias values, respectively, and $\hat{y}$ is the resulting class probability distribution of the input ECG.

The reason for choosing a very simple classifier is that the proposed alpha-based blending approach provides an efficient and discriminative feature representation by combining both spectral and structural features. In other words, a straightforward classifier is enough to achieve good results since the obtained feature representation is classifier independent and can

be easily used with traditional classifiers like SVM and DTs.

5.3.6 Computational Considerations

One key property of the developed blending scheme is its capability to incorporate multiple ECG representations complementing each other without additional computational overhead. The alpha-controlled blending scheme represents a weighted element-wise addition of feature vectors with the complexity of

$$\mathcal{O}(n) \tag{5.7}$$

where n is the size of the feature vector.

The fact that the blending operation is applied on top of learned features means that the additional computational overhead introduced by the blending process is negligible compared to the feature extraction procedure. Thus, the major part of computational overhead in terms of time complexity in this framework is related to the representation generation process and CNN-based feature learning.

As it was stated before, the main idea behind the developed blending scheme is the possibility to generate richer feature representations using a simple blending procedure without any additional attention mechanisms or complicated fusion modules.

5.4 Experimental Setup

The following paragraphs will provide information regarding the experimental setup used for assessing the performance of the proposed BlendNet framework in classifying ECG signals. Information about ECG signal databases that were used, the process of pre-processing and representation generation, the setup of the model implementation and training, as well as the metrics used for performance evaluation, will be provided.

The design of the experiment is aimed at systematic study of the proposed representation-aware approach to fusing multiple ECG representations. Specifically, the effects of using complementary ECG representations, which include scalogram-based spectral features and binary structural features on the classification performance will be evaluated. Moreover, the effects of using the alpha-controlled Blending scheme on the classification performance will be studied.

5.4.1 Dataset Description

The following experimental study uses ECG signals from three different clinical classes, namely arrhythmia (ARR), congestive heart failure (CHF), and normal sinus rhythm (NSR). For the purpose of collecting data, three public datasets provided by PhysioNet, namely the MIT-BIH Arrhythmia Database, MIT-BIH Normal Sinus Rhythm Database, and BIDMC Congestive Heart Failure Database (Moody and Mark, 2001; Goldberger *et al.*, 2000; Baim *et al.*, 1986) were used. The total number of recordings is equal to 162, including 96 ARR, 30 CHF, and 36 NSR recordings.

All ECG recordings are resampled at 128 Hz, and the first 1,000 samples of each recording are used. Resampling and cutting the recordings to the same size provide standardized input data with preserved necessary morphological properties.

Even though the dataset size is rather limited and there is an imbalance among classes, the dataset can serve as a reasonable benchmark for the evaluation of the representation-learning framework that is proposed. Instead of evaluating large-scale performance, the experiments will be aimed at evaluating the effectiveness of the proposed alpha-controlled feature blending.

5.4.2 Data Preprocessing and Segmentation

The ECG segments were normalized using standard score normalization to ensure uniformity between subjects and classes. In order to minimize bias brought on by amplitude changes between recordings and guarantee consistent convergence during model training, each extracted ECG segment was specifically normalized to zero mean and unit variance.

We incorporated imbalance handling during training because the database contained unequal sample distributions across the ARR, CHF, and NSR classes. To mitigate class dominance and promote balanced learning, the `class_weight` strategy was employed during model training instead of using data augmentation techniques.

5.4.3 Signal-to-Image Representation Generation

A scalogram and its associated binary structure map are two complementary image representations that must be created from the 1D ECG signal in order to use the suggested architecture.

(a) Scalogram Image Formation

A time-frequency representation has been created through CWT. The CWT has been created through MATLAB function `cwtfilterbank` to create an image of the 2D scalogram showing the evolution of spectrum of ECG signal with respect to time. These images have been scaled down and normalized to the resolution of 227×227 pixels so that these images could be useful for the process of feature extraction through convolution.

(b) Binary Structural Representation

To better emphasize the characteristics of the ECG signal, the scalogram was further converted to a binary image through a threshold binarization method. To differentiate between the important signal portions and the background, the threshold level of 128 was adopted since it is halfway of the grayscale level (from 0 to 255). This process allows the binary image to convey additional structural information that may not be visible in the scalogram by taking advantage of the simplified visualization.

5.4.4 Training Configuration and Implementation details

The design of the BlendNet was assessed through multiple experimental setups to provide a thorough evaluation of its robustness and generalization capabilities. To assess performance with respect to various data availability scenarios, the experiments were performed by using four combinations of train-test splitting ratios: 60:40, 70:30, 80:20, and 90:10.

In order to evaluate the effectiveness of the proposed representation-aware fusing approach, the blending coefficient α was systematically varied from 0.1 to 0.9 with the step 0.1. Additionally, this experiment makes it possible to estimate the importance of the contribution of the binary structural features and scalogram spectral features to the overall model.

Training of the model was performed through use of the Adam optimiser and the objective function used was the sparse categorical cross entropy. The process of training the neural network continued for 20 epochs through use of a batch size of 64. Dropout was also used during training as a means of improving generalization and reducing overfitting.

5.4.5 Evaluation Metrics

The suggested blending-based ECG classification approach was examined based on quantitative criteria which measure both prediction performance and the feasibility of the algorithm

implementation in practice.

Accuracy

The accuracy score defines the number of classified properly ECG samples within the ARR, CHF, and NSR groups. Thus, it is a measure of the whole prediction ability of the model. The accuracy is calculated by the following formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.8}$$

Herein, TP, TN, FP, and FN are the true positive, true negative, false positive, and false negative cases, respectively. Though the accuracy criterion helps to estimate the performance of the classification process, it is used along with other measures of performance per each class.

Execution Time

The execution time is the computational time needed to extract features, perform the fusion step, and classify ECG signal segments. It shows the real possibility of using the suggested framework in practice, especially in case of continual heart activity monitoring.

Because the blending approach is based on an alpha-controlled operation that does feature-level integration via a straightforward element-wise operation, it requires little computation overhead while allowing efficient interaction between spectral and structural representations. Time required for execution becomes an important measure for determining the framework's suitability for practical ECG analysis applications.

5.5 Results and Discussion

This section presents a comprehensive evaluation of the proposed blending-based DL architecture for ECG signal classification. The experimental analysis aims to:

1. Evaluate the classification effectiveness of the proposed architecture,
2. Analyze the influence of the alpha-controlled blending mechanism, and
3. Compare the proposed method with conventional single-representation and non-blended models.

Performance is assessed using classification accuracy and execution time under differ-

ent experimental configurations, including alternative backbone architectures, classifier selections, and blending proportions.

5.5.1 Baseline Comparison with Single-Representation Models

The proposed blending-based architecture is initially contrasted with conventional CNN-based models trained with a single representation in order to provide a point of comparison, specifically: only scalogram pictures, and only binary structure pictures.

- Scalogram-only representations, and
- Binary structural representations only.

Table 5.1 Scalogram vs Binary CNN Baseline Performance

Input Representation	**Modality**	**Accuracy (%)**	**Execution Time (s)**
Scalogram	2D CNN	96.29	272.17
Binary	2D CNN	91.11	263.24

The classification performance of different baseline methods is shown in Table 5.1. The findings show that neither representation by itself performs at its best, even while scalogram-based CNNs perform better than binary-only representations because of their greater spectral richness. Instead than replacing representations, this insight encourages feature-level integration.

5.5.2 Effect of Alpha-Controlled Feature Blending

By changing α between 0.1 and 0.9, the impact of the blending proportion α on classification performance is examined. For the majority of α values, blending-based learning consistently performs better than single-representation baselines, as seen in Fig 5.3. Peak performance is seen at α = 0.7, suggesting that more discriminative ECG representations result from the balanced integration of binary structure characteristics and scalogram-derived spectral features. Extreme α values, on the other hand, cause the model to be biased toward a single representation, which lowers classification performance. These findings demonstrate that in order to properly utilize the complementary nature of spectral and structural ECG data, regulated feature blending is essential.

As seen from the results of the experiment, the value of the blending coefficient α greatly influences the learned quality of the ECG representations.

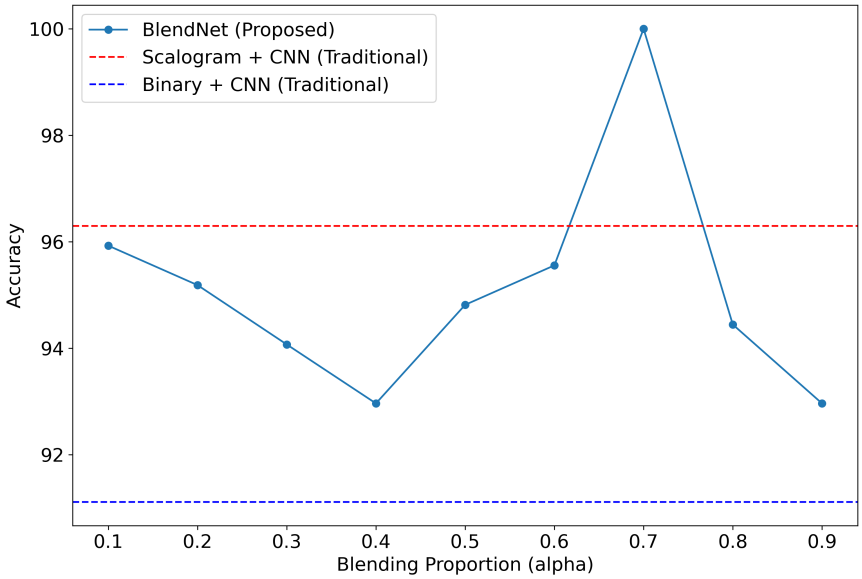


Fig. 5.3 BlendNet Accuracy Across Alpha-Blending Configurations

- **Small blending coefficients ($\alpha = 0$):** In case $\alpha = 0$, the learned representation of ECG is completely binary, i.e., the only used representations are structural ones. In spite of the fact that such representations allow capturing the morphological transitions well, there is no spectral data involved which makes the representation of the dynamic processes less discriminative.
- **Large blending coefficients ($\alpha = 1$):** If $\alpha = 1$, then the learning process uses only scalogram-based features having rich time-frequency structure. However, since there is no contribution to the process from the structural characteristics (binary representation), there is no possibility to make use of complementary information about the ECGs.
- **Intermediate blending coefficients:** It is clear that the highest quality of the representation can be achieved when the value of the blending coefficient is intermediate, and the best result is obtained when $\alpha = 0.7$. Thus, spectral and structural representations contribute with complementary information, and, hence, provide better feature representations for the classification task.

5.5.3 Backbone Network Analysis

A regular CNN, InceptionV3, ResNet152V2, DenseNet169, and MobileNetV2 were among the CNN backbone architectures used to adopt the architecture in order to assess the robustness

Table 5.2 Influence of CNN Backbones on BlendNet Performance Under Different α Values

α	**InceptionV3**		**ResNet152V2**		**DenseNet169**		**MobileNetV2**	
	Acc (%)	**Time (s)**	**Acc (%)**	**Time (s)**	**Acc (%)**	**Time (s)**	**Acc (%)**	**Time (s)**
0.1	90.37	332.90	89.25	1676.70	94.07	1287.10	93.70	287.48
0.2	83.33	317.10	90.74	1704.80	91.85	1068.00	90.74	286.21
0.3	85.92	637.90	91.11	2225.60	89.62	1159.50	91.85	327.62
0.4	85.18	326.60	88.51	1075.90	91.48	1298.80	91.48	341.19
0.5	87.03	361.00	91.85	1705.80	89.99	1109.40	86.29	294.97
0.6	89.99	315.40	93.70	5260.90	90.74	1149.10	89.25	293.13
0.7	**97.47**	312.60	**97.77**	1908.20	**99.62**	1112.30	**99.25**	304.03
0.8	88.88	560.90	92.22	1785.10	88.51	1145.40	91.48	332.05
0.9	88.51	374.25	91.85	2992.90	92.22	1174.30	92.22	345.90

Note: Acc – Classification Accuracy, Time – Execution Time.

and architecture-independence of the suggested blending method. Fig 5.4–5.7 show how each backbone network's categorization accuracy varies with the blending proportion α.

The proposed blending-based models consistently outperform their corresponding single-representation baselines across all evaluated backbone architectures, indicating that the observed performance improvement primarily originates from the representation integration strategy rather than from backbone-specific architectural design choices. Furthermore, all architectures exhibit a consistent performance trend, where intermediate blending coefficients—particularly $\alpha = 0.7$—produce the highest classification accuracy.

The accuracy and execution time achieved with various backbone networks under various blending parameters are compiled in Table 5.2. Even if deeper designs like DenseNet169 and ResNet152V2 get excellent absolute accuracy, feature blending also greatly benefits lightweight models like MobileNetV2, which reach competitive performance at a far reduced computational cost. These outcomes demonstrate that the suggested blending technique works well across a broad spectrum of network complexity and is independent of architecture.

Importantly, feature blending also shows significant performance gains in lightweight systems like MobileNetV2. At optimal α setting, MobileNetV2 obtains accuracy levels equivalent to deeper networks yet having a far lower processing overhead. This discovery emphasizes that the suggested mixing method may be suitable for resource-constrained settings (latency was not benchmarked) because it does not require extremely complicated systems to function

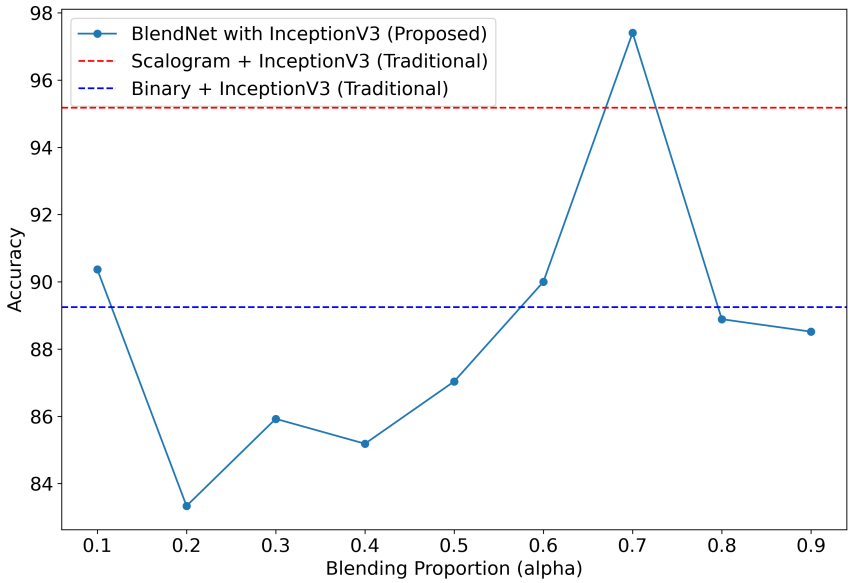


Fig. 5.4 Alpha-Blending Performance Analysis Using InceptionV3

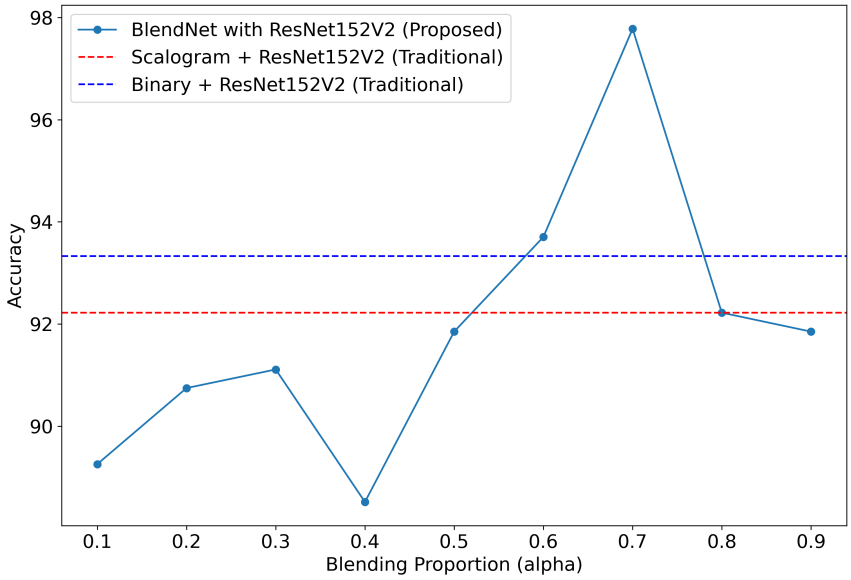


Fig. 5.5 Alpha-Blending Performance Analysis Using ResNet152V2

well.

5.5.4 Influence of Training Data Split Ratios

To further explore the robustness of the proposed methodology, various split combinations were used, such as 60:40, 70:30, 80:20 and 90:10 splits. As shown in Table 5.3, the suggested BlendNet architecture shows consistent accuracy for all split ratios with the best accuracy be-

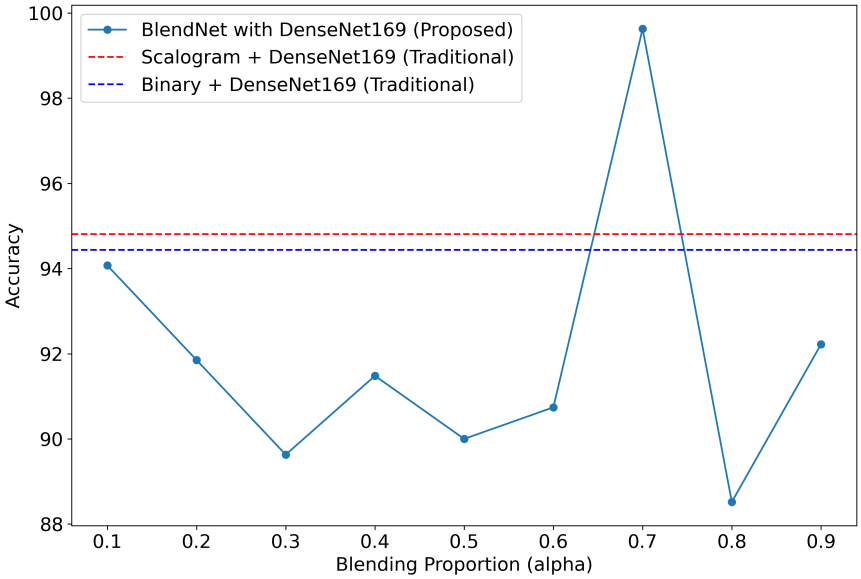


Fig. 5.6 Alpha-Blending Performance Analysis Using DenseNet169

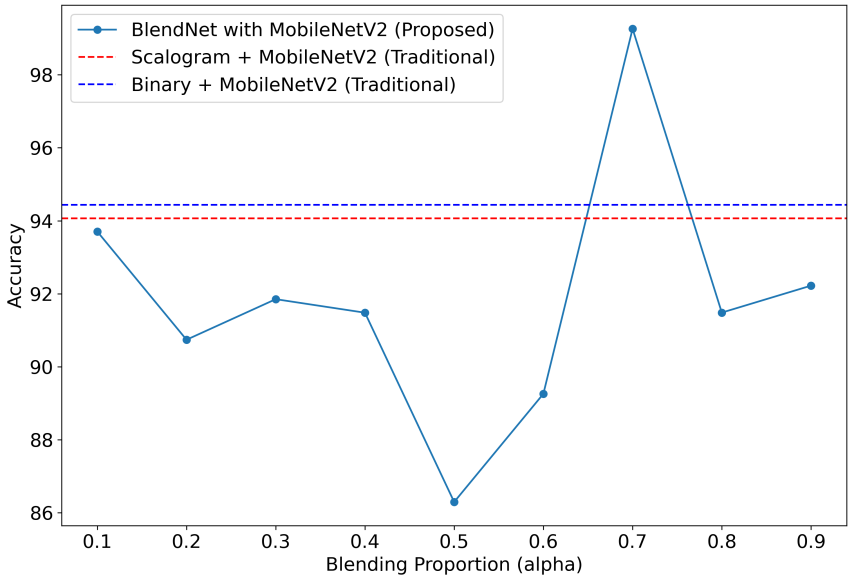


Fig. 5.7 Impact of Alpha-Blending on MobileNetV2-Based BlendNet Performance

ing achieved by 70:30 and 80:20 ratios. This uniformity suggests that the blending process shows consistent accuracy across split ratios of the blended feature space for various classifiers.

Table 5.3 Effect of α Under Different Train–Test Splits

α	60:40		70:30		80:20		90:10	
	Acc (%)	**Time (s)**	**Acc (%)**	**Time (s)**	**Acc (%)**	**Time (s)**	**Acc (%)**	**Time (s)**
0.1	96.11	173.51	95.18	203.12	92.22	212.73	96.66	240.01
0.2	94.07	119.99	95.18	130.97	94.44	144.56	93.33	159.47
0.3	92.59	114.36	94.07	131.35	93.88	146.29	95.55	156.48
0.4	95.92	192.70	97.40	195.70	95.00	214.65	94.44	251.64
0.5	93.33	178.49	94.81	190.41	91.66	211.68	96.66	234.81
0.6	94.07	183.30	95.55	206.17	93.33	214.13	96.66	239.09
0.7	**100**	177.43	**100**	197.02	**100**	219.98	**100**	231.67
0.8	95.55	170.02	90.70	192.78	89.44	218.84	96.66	277.94
0.9	93.33	174.75	92.22	214.62	92.77	225.36	93.33	254.37

Note: Acc – Classification Accuracy, Time – Execution Time.

5.5.5 Evaluation with Different Classifiers

Classical ML classifiers including SVM, RF, KNN, and XGBoost were used to assess the blended feature embeddings in addition to the deep classification head.

Each classifier's classification accuracy and execution time under various α values are shown in Table 5.4. Further proving the representation quality of the blended feature space, the results indicate that the blended features maintain their strong discriminative power even when combined with conventional classifiers.

Among the evaluated classifiers, SVM and XGBoost achieve competitive accuracy with reasonable computational cost, making them suitable alternatives for deployment in constrained environments.

5.5.6 Comparison with Existing ECG Classification Methods

The classification performance of the suggested blending-based architecture is compared with representative ECG classification techniques documented in recent literature in order to further evaluate its efficacy. A fair comparison of various representation strategies is made possible by the inclusion of both signal-based and image-based approaches in the chosen research.

When compared to representative signal-based and image-based ECG classification techniques, the suggested blending-based architecture produces better classification performance on the evaluated dataset; protocols differ across studies, as Table 5.5 illustrates. In contrast

Table 5.4 Effect of α on Classical Classifier Performance

α	SVM		RF		KNN		XGB	
	Acc (%)	**Time (s)**	**Acc (%)**	**Time (s)**	**Acc (%)**	**Time (s)**	**Acc (%)**	**Time (s)**
0.1	97.03	197.31	96.66	200.83	96.29	187.16	96.29	192.24
0.2	98.88	133.22	**100**	130.22	98.88	135.51	99.44	134.71
0.3	97.22	134.76	97.22	161.29	97.22	134.64	97.22	136.10
0.4	95.18	193.09	95.18	201.13	95.18	220.44	94.81	198.64
0.5	96.66	193.34	96.66	202.10	96.66	228.21	97.03	209.98
0.6	94.81	193.03	94.81	194.66	94.81	214.45	94.44	204.43
0.7	92.22	199.23	92.59	201.26	91.85	198.79	92.22	200.42
0.8	93.33	194.79	93.33	204.68	93.33	227.76	93.33	204.98
0.9	93.33	194.28	92.96	200.41	93.33	225.55	93.33	205.87

Note: Acc – Classification Accuracy, Time – Execution Time.

Table 5.5 Comparative Performance Analysis of BlendNet and Prior ECG Models

Study / Method	**Model / Approach**	**Representation Type**	**Performance**
Saeed and Yousif (2021)	Slant Transform + SVM Classifier	Signal-based	Acc: 92.9%
Yoon and Kang (2023)	SIFT Feature Extraction + CNN Classifier (5-fold CV)	Image-based	Acc: 97.8%
Proposed	BlendNet with $F = \alpha F_s + (1-\alpha)F_b$	Fusion (Scalogram + Binary)	**Acc: 99.21% $\pm$ 1.05% (5 Fold – Cross Validation)**

Note: Acc – Classification Accuracy

to previous methods that only use one representation domain; the suggested method uses controlled feature blending to explicitly incorporate complementing spectral and structural information. The performance boost that results emphasizes how representation-level fusion works better than manually created feature extraction or deep learning models that only use one domain. The mean accuracy of $99.21 \pm 1.05\%$ obtained through five-fold cross-validation (CV) is considered the more reliable performance estimate. The 100% accuracy reported at $\alpha = 0.7$ in Table 5.3 corresponds to specific data splits in a relatively small dataset comprising 162 recordings and should therefore be interpreted with caution.

5.5.7 Feature Space Visualization and Interpretability

The blended feature representations for various α values were subjected to t-SNE visualization in order to obtain more understanding of the impact of feature blending. The distribution of samples in the reduced feature space is shown in Fig 5.8.

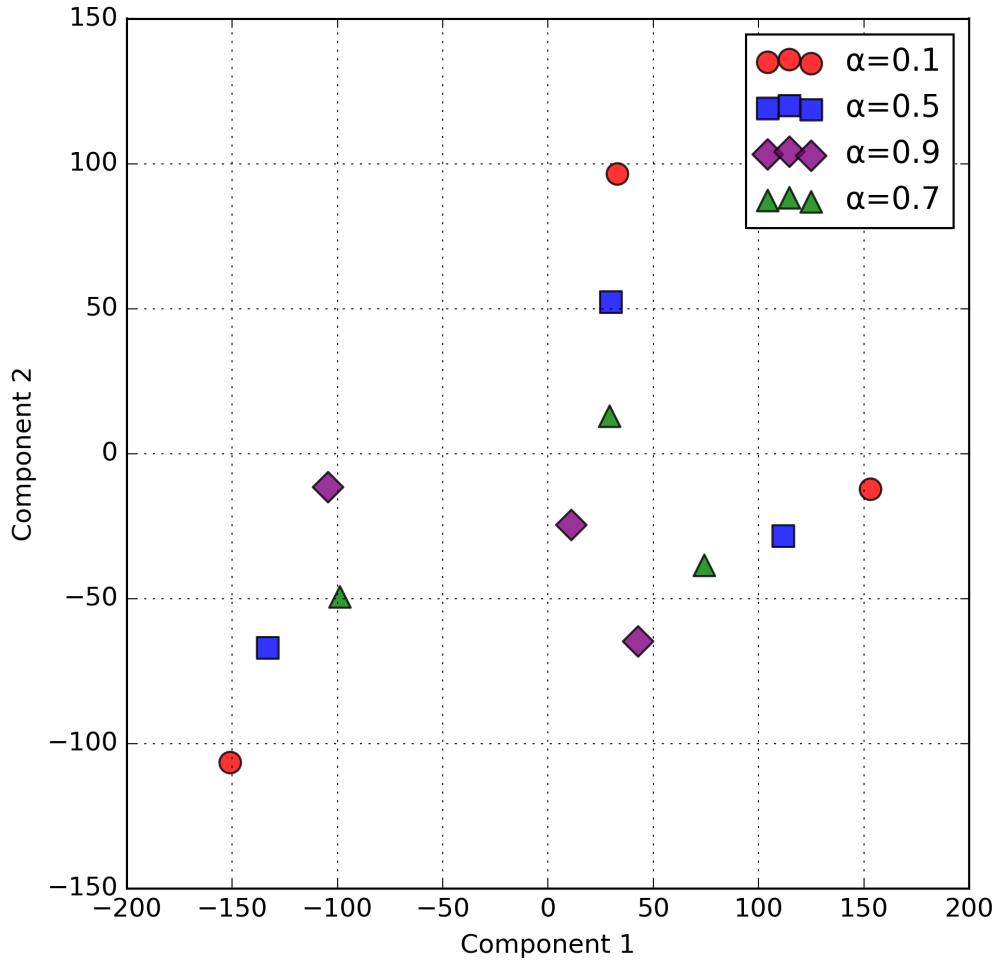


Fig. 5.8 t-SNE Visualization of Feature Separability Under Varying α

Samples from various ECG classes form distinct clusters at optimal α values, while inferior α settings lead to more overlap. Class separability in the learned feature space is enhanced by alpha-controlled blending, as demonstrated by this visual evidence, which also validates the quantitative findings.

The interpretability of the blending mechanism is further supported by mean activation maps for various α values (Fig 5.9), which show a steady change in feature emphasis from structural to spectral regions as α grows.

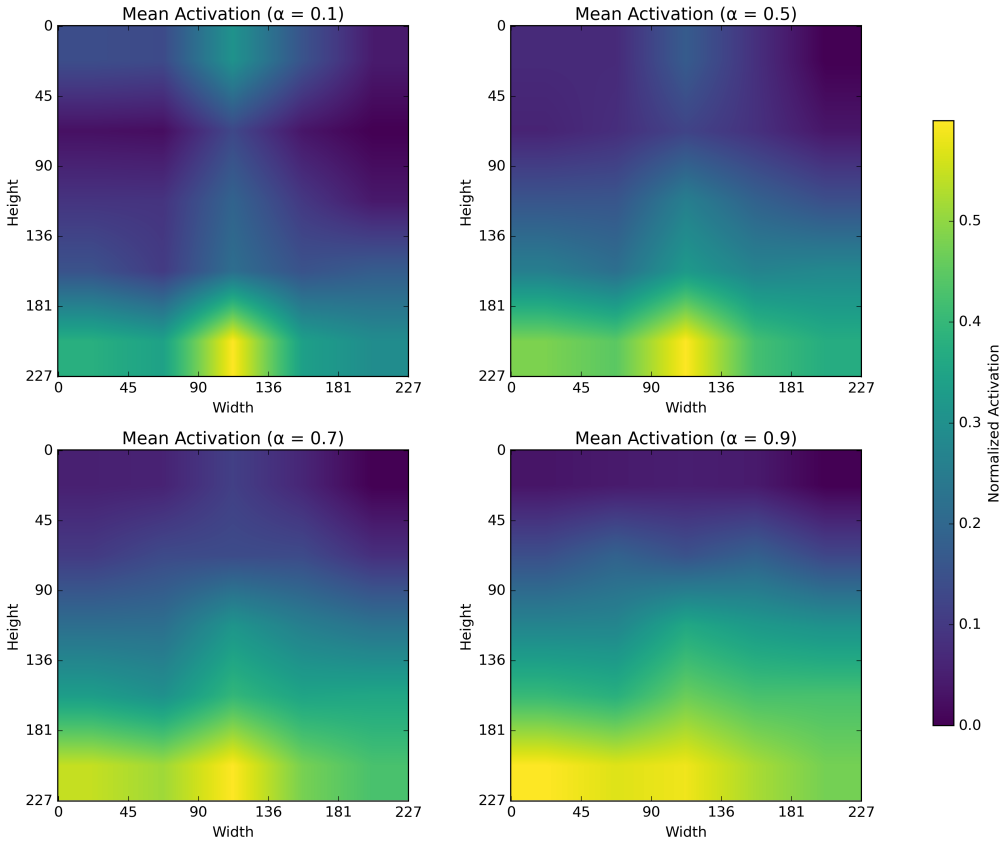


Fig. 5.9 Mean Activation Maps of Blended Features Under Different α Configurations

5.6 Chapter Summary

Motivated by the need to effectively integrate complementary ECG representations without relying on complex fusion architectures, this chapter presented BlendNet, a blending-based deep learning framework for automated ECG classification. Unlike conventional fusion approaches that often depend on feature concatenation, attention mechanisms, or multi-stage aggregation strategies, the proposed framework introduces a simple yet powerful representation-aware fusion paradigm based on the observation that different ECG representations contribute unequally to diagnostic decision-making.

The proposed architecture combines time–frequency scalogram representations, generated using the CWT, with binary structural representations that emphasize waveform morphology and transition patterns. To preserve the unique characteristics of each representation, parallel CNN streams are employed to independently learn discriminative feature embeddings. These embeddings are subsequently integrated through an alpha-controlled blending mechanism, where the blending coefficient α explicitly regulates the relative contribution of spectral and structural information during feature fusion.

Experimental evaluation on ARR, CHF, and NSR ECG classes demonstrated that the proposed representation-aware fusion strategy consistently outperformed single-representation learning approaches. The results also showed that the blending coefficient has a significant impact on classification performance, underscoring the significance of regulated feature interaction. Specifically, a 5-fold cross-validation accuracy of 99.21% was attained by the ideal blending configuration ($\alpha = 0.7$), demonstrating that a suitable balance between spectral and structural representations results in more discriminative ECG feature learning.

Overall, this chapter extends the contributions of the thesis beyond feature extraction and temporal modeling toward intelligent representation fusion. By demonstrating that complementary ECG representations can be integrated through a transparent, adaptive, and interpretable blending strategy, BlendNet introduces a new perspective on multimodal ECG representation learning and establishes a strong foundation for future research on adaptive fusion frameworks, multi-lead ECG analysis, and explainable cardiac diagnostic systems.

Chapter 6

CONCLUSION AND DIRECTIONS FOR FUTURE RESEARCH

6.1 Research Overview

CVD are still the leading cause of death globally and ECG is the gold standard non-invasive imaging tool to access the heart. The number of ambulatory monitors, implantable loop recorders and wearable sensing platforms is increasing, and the volume of ECG data which is being created as a result is too large to interpret manually. This general clinical fact inspired the central research question of this thesis: how to create DL architectures to represent, model and classify ECG waveforms with sufficient representational richness, temporal modelling depth, and feature diversity without incurring excessive computational costs?

The three objectives of our thesis were related to each other: increasing AF diversity in convolutional architectures, including long-range temporal dependencies across cardiac cycles, and utilizing complementary ECG representations from different signal domains. The three developed architectures – CHAF, ECG-ResViT and BlendNet add together a coherent and progressive contribution to automated ECG analysis.

6.2 Summary of Research Contributions

This section revisits the three proposed architectures and considers the contribution of each in terms of performance numbers, but also, in terms of conceptual and methodological developments they represent.

6.2.1 CHAF - Convolution-Based Heterogeneous Activation Framework

The idea for the CHAF architecture was inspired by the fact that ECG signals contain a variety of components, whose morphologies are each affected differently by nonlinear transformations, such as P-wave, QRS complex, and T-wave. The conventional CNNs use a single AF for all layers which is not suitable for handling the segments with widely varying characteristics.

CHAF proposes a systematic empirical procedure to assign heterogeneous AFs (Tanh,

Linear, Softsign, ELU, CReLU, ReLU-6) to successive convolutional blocks according to their appropriateness for specific morphological properties. Then the extracted feature vectors are passed to a suite of traditional ML classifiers, without the computational burden of the fully connected dense layers. This hybrid design not only enhances representational diversity but also significantly cuts down the inference time to 0.008 seconds on the PTB Diagnostic ECG Database.

CHAF outperforms baseline CNNs and hybrid models with 99.55% and 99.08% accuracy on PTB and MIT-BIH, respectively. The overarching contribution of this approach is that AF diversity is used as a powerful architectural feature for learning ECG features, a feature which has not been previously explored to its full extent before this work, and that performance and computational efficiency can be achieved by design choices that are guided by the signal characteristics of the specific application domain.

6.2.2 ECG-ResViT: Hybrid CNN–Vision Transformer Architecture

Convolutional operations are limited by their receptive field, while CHAF enriches local morphological features; when rhythm level diseases appear in several cardiac cycles, the receptive field of the convolution is a critical limitation. Gradient instability and limited parallelisability are issues for recurrent architectures that address sequential dependencies.

ECG-ResViT fuses a multi-scale temporal convolutional component (residual-dilated convolutional module) with a MHSA module for global temporal modeling to capture the morphological features at multiple temporal scales. Dilated convolutions are used to expand the receptive field without increasing the number of parameters, while CNN-derived token embeddings are used with positional encodings to capture inter-beat dependencies in the ViT.

Finally, the ablation study on ECG-ResViT shows the value of each architectural component from the base CNN-ViT model that obtained 96.30% accuracy, an improvement to 97.78% when residual dilated convolutions and label smoothing are introduced to the model, up to the fully tuned model with 99.13% accuracy and 99.94% ROC-AUC achieved on the three-class PhysioNet benchmark. This systematic evaluation supports the view that the gains arise from architectural innovations and not to other factors. The 1D signal classification problem can be effectively reformulated as a combined local-global modeling problem, with the complementary strengths of convolutional and attention-based models used explicitly, as is

demonstrated in ECG-ResViT.

6.2.3 BlendNet: Alpha-Controlled Feature Blending Architecture

CHAF works directly with the raw 1D ECG signal, and ECG-ResViT works on the 1D ECG signal. BlendNet, on the other hand, has a multi-domain representation method, combining both the temporal and frequency domain, and structural domain representations, each of which brings complementary diagnostic information. In particular, the frequency-domain scalogram obtained by means of CWT is very informative about the evolution of spectral energy and the dynamics of rhythm, characteristics adequate for recognizing arrhythmic variations. At the same time, important morphological elements like the T-wave transitions and QRS complex onsets are preserved in the binary structural representation.

BlendNet uses a dual-stream 2D CNN architecture to maximize the use of these complementary representations; each arm processes a single representation, either the scalogram or the structure image. The extracted feature embeddings are then fused together using the following alpha-blending mechanism that is transparent and parameter-free:

$$F = \alpha F_s + (1 - \alpha) F_b \tag{6.1}$$

In which, F_s and F_b represent spectral and binary structural feature embeddings, respectively, and α represents the degree of the fusion. It is an intuitive, efficient and understandable feature combination strategy which is different from attention-based fusion modules or complex multi-branch aggregation modules.

Experimental tests show that BlendNet works best when $\alpha = 0.7$, which means that 70% of the weighting is done on the spectral scalogram features and 30% on the binary structural features. With this mode, the proposed model correctly classifies the PhysioNet 3-class benchmark database with 100% accuracy, and a mean accuracy of $99.21\% \pm 1.05\%$ using 5-fold Stratified Cross-Validation.

The optimal blending coefficient also gives quantitative information related to the relative importance of the spectral representation with respect to the structural representation for ECG classification. Furthermore, the best intra-class compactness and separation of inter-classes is observed in the t-SNE visualization at $\alpha = 0.7$ by the lowest value for the within-class sum of squares (WCSS $= 4117.38$).

Furthermore, BlendNet exhibits excellent architectural flexibility and modularity, compatible with various popular CNN backbones, such as InceptionV3, ResNet152V2, DenseNet169 and MobileNetV2. The results show that the proposed architecture, which uses alpha blending, is compatible with several CNN backbones.

The three architectures form a coherent progression across complementary design axes. CHAF introduces intra-layer representational diversity through heterogeneous activation functions. ECG-ResViT builds on this at the system level, combining transformer and convolutional blocks for joint local-global modeling. BlendNet extends this further into representational innovation, fusing information across distinct signal domains entirely.

Collectively, they present a trajectory that starts from micro-design (activation selection) to meso-design (local-global design) to macro-design (cross-domain fusion), as captured in Table 6.1. Interestingly, the 0.008 seconds inference time of CHAF, the potential for efficient inference ViT modules in ECG-ResViT, and the linear cost fusion in BlendNet indicate one important design aspect: clinical viability over any marginal accuracy improvement.

While the accuracy improvement may be marginal compared with the existing architectures, their true merit is speed and efficiency. Rapid and efficient inference ensures that they can serve as decision support systems to assist cardiologists in time-sensitive cases, such as triage and continuous monitoring.

6.3 Research Limitations and Future Directions

Though there are good results from the proposed frameworks, some limitations need to be mentioned.

- **Diversity of training data:** The proposed methods were tested on benchmark ECG databases which facilitate fair comparison, yet they cannot reflect the diversity of real-life patient ECGs including noise, movement, and various patients.
- **Manual choice of activation functions in CHAF:** The choice of heterogeneous AFs in CHAF was done empirically and based on domain knowledge, although it worked well, it is not necessarily the best option in other scenarios.
- **Fixed feature blending in BlendNet:** The proposed blending coefficient in BlendNet (α) is fixed and chosen using grid search technique, it does not adjust to the changes in

ECG shape, quality, and other factors.

- **Attention mechanism overhead in ECG-ResViT:** Although ECG-ResViT is fast enough for small segments of ECG signals, the use of self-attention can lead to computational problems in the case of prolonged ECG recordings.
- **Absence of clinical trials:** All the proposed models have been tested only on publicly available annotated datasets and their practical applications require prospective testing on patient data and validation by cardiologists.
- **Single-lead only:** CHAF uses Lead II, while ECG-ResViT and BlendNet use single-channel segments. Their performance on multi-lead ECG recordings has not been established.
- **Noise robustness not evaluated:** CHAF depends on R-peak detection, while BlendNet uses a fixed binarisation threshold and a fixed α. Future work could evaluate robustness using the MIT-BIH Noise Stress Test Database at graded signal-to-noise ratios (SNRs).
- **Cross-device transferability not tested:** Generalisation across ECG acquisition devices remains unverified, as no train-on-one-device and test-on-another evaluation was performed.
- **Patient-level partitioning and external validation:** Patient-level partitioning was not used in some chapters, and external validation using independent datasets or hospital recordings has not been conducted.
- **Interpretability of individual predictions:** Future work could explore Grad-CAM or integrated gradients, KNN nearest-beat explanations, and attention visualisation to improve the interpretability of individual predictions.
- **Regulatory and clinical deployment:** The proposed models remain at the research stage. Further work is required to assess sensitivity, specificity, and calibration, conduct prospective clinical validation, and address applicable medical-device regulatory requirements.
- **Computational cost of ECG-ResViT:** FLOPs, inference latency, and memory consumption were not measured. These metrics should be evaluated in future work to establish computational requirements and enable fair comparisons with lightweight architectures.

These novel architectures offer a solid base for future research in the field of intelligent ECG diagnosis. In light of the limitations mentioned above, here are some potential research directions.

- **Automated activation function optimization:** Future research may employ NAS or meta-learning techniques to replace manual activation function optimization in CHAF for adaptation of activations to various settings and data samples.
- **Multi-lead ECG processing:** Extension of proposed architecture to 12-lead ECG will make it possible to leverage spatial information from heart. This can be done by designing lead-aware activation functions and attention mechanism over leads.
- **Adaptive feature blending:** Future works may use attention mechanism for learning of the adaptive feature blending parameter in BlendNet instead of a static value.
- **Edge-efficient processing:** Future research can explore model compression techniques like pruning, quantization and knowledge distillation as well as efficient attention mechanism for reduction of the computational cost.
- **Explainable AI:** Utilizing explainability methods, such as saliency maps and attention visualization, could increase transparency by emphasizing clinically important ECG areas and increasing clinician trust.
- **Federated learning:** Tailoring the proposed architectures to federated learning would allow for privacy-preserving collaborative learning among various healthcare organizations and improved generalization of models.
- **Extension to other biosignals:** The proposed architectures could be applied to physiological signals like EEG and PPG as well, allowing for wider use in biomedical signal analysis.

This thesis provided three solutions to the problem of ECG classification using deep learning architectures. CHAF proved that heterogeneous activation functions improve feature extraction at a low computational cost. ECG-ResViT showed the ability to efficiently combine CNNs and Vision Transformers in order to capture both local morphology and global temporal dependency in the signal. BlendNet illustrated the benefit of combining complementary signal representations in improving classification performance.

In conclusion, the presented research has contributed to the development of precise and computationally efficient, with explainability identified as future work ECG analysis systems. Although further validation is needed, the developed architectures provide a scalable approach for next-generation intelligent cardiac diagnostics systems and biomedical signal analysis.

Table 6.1 Comparison of the Three Research Contributions

S.No	Factor	Contribution 1 (CHAF)	Contribution 2 (ECG-ResViT)	Contribution 3 (BlendNet)
1	Signal Representation	1D ECG Raw Signal	1D ECG Raw Signal	2D ECG – Scalogram and Binary Images
2	Proposed Architecture	Convolution-based Heterogeneous Activation Facility with Machine Learning Classifier (CHAF-ML)	Hybrid CNN with Residual and Dilated Convolutions integrated with Vision Transformer (ECG-ResViT)	Alpha Blending of Scalogram Feature Maps and Binary Image Feature Maps
3	Novelty	Heterogeneous activation functions across multiple CNN blocks for richer feature extraction compared to homogeneous activation functions	Hybrid ECG architecture where CNN extracts local features and ViT captures global dependencies from learned representations	Blending complementary features from scalogram and binary representations to generate a richer composite feature set
4	Datasets Used	PTB Diagnostic ECG Database (290 subjects), MIT-BIH Arrhythmia Database (96 subjects)	MIT-BIH Arrhythmia Database (96 subjects), BIDMC Congestive Heart Failure Database (30 subjects), MIT-BIH Normal Sinus Rhythm Database (36 subjects)	MIT-BIH Arrhythmia Database (96 subjects), BIDMC Congestive Heart Failure Database (30 subjects), MIT-BIH Normal Sinus Rhythm Database (36 subjects)
5	Classification	Binary Classification: Normal vs. Abnormal	Multi-class Classification: ARR vs. CHF vs. NSR	Multi-class Classification: ARR vs. CHF vs. NSR
6	Accuracy Metrics	99.55% (PTB), 99.08% (MIT-BIH)	99.13% (ARR–CHF–NSR)	99.21% $\pm$ 1.05% (5-Fold Cross Validation) (ARR–CHF–NSR)
7	Key Findings	Heterogeneous AFs (tanh, linear, softsign, ELU, CReLU, ReLU6) extract richer and more diverse features than a fixed AF, improving accuracy with comparable inference speed	Combining CNN-based local feature extraction with ViT-based global attention achieves superior performance (99.13% accuracy and 99.94% ROC-AUC) compared to CNN-only and Transformer-only models	$\alpha = 0.7$ blending (70% scalogram + 30% binary features) consistently achieves the best classification performance across all dense-layer and machine learning classifier settings
8	Limitations	AF combinations were selected empirically; no hyperparameter optimization performed; scalability with larger datasets not investigated	ViT self-attention introduces quadratic computational complexity and high memory requirements; evaluation limited to 162 recordings	Optimal blending coefficient α is dataset-dependent and requires manual search; no automated α selection mechanism
9	Applicability	Applicable for wearable and point-of-care ECG applications requiring fast and lightweight inference on raw 1D signals, where CNN-based feature extraction and flexible ML classifiers enable efficient cardiac condition classification.	This model is particularly useful for CVDs which have both anatomical differences and rhythm problems, like Arrhythmia, CHF, and AF, wherein the local ECG characteristics, along with their temporal relationships, need to be captured.	This method is applicable for diagnosing heart ailments through multimodal imaging-based ECG analysis, in which scalogram representations are used along with complementary structural attributes, which makes it relevant for offline diagnostic purposes.

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